



Four decades of chemotherapy-induced cognitive dysfunction: comprehensive review of clinical, animal and in vitro studies, and insights of key initiating events

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Abstract

Cognitive dysfunction has been one of the most reported and studied adverse effects of cancer treatment, but, for many years, it was overlooked by the medical community. Nevertheless, the medical and scientific communities have now recognized that the cognitive deficits caused by chemotherapy have a strong impact on the morbidity of cancer treated patients. In fact, chemotherapy-induced cognitive dysfunction or ‘chemobrain’ (also named also chemofog) is at present a well-recognized effect of chemotherapy that could affect up to 78% of treated patients. Nonetheless, its underlying neurotoxic mechanism is still not fully elucidated. Therefore, this work aimed to provide a comprehensive review using PubMed as a database to assess the studies published on the field and, therefore, highlight the clinical manifestations of chemobrain and the putative neurotoxicity mechanisms.

In the last two decades, a great number of papers was published on the topic, mainly with clinical observations. Chemotherapy-treated patients showed that the cognitive domains most often impaired were verbal memory, psychomotor function, visual memory, visuospatial and verbal learning, memory function and attention. Chemotherapy alters the brain’s metabolism, white and grey matter and functional connectivity of brain areas. Several mechanisms have been proposed to cause chemobrain but increase of proinflammatory cytokines with oxidative stress seem more relevant, not excluding the action on neurotransmission and cellular death or impaired hippocampal neurogenesis. The interplay between these mechanisms and susceptible factors makes the clinical management of chemobrain even more difficult. New studies, mainly referring to the underlying mechanisms of chemobrain and protective measures, are important in the future, as it is expected that chemobrain will have more clinical impact in the coming years, since the number of cancer survivors is steadily increasing.

Keywords Chemobrain · Chemofog · Neurotoxicity · Cognitive impairment · Chemotherapy

Abbreviations

5-FU	5-Fluorouracil	BDNF	Brain-derived neurotrophic factor
3-NT	3-nitrotyrosine	CNS	Central nervous system
BBB	Blood-brain barrier	COX-2	Cyclooxygenase type 2
		CYA	Cyclophosphamide

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DOX	Doxorubicin
DTI	Diffusion-tensor imaging
EEG	Electroencephalogram
EORTC QLQ-C30	European organization for research and treatment of cancer quality of life questionnaire C30
ERP	Event-related potential
fMRI	Functional magnetic resonance imaging
GCEE	γ -glutamyl cysteine ethyl ester
GFAP	Glial fibrillary acidic protein
GSH	Glutathione
HNE	4-hydroxynonenal
HSCS	High-sensitivity cognitive screen
iNOS	Inducible nitric oxide synthase
MMSE	Mini mental state examination
MRI	Magnetic resonance imaging
MTX	Methotrexate
NF- κ B	Nuclear factor kappa B
PET	Positron emission tomography
RONs	Reactive oxygen and nitrogen species
ROS	Reactive oxygen species
TNF- α	Tumor necrosis factor alpha
VBM	Voxel-based morphometry

Introduction

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Cancer is the second leading cause of death worldwide, after cardiovascular diseases (Fitzmaurice et al. 2019). In 2018, 18.1 million new cases of cancer were diagnosed, and 9.6 million deaths occurred due to cancer. Over the years, cancer incidence has been increasing, mainly due to the growing and ageing population (Bray et al. 2018). Nevertheless, survival rate in most common cancer types has steadily increased over the years, mainly due to diagnosis at earlier stages, new and improved therapies and through the implementation of combined therapies [surgery, chemotherapy and radiotherapy (Aziz 2002)]. In fact, the American Cancer Society reported that in 2019 in the United States of America alone, there were 16.9 million cancer survivors and this number is expected to rise to more than 22.1 million by 2030 based on the 5-year survival rate (Miller et al. 2019). Nevertheless, cancer survivors are at risk for long-term physical and psychosocial side-effects, either because of cancer itself or the anticancer treatment (Arndt et al. 2017). Hence, the quality of life of cancer survivors has been gaining more attention, and the goal nowadays is not only to treat cancer, but also to ensure minimal long-term effects allowing an easier adjustment to life after treatment (Mayer et al. 2017).

Current approaches to treat cancer include radiotherapy, immunotherapy, surgery, hormonal therapy and chemotherapy (Urruticoechea et al. 2010). Despite the numerous and more

recent therapies available, chemotherapy is still the most used due to its high rate of success against a wide range of cancers and proven efficacy over the years (Shih et al. 2015). Chemotherapy uses drugs that have cytotoxic effects and mainly targets the dividing cells like the neoplastic cells by interfering with DNA synthesis, transcription or/and duplication or mitosis, being often used in combination (Luqmani 2005). Even so, chemotherapy is not specific for tumor cells and can affect healthy cells, having an extended list of short- and long-term adverse effects. The most common immediate adverse effects of chemotherapy are nausea and vomiting (Richardson et al. 2007), alopecia, anemia, myelosuppression (Olver et al. 2018) and haematotoxicity (Grisold et al. 2017). Chemotherapy can also cause cardiac toxicity, secondary leukemia (Seiter 2005), chronic fatigue and sleep disorders (Palesh et al. 2018). Of relevance, in the past decades, more attention has been given to the fact that chemotherapy can cause neurotoxicity (Becker et al. 2015; Wefel et al. 2004a).

The impact of cancer treatment-related adverse effects on the long-term patients' life quality, and also the burden that it may cause at the social and economic levels, have gained more relevance in the past years (Gutmann 2019). Forty to 60% of cancer survivors are younger than 65 years old and still a part of the labor force (Duijts et al. 2017) with high life expectancy post cancer. However, many cancer survivors struggle to return to work or to a normal life after cancer and cancer treatment (de Boer 2014). On that matter, the reported cognitive limitations (Becker et al. 2015; Wefel et al. 2004a) take an important aspect of post-cancer life and will be addressed next.

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Neurotoxicity of anticancer therapies

The past decades brought a new awareness for the neurotoxic effects of systemic chemotherapy, mainly due to the increasing use of chemotherapy in the treatment of cancer and the increasing number of long-term cancer survivors. However, that was not always the case and despite the recurrent complaints of memory and attention/concentration problems of the chemotherapy-treated patients since earlier reports, stress, anxiety and even memory loss were often linked to cancer diagnosis, not being much valorized as a relevant issue. Moreover, the conviction that chemotherapy agents could not cross the blood brain barrier (BBB) and, therefore, do not harm the brain, or even poor after-treatment follow-up, contributed to the overlook of these concerns. Actually, the systematic research on the cognitive side effects of chemotherapy only started effectively in the 1980s (Oxman and Silberfarb 1980). Nevertheless, the first papers revealing the putative link between cognitive impairment and neurotoxicity induced by chemotherapeutic agents were published in 1974 by Weiss and co-workers, highlighting the correlation

between the increase of neurologic problems in cancer patients and chemotherapy (Weiss et al. 1974a, b). Levine et al. also accentuated the need for assessment of the mental status in cancer patients throughout the treatment, especially the cognitive function, since they considered it to be the most neglected part of the patient care (Levine et al. 1978).

To the best of our knowledge, the first clinical study published evaluating putative cognitive deficits on cancer patients after chemotherapy was performed by Oxman and Silberfarb. The study compared nine women and one man that underwent cognitive testing before chemotherapy, 24 h and 1 month after the end of chemotherapy. The study found no differences between the three time points analyzed, but authors acknowledged that the small sample size and the practice effect due to the patients performing the same tests in a small-time frame, may have influenced the results. However, they also recognized the possibility that chemotherapy could induce cognitive dysfunction and the need for further studies to evaluate cognitive function (Oxman and Silberfarb 1980).

Ever since, numerous studies were done to assess the effects of chemotherapy on cognition, using standardized neuropsychological testing. However, at first, the terminology used to describe the possible cognitive dysfunction on treated cancer patients was very broad, such as “dementia” “delirium” (Folstein et al. 1984; Zimberg and Berenson 1990) or even unspecific like “organic brain disorder” (Davis et al. 1987). This terminology did not clarify what domains of cognition were affected. Additionally, only one or two tests were done to patients, usually for general cognition and not for specific domains (Joly et al. 2000; Pereira et al. 1997). Moreover, the tests first applied were not very sensitive and most of the times the simple observation of patients was used to detect cognitive changes (Oxman et al. 1986), thus were of subjective nature. The ‘practice effect’ that results from patients knowing and repeating the same tests at different time-points also caused apparent improvements overtime, not being an actual vision of reality (Cerulla et al. 2019; Freeman and Broshek 2002). Conversely, some studies started to assess stress and anxiety levels among cancer patients and found out that there was no correlation between the levels of stress and anxiety and the cognitive problems displayed by the patients (Fan et al. 2005; Wefel et al. 2004b).

Self-reported cognitive problems by chemotherapy-treated patients included difficulties in memory and concentration (Shilling and Jenkins 2007) and a feeling of “fog-iness” in their mind after receiving chemotherapy, which has been named “chemofog” or “chemobrain” (Weiss 2008). These terms started to be used among patients and in the lay press, being after adopted by the scientific community (Wefel et al. 2004a). The term “chemobrain” appeared in the scientific literature for the first time in 2004 in an article

by Wefel et al. (Wefel et al. 2004a). From that moment on, chemobrain or “chemofog” were used to describe the long-term cognitive deficits induced by chemotherapy (Gutmann 2019).

In the adult brain, the effects of chemotherapy were less considered and studied. In fact, it was assumed that several of the chemotherapeutic agents related to chemobrain could not cross the mature BBB. On the other hand, those that can cross the barrier were assumed that could not harm the brain since the neurons are post-mitotic cells and chemotherapy acts mostly on fast dividing and proliferative cells (Ahles and Saykin 2007; Morse et al. 2003). Despite neuroglial cells being as abundant as neurons and are able to undergo cell division in the adulthood, little attention was given to the effects of cytostatic agents on these cells (Kettenmann and Verkhratsky 2013). Additionally, the long-term neurotoxicity of chemotherapy was not a significant concern since a few decades back the life expectancy of cancer patients was generally short. But after systematic studies with well-controlled analysis, and because of the vast improvements in cancer treatment that lead to an increase in patient’s survival, that paradigm has shifted and quality of life of cancer survivors is now pivotal (Castellon and Ganz 2009) and chemobrain is in the frontline on that quest.

Bearing this in mind, the present comprehensive review aims to approach the issue of chemobrain on several aspects mainly to: (i) understand the effects of systemic chemotherapy on the central nervous system (CNS) and the clinical manifestations of its neurotoxicity and (ii) elucidate on the underlying mechanisms of chemotherapy-induced cognitive dysfunction or chemobrain. To meet these goals, the current literature available was reviewed, focusing on the progressive interest on this matter by the scientific community, clinicians and lay public.

Search strategy and selection criteria

To address chemobrain, its effects and mechanisms, a search on PubMed database was performed on February 2021 and included references from 1980 until January 2021 with the following keywords: chemobrain OR chemofog OR (cognitive dysfunction AND cancer) OR (cognitive impairment AND cancer). These keywords generated more than 4500 articles and not all of them were related to the effects of systemic chemotherapy on the brain. To narrow the number of references to the ones related to the topic, we excluded those related to smoking, neurologic or neurodegenerative diseases, anesthetics, encephalitis, seizures, epilepsy, diabetes, neurofibromatosis and patients only undergoing surgery. Moreover, depression, fatigue and articles analyzing the life quality of the cancer-treated patients that measured the cognitive function but

Fig. 1 Schematic representation of the workflow of the comprehensive review

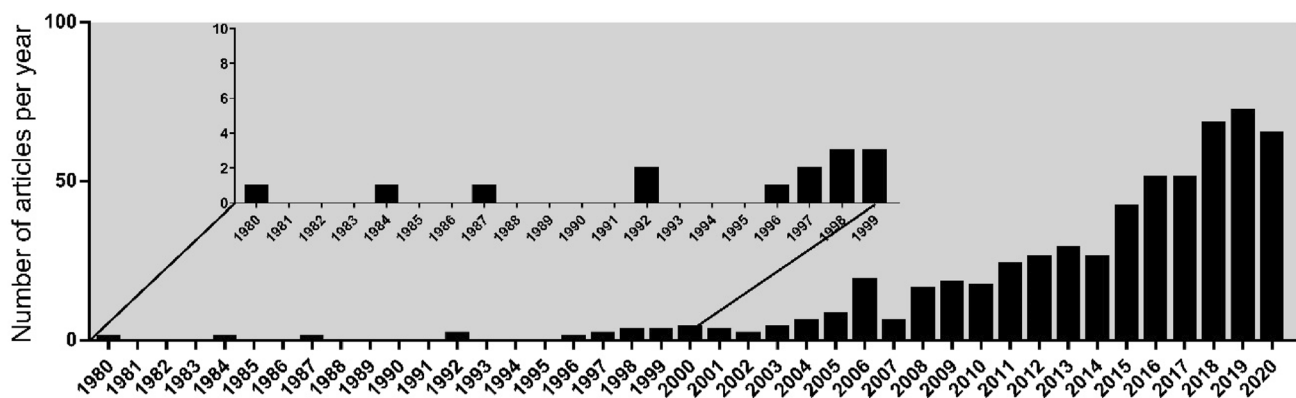
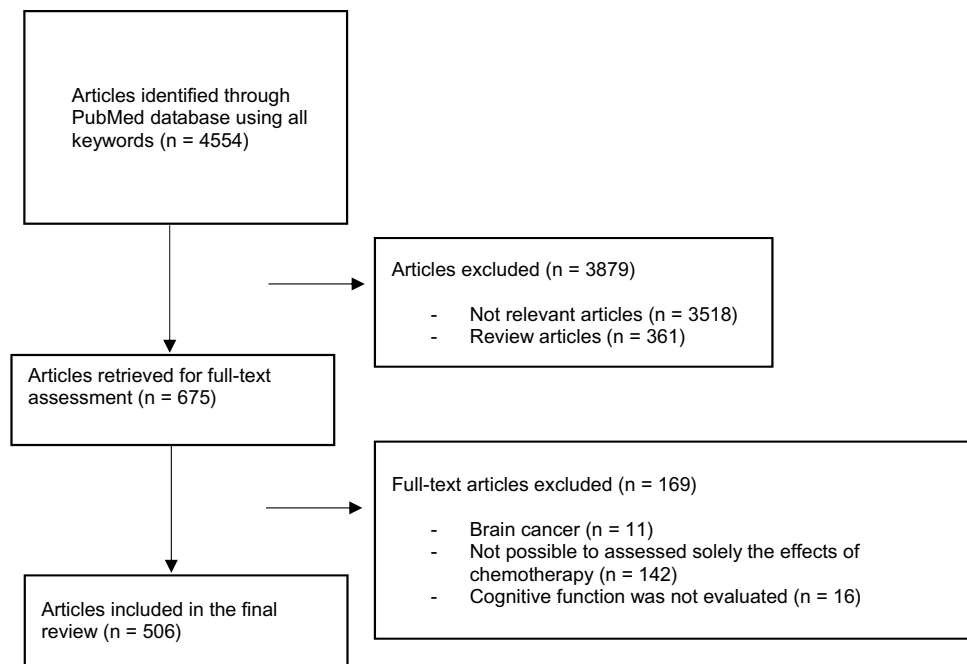


Fig. 2 Literature published in PubMed database per year on the effects of chemotherapy on the brain between 1980 and 2020

were not related to the therapy given were excluded. We also excluded studies related to cerebral cancer and brain metastases since these conditions could per se interfere with the cognitive function and not conclusively relate with any anticancer treatment undertaken (Gallego Perez-Larraya and Hildebrand 2014; Gerstenecker et al. 2014; Noh and Walbert 2018). Studies with patients treated with hormonal therapy or/and radiation were included only when it was possible to assess solely the effects of chemotherapy, as the effects of radiation are out of the scope of this review. A diagram representation of the workflow is presented in Fig. 1. Furthermore, only articles in English were used. After this initial selection, there were 506 references of interest (Fig. 1).

By analyzing the number of articles published each year regarding the subject matter in question, we can observe that before the year 2000 there were few articles published (14 articles) (Fig. 2). However, after 2000 an exponential increase in the number of articles published about this topic was seen. This tendency highlights the growing importance that chemotherapy-induced cognitive dysfunction has gained over the years and how new perspectives and interest has arisen in the clinical and scientific community (Fig. 2).

Regarding the type of articles, clinical studies represent more than half of the published articles about chemotherapy-induced cognitive dysfunction, highlighting not only the impact of this subject for cancer patients and physicians, but also the need for more mechanistic studies to disclose the underlying neuronal toxicodynamics (Fig. 3). The

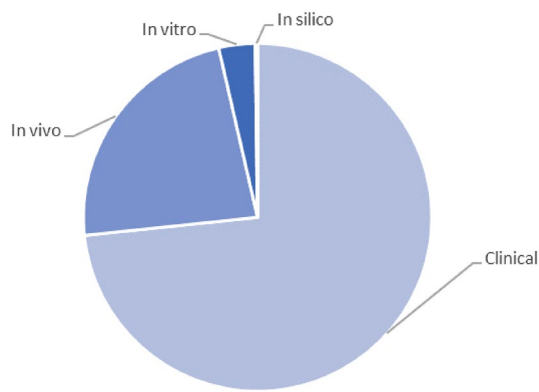


Fig. 3 The number of articles of interest published in each category following a systematic selection in PubMed from 1980 until 2020

in vivo studies outdo by far the in vitro studies. The clinical studies comprise predominantly neuropsychological studies, electrophysiological and neuroimaging techniques. The most relevant studies in each category will be analyzed next to understand the effects of chemotherapy on the brain and unveil some of the underlying neurotoxicity mechanisms.

Neuropsychological testing in humans

Chemotherapy-induced cognitive dysfunction can be defined as changes in cognitive domains such as memory, attention, processing speed and executive function that appear during the chemotherapeutic treatment and last longer after the cessation (Deprez et al. 2013). A large bulk of literature emerged, trying to evaluate the prevalence of chemotherapy-induced cognitive dysfunction on cancer patients. To evaluate the changes in cognition, physicians used mainly standard batteries of neuropsychological tests and electrophysiological and neuroimaging techniques. These studies are predominantly longitudinal and have cross-sectional designs.

Actually, a multitude of neuropsychological tests have been developed throughout the years to assess the various domains of cognitive function, such as attention and concentration, executive function, information and processing speed, language, motor function, visuospatial skill, learning and memory (Myers 2010). Nevertheless, the use of objective and standard tests requires trained professionals, usually either a neuropsychologist or a psychometrician, and, depending on the battery of tests used, they could take between 4 and 7 h per patient (Freeman and Broshek 2002). Moreover, the tests chosen to evaluate cognitive dysfunction need to be validated, sensitive and specific to the domain being studied, and also replicable and feasible (Jansen et al. 2007). Only objective evaluation can in fact confirm or

dismiss the effect of chemotherapy on the patients' cognitive function.

Summary data and results extracted from studies that evaluated cognitive function through neuropsychological testing on chemotherapy-treated patients are presented in Table 1, highlighting the disparities among testing strategies, results and groups matching. The data summarized in Table 1 highlights several aspects as follows: (1) the number of patients per study is very diverse and clearly influences its conclusions, with larger studies being more prone to detect cognitive dysfunction; (2) women (with breast cancer) are the large majority of patients included in the studies; (3) anthracyclines [e.g. doxorubicin (DOX), epirubicin] are the most represented and studied drugs; (3) the studies that undergo baseline evaluations (before treatment) are more prone to detect cognitive dysfunctions in the course or after treatment; (4) so far, it is not clear if age of diagnosis or age of treatment impacts on cognitive dysfunction outcome; (5) although hormonal therapy has been described (Eberling et al. 2004) as causing cognitive dysfunction, when placed with chemotherapy, the effects of the latter seem to cause more cognitive dysfunction; (6) the effect of practice (or education on neurophysiological tests) is commonly seen at baseline evaluation and at longer periods of evaluation, being a bias because it may mask underlying and/or persisting cognitive dysfunction; and (7) whether chemobrain is long lasting is still not clear and other studies are required. Despite some doubts regarding the chemotherapy-induced cognitive dysfunction, it is unquestionable, at this point, that chemotherapy affects patients' cognition.

Before chemotherapy-induced long-term cognitive impairment was proven, only a few of the clinical trials done in cancer patients incorporated neuropsychological assessment as part of the routine evaluation. One of the first studies was performed by van Dam and colleagues (van Dam et al. 1998) who found that chemotherapy-treated patients had more subjective complaints of memory and concentration loss. In the neuropsychological testing, patients treated with high-dose chemotherapy had higher cognitive impairment, 8.2 higher than the control group (consisting of breast cancer patients not treated with chemotherapy) and 3.5 higher than the standard dose chemotherapy-treated patients. In fact, the high-dose group had impaired attention/concentration, speed of information processing, visual memory, visuospatial and motor function. Moreover, the authors found that fatigue, stress and depression tests scores did not correlate with the cognitive deficits observed (van Dam et al. 1998). This kind of matching, with cancer-diagnosed patients with no treatment other than surgery, eliminates the question of stress and anxiety regarding cancer diagnosis.

The longitudinal studies that started to emerge mainly in the last decades tried to bridge some limitations of the early studies and provided a deeper insight into the cognitive

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Table 1 Clinical studies with neuropsychological testing and major findings

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
10 patients admitted for chemotherapy treatment	Patients received different regimens that included doxorubicin (DOX) with varying combinations of cyclophosphamide (CYA), cis-platinum and vincristine	Patients were evaluated before the start of chemotherapy (T1) with vocabulary, reading and information subtests of the Wechsler Adult Intelligence Scale, the Halstead-Reitan Battery, 24 h (T2) after chemotherapy and a month after (T3), the patients did only the Halstead-Reitan Battery tests	No differences were found between the 3 time points	Oxman and Silberfarb (1980)
91 patients diagnosed with lymphoma (Hodgkin's and non-Hodgkin's) that were cancer and treatment-free for more than 6 months. The patients were divided in 2 groups: the ones that complaint of difficulties in memory and concentration or no-complainers	Chemotherapy single agent (11 patients) or combination of more than one agent (51 patients). Twenty-seven patients did not received chemotherapy	Subjective reports: European Organization for Research and Treatment of Cancer Quality of Life Questionnaire C30 (EORTC QLQ-C30), Hospital Anxiety and Depression scale, Multidimensional Fatigue Inventory, Brief Mental Fatigue Questionnaire, Memory Aids Objective tests: The National Adult Reading Test, Paced Auditory Serial Addition Task, and Rivermead Behavioural Memory Test Mini mental state examination (MMSE) test in average every 4 days until they were discharged	Complainers and non-complainers did not have significant differences in the neuropsychologic test scores. Complainers score higher on measures of anxiety, depression, and fatigue No comparisons between the patients that did or did not received chemotherapy were made	Cull et al. (1996)
321 patients in palliative care with non-CNS cancer	Not described	Mini mental state examination (MMSE) test in average every 4 days until they were discharged	142 patients presented irregular MMSE scores and 176 patients had altered scores at time of death or dismissal Of the 240 patients that did more than 2 MMSE tests, 54 presented normal scores in the first test but the final scores were abnormal	Pereira et al. (1997)
Breast cancer patients treated with high-dose chemotherapy plus tamoxifen ($n = 34$) or treated with standard-dose chemotherapy ($n = 36$) 34 control patients with stage I cancer and no chemotherapy (all patients admitted to the study were off hormonal treatment for at least 6 months)	Standard dose consisted of 4 cycles of 5-fluorouracil (5-FU), epirubicin and CYA and the high-dose regimen included a fifth course of CYA, thiotepa and carboplatin	Battery of 19 neuropsychologic tests	Patients treated with high dose chemotherapy had higher cognitive impairment (8.2 higher fold comparing to the control group and 3.5 higher fold than the standard dose chemotherapy patients)	van Dam et al. (1998)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
Group A: 39 breast carcinoma patients treated with adjuvant chemotherapy (mean time since last treatment was 2 years) Group B: 34 control patients with axillary lymph node negative breast carcinoma not treated with adjuvant chemotherapy: patients only underwent surgery	Group A treatment: 6 cycles of CYA, methotrexate (MTX), 5-FU followed by 20 mg of tamoxifen daily for 3 years (20 patients) or not (19 patients)	Battery of 14 neurologic tests screening for a wide range of functions mainly focusing on the cognitive function	Group A patients presented significantly more concentration and memory problems than group B Cognitive impairment was found in 28% patients of group A versus 12% in group B No differences in performance were found in cancer patients treated with chemotherapy alone or chemotherapy plus tamoxifen	Schagen et al. (1999)
Breast cancer patients receiving adjuvant chemotherapy ($n = 31$, group A) and patients that had completed the treatment at least 1 year before the study ($n = 40$; Group B) and 36 healthy control women (Group C)	Group A received every 4 weeks cycles of CYA, epirubicin and 5-FU for at least 8 weeks Group B completed a full course of adjuvant chemotherapy at least a year before	High Sensitivity Cognitive Screen (HSCS) and Profile of mood states	Group A presented cognitive impairment when compared to group C Group B had intermediate scores but with no substantial difference when compared with group A and C	Brezden et al. (2000)
5 years post-systemic chemotherapy cancer survivors (breast cancer $n = 35$; lymphoma $n = 36$) and patients who received local therapy only (breast cancer $n = 35$; lymphoma $n = 22$)	Systemic chemotherapy was indicated as 'standard dose chemotherapy', while no specific regimens were mentioned Local therapy includes radiation and surgery	Battery of neuropsychologic tests and self-reported measures of cognitive and psychologic function	Patients treated with systemic chemotherapy presented dramatically lower scores on the neuropsychologic tests than the patients with local treatment, mainly in the domains of verbal memory and psychomotor function	Ahles et al. (2002)
109 patients on 3 different cancer treatment regimens and 27 control patients with stage I breast cancer	5 cycles of chemotherapy (5-FU, CYA and epirubicin), radiotherapy and tamoxifen ($n = 36$) or high-dose chemotherapy where the fifth course of chemotherapy was replaced by CYA, thiotepa and carboplatin with peripheral blood progenitor cell transplantation ($n = 34$) 6 cycles of CYA, MTX and 5-FU ($n = 39$) The control group, after surgery, was only treated with radiotherapy	Series of neuropsychological tests 4 years post-therapy (T1) and 2 years after the last evaluation (T2)	At T1, 28% of the patients had cognitive dysfunction comparing to 12% in the control group. However, at T2 there were no differences among the set groups	Schagen et al. (2002)
Long-term survivors of breast cancer with the allele E4 of the Apolipoprotein E (ApoE) ($n = 17$) or absence ($n = 63$)	Standard-dose systemic chemotherapy	Series of neuropsychological tests ApoE genotyping	Patients with at least one allele E4 had lower scores on the domains of visual memory and spatial ability when compared to no-carriers	Ahles et al. (2003)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
100 women treated with adjuvant chemotherapy for breast cancer, each one matched by age and education with a healthy control person	Several chemotherapy regimens including: 6 monthly cycles of CYA, MTX and 5-FU or 6 cycles of CYA, epirubicin and 5-FU or 4 cycles of DOX and CYA sometimes followed by paclitaxel	Several neuropsychological tests including the MMSE	Cancer patients presented higher levels of cognitive dysfunction when compared with the matched control. In addition, the patients experienced more fatigue, menopausal symptoms and poorer quality of life	Tchen et al. (2003)
53 breast cancer survivors (2–5 years post-diagnosis): 36 received adjuvant chemotherapy either alone ($n = 18$) or plus tamoxifen ($n = 18$) and 17 did not received chemotherapy 19 healthy controls were used as comparison	Chemotherapy regimens included several combinations: CYA, MTX and 5-FU; a DOX-containing regimen along with CYA alone or with CYA, MTX and 5-FU; DOX, CYA and taxane	Neurocognitive performance tasks, self-reported cognitive function and assessment of depression, anxiety and fatigue	Among the cancer patients, those who received adjuvant chemotherapy scored significantly lower in the domains of visual memory, visuospatial and verbal learning compared to cancer patients not treated with chemotherapy Cancer patients that received chemotherapy plus tamoxifen had the worse scores among all groups	Castellon et al. (2004)
84 patients with breast cancer	Patients received 5-FU, DOX and CYA	Battery of neuropsychologic tests before chemotherapy (baseline), short-term post-chemotherapy (3 weeks) and long-term post-chemotherapy (1 year)	Before chemotherapy, 33% of the patients presented cognitive impairment, when comparing to published normative values. Short-term post-chemotherapy assessment revealed cognitive impairment in the domains of verbal learning (18%) and memory function (25%) when compared to the baseline evaluation. Furthermore, nonverbal memory (17%), psychomotor processing speed and attention (13%), confrontation naming (13%), visuospatial construction (13%), and upper-extremity fine motor dexterity (12%) were significantly impaired At the long-term evaluation, 50% of the patients that were previously cognitively impaired demonstrated improvement	Wefel et al. (2004b)
46 patients with advanced pancreatic adenocarcinoma	Every two-weeks patients received oxaliplatin and irinotecan, leucovorin and 5-FU	Quality of life was assessed using EORTC QLQ-C30 questionnaire	Between baseline and end of treatment, patients had improvement in all functional scales of the EORTC QLQ-C30, except cognitive function, that decreased	Conroy et al. (2005)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
Women diagnosed with early-stage breast cancer received chemotherapy plus radiotherapy ($n=60$) or radiotherapy only ($n=83$)	Not described	Battery of neuropsychologic tests	No significant differences were found between the women treated with chemotherapy plus radiotherapy or radiotherapy alone regarding their cognition	Donovan et al. (2005)
104 breast cancer patients did a first evaluation at the end of 3 cycles of chemotherapy One-year post-chemotherapy, 91 of these patients were evaluated. Two-years post-chemotherapy, 81 of the patients were evaluated again Cancer patients were matched with an equivalent healthy control	6 cycles of CYA, epirubicin, and 5-FU or 4 cycles of DOX and CYA	Battery of tests to evaluate cognitive function, levels of depression and fatigue	At the initial assessment, 16% of cancer patients presented moderate-severe cognitive dysfunction comparing to 5% in the control group. One year after, the patients with cognitive dysfunction decreased to 4.4% and at the 2-year follow up, decreased to 3.8%, versus 3.6% and 0% respectively in the control group	Fan et al. (2005)
702 cancer survivors over the age of 65 years were compared with their cancer-free twins	The authors did not have access to the cancer treatments received by the patients	Cognitive screening assessing orientation, short-term memory, general knowledge, verbal recall and verbal abstract reasoning Dementia screening using Diagnostic and Statistical Manual-IV criteria	Long-term cancer survivors presented higher rates of cognitive dysfunction than their respective twin	Heflin et al. (2005)
54 patients with acute myelogenous leukaemia or myelodysplastic syndrome evaluated before the start of the treatment, of which 26 of these patients were also evaluated after 1 month after treatment	Lipodaunocin plus CYA or topotecan with or without thalidomide	Battery of neuropsychologic tests Serum levels of multiple cytokines	Before starting the treatment, a significant number of patients displayed cognitive dysfunction. They also had elevated levels of cytokines when compared with normal controls At 1-month follow-up, in the 26 patients analyzed, there was significantly worsen of fine motor control and fatigue, when compared to baseline. The cytokines levels were not measured at this time point	Meyers et al. (2005)
100 women with breast cancer scheduled to received adjuvant chemotherapy 50 cancer patients not receiving chemotherapy 59 healthy controls	Not described	Battery of cognitive tests at 2 time points: T1 (baseline evaluation 4 weeks after the final session of chemotherapy) and T2 (18 months post-chemotherapy)	Overall, chemotherapy treated patients were 2.5 more likely to have cognitive impairment than the healthy controls and the patients not receiving chemotherapy	Shilling et al. (2005)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
Group 1: 19 women with hormone receptor negative breast cancer received only chemotherapy	In group 1, 20% received CYA, MTX and 5-FU and 80% received CYA and DOX	Cognitive function measures at baseline, prior to chemotherapy (T1), 6 months after chemotherapy (T2) and 18 months (T3)	There were no significant differences between groups in the baseline evaluation Group 3 improved scores from T2 to T3 At T3, groups 1 and 2 performed poorly when compared to group 3	Bender et al. (2006)
Group 2: 19 women with hormone receptor positive breast cancer received chemotherapy plus tamoxifen	In group 2, 75% received CYA and DOX with (56%) or without (44%) a taxane. Three women in group 2 received DOX plus docetaxel			
Group 3: 19 control women with ductal carcinoma in situ that did not received chemotherapy nor tamoxifen				
34 old patients (> 60 years) and 43 younger patients (> 18 and < 60 years). All the ages had either malignant hematological diseases or cancer in the gastrointestinal tract	Not described	Battery of neuropsychological tests to assess the cognitive function for baseline evaluation and then a 6-months follow-up	Six months after diagnosis, the scores on the tests improved considerably for all cognitive domains, which was not correlated to the age of the patients	Eberhardt et al. (2006)
798 patients with ovarian cancer	Patients were randomly assigned to receive either paclitaxel/carboplatin or paclitaxel/cisplatin	Quality of Life Questionnaire before treatment, within 3 days before, the 2nd and the 4th chemotherapy cycles, 3 weeks after six cycles of chemotherapy and at every 6 months follow-up	Patients in the paclitaxel/carboplatin regimen presented overall better cognitive function than those on the paclitaxel/cisplatin regimen in the follow-up assessments	Greimel et al. (2006)
45 patients ≥ age 65 with stage I to III breast cancer	All patients received adjuvant chemotherapy. 31% received an anthracycline-based regimen and the others received CYA, MTX, and 5-FU	The Squire Memory Self-Rating Questionnaire (pre-chemotherapy baseline evaluation and post-chemotherapy evaluation)	51% of the patients perceived a decline in memory when comparing to the baseline evaluation	Hurria et al. (2006a)
28 patients ≥ age 65 with stage I to III breast cancer	71% received CYA, MTX, 5-FU every 3 weeks for 8 cycles; 7% received DOX and CYA every 2 or 3 weeks for 4 cycles; 18% received DOX and CYA followed by paclitaxel and 4% DOX, CYA and paclitaxel followed by trastuzumab	All patients completed age-adequate cognitive assessments before chemotherapy (T1), at the end of chemotherapy (T2), and 6 months after chemotherapy (T3)	Before chemotherapy, 11% of the patients scored 2 standard deviation (SD) below the standard values on two or more neuropsychological tests. Six-months after chemotherapy (T3), 29% of the patients had 2 SDs below the standard values on two or more neuropsychological tests	Hurria et al. (2006b)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
85 women scheduled to receive chemotherapy 43 women received either hormonal therapy or/and radiotherapy 49 healthy controls	Most of the treated patients received 6 cycles of 5-FU, epirubicin and CYA	Cognitive assessments were made at baseline (T1), 4 weeks after the final chemotherapy session (T2) and 12 months after the last session of chemotherapy (T3)	After normalizing the groups for age and intelligence, repeated measure analysis found no relevant differences between the groups On an individual level, there was a decline in more than 2 measures, from T1 to T2, in 20% of the chemotherapy-treated group, versus 25.65 and 18.4% in the non-chemotherapy and control group, respectively. In T3, the chemotherapy group had a decline on cognitive function of 18.1% versus 14.6% and 10.6% in the non-chemotherapy and control group, respectively	Jenkins et al. (2006)
35 patients with metastatic gastric cancer admitted for second-line chemotherapy	Most patients treated received epirubicin, cisplatin, 5-FU or capecitabine The first line of treatment, in most cases, consisted of cisplatin-based chemotherapy and surgery	EORTC QLQ-C30 questionnaire before the start of the treatment and at the start of each cycle	There was an improvement in the cognitive parameters from the start of the chemotherapy until the last cycle of treatment	Park et al. (2006)
71 patients with gastric adenocarcinoma randomized to receive one of two regimens	Regimen A ($n = 35$): epirubicin, cisplatin and 5-FU Regimen B ($n = 36$): docetaxel, cisplatin and 5-FU	EORTC QLQ-C30 questionnaire before the start of the treatment and after the third treatment	Before the start of the treatment, there were no differences among the groups. After treatment, there was improvement, in both groups, in all domains except the cognitive domain that deteriorated Both treatments affected the patients in a similar way	(Sadighi et al. 2006)
39 cancer patients treated with regimen A 28 cancer patients treated with regimen B 57 cancer patients that did not received chemotherapy 60 healthy controls	Regimen A: 5 cycles of 5-FU, epirubicin and CYA Regimen B: 4 cycles of regimen A followed by CYA, thiotepa and carboplatin	Neuropsychologic examination before the start of the treatment and 6 months after the end of treatment	No differences between groups were observed in the first assessment Patients in regimen B (high-dose chemotherapy) presented a decline in cognitive performance when compared to the healthy control group 6 months after treatment. This decline was not observed in any other group	Schagen et al. (2006a,b)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
100 women scheduled to receive adjuvant chemotherapy and 53 patients scheduled to receive radiotherapy and/or endocrine therapy or no further treatment	Not described	Cognitive assessments at baseline (T1), 4 weeks after the final chemotherapy session (T2), and 12 months after the final chemotherapy session (T3)	Women in the chemotherapy-treated group self-reported more memory and concentration problems than the women in the non-chemotherapy group There was no correlation between self-reported cognitive problems and objective measurements	Shilling and Jenkins (2007)
18,360 women diagnosed with breast cancer (age ≥ 65 years) of whom 6289 received chemotherapy and 12,071 did not received chemotherapy	Not described	International Classification of Diseases, Ninth Revision (ICD-9) diagnoses of dementia to evaluate the patients during the 5-years period post-treatment	In the first years after breast cancer diagnosis, the higher number of dementia diagnosis occurred in group not treated with chemotherapy; however, after this time point, there was a shift and the group who received chemotherapy had higher rates of dementia	Heck et al. (2008)
50 patients with solid tumors and hemoglobin levels higher than 11.0 g/dL	Erythropoietin 40 000 U once a week for 12 weeks and standard-dose chemotherapy	MMSE and EORTC QLQ-C30 at the beginning of the erythropoietin treatment and at week 12	At admission, 16% of the patients were diagnosed with cognitive impairment. Throughout the evaluation period, the improvement in cognition was very little, but there was an increased in the quality-of-life scores There was no association between hemoglobin levels and cognition deficits	Iconomou et al. (2008)
30 women with breast cancer	Regimen with CYA and DOX alone or followed by taxane	Repeatable Battery of Adult Neuropsychological Status prior to the treatment (T1), 1 week after the fourth cycle of treatment (T2) and 6 months after the completion of chemotherapy (T3)	At T1, 4 women had cognitive impairment namely in memory, language and visuospatial skill Overall, in T2, visuospatial skill and total cognition scores decreased over-time. On contrary, executive function improved over-time On an individual level, 10 women (33%) decreased one or more SDs on two or more tests in T3	Jansen et al. (2008)
57 cancer patients scheduled to received chemotherapy	Patients treated with chemotherapy were randomized to receive methylphenidate ($n = 29$) or placebo ($n = 27$)	MMSE HSCS Revised Hopkins Verbal Learning Test The tests were performed before the treatment (T1), at the end of the treatment (T2) and 6 months after the treatments' end (T3)	There were no significant differences in cognition in the 3 time points evaluated between the methylphenidate group and the placebo group	Mar Fan et al. (2008)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
Cancer patients aged more than 65 years divided into 3 groups: no-chemotherapy (control group; $n=32$); chemotherapy group ($n=16$) and hormonal therapy group ($n=13$)	Chemotherapy treatments differed among patients but included 5-FU, CYA, MTX, DOX, paclitaxel and epirubicin The hormonal therapy was anastrozole or tamoxifen	Cambridge Cognitive Examination performed before chemotherapy or endocrine therapy, (T0) and after 3 (T1) and 6 months (T2)	Overall, there was no deterioration in cognitive function in the 3 time points tested for the 3 groups. However, in the chemotherapy group, more patients displayed reduction of the scores in the domains of learning memory and total memory, from T1 to T2	Minisini et al. (2008)
Patients with testicular cancer divided into 3 groups: group 1 ($n=70$) received chemotherapy after surgery; group 2 ($n=57$) were only treated with radiotherapy after surgery; group 3 ($n=55$) patients received surgery only	4 courses of bleomycin, etoposide, and cisplatin	Battery of neuropsychological tests approximately 3 years after the finalization of treatment	Both in group 1 and group 2, 32% of the patients reported cognitive problems. However, in the neuropsychological tests done no differences between groups were seen. In individual scores, patients in group 1 presented more cognitive dysfunction when compared to group 3, but no differences were observed when compared to group 2	Schagen et al. (2008)
61 breast cancer patients scheduled to receive chemotherapy and 51 control women receiving hormonal therapy	Most women received anthracycline-based chemotherapy	Neuropsychological testing after surgery and prior to initiation of any chemotherapy (T1) and after the last chemotherapeutic treatment (T2)	Mean scores for both groups were similar and within the norms for cognitive tests; however, on individual scores in T2, chemotherapy-treated patients were 3.3 times more likely to had cognitive decline than the hormonal therapy group	Stewart et al. (2008)
38 women who received chemotherapy 5–39 months prior to the study (39–71 years old) and 55 control healthy women aged matched	Most women received regimens that included CYA plus MTX and 5-FU, or anthracycline-based regimens	Useful Field of View assessment program	Older (more than 54 years old) chemotherapy-treated women performed worse on speed of processing than controls and younger chemotherapy-treated women	Adams-Price et al. (2009)
87 patients 12–30 months post-chemotherapy of whom 45 received epoetin-alfa and 42 received standard care	Most patients received an anthracycline-based regimen or received CYA, 5-FU, MTX and taxane. Besides the chemotherapy treatment, the patients were randomized to receive epoetin-alfa	MMSE HSCS Revised Hopkins Verbal Learning Test Functional Assessment of Cancer Therapy questionnaires assessing fatigue and general life quality Hospital Anxiety and Depression Scale	Although there were no significant differences in cognition with or without epoetin-alfa, those whom received epoetin-alfa reported better quality of life	Fan et al. (2009)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
187 breast cancer survivors of whom 97 patients were treated with chemotherapy only or chemotherapy plus radiotherapy and 90 patients received radiotherapy only	Most chemotherapy regimens included DOX and CYA	Battery of neuropsychological tests 6 months after completion of treatment	Small differences in cognition were found in women treated with chemotherapy when compared to the age-matched controls No differences were observed among the other groups	Jim et al. (2009)
187 aged-matched healthy controls				
76 breast cancer patients that completed chemotherapy only or with radiation and presented cognitive dysfunction took modafinil for 4 weeks, and then were randomized to continue the medication for another 4 weeks or take placebo pills	Not described	Tests of memory and attention selected from the Cognitive Drug Research before modafinil, 4 weeks during treatment and 8 weeks after the initiation of the treatment	On the first 4 weeks of the treatment, modafinil improved speed of memory and quality of episodic memory. Patients that continued to take the medication improved when comparing to the patients on placebo	Kohli et al. (2009)
36 female breast cancer patients receiving chemotherapy				
14 cardiac patients (cancer-free)	Chemotherapeutic regimen consisted of CYA, epirubicin and 5-FU	Battery of neuropsychological tests performed before the beginning of chemotherapy, 4–6 weeks after the last treatment and 25 weeks after the first test. Cardiac patients were tested 4 days after hospitalization and at a 3-month follow up. Healthy controls were tested within an interval of 12–16 weeks	No differences were found between groups tested. Cancer patients did not differ from healthy controls neither before nor after chemotherapy	Mehlsen et al. (2009)
17 healthy controls				
72 men treated for testicular cancer 2 to 7 years prior the assessments were divided in 2 groups: Group A: 36 men treated with chemotherapy Group B: 36 men treated with radiotherapy or surgery	The chemotherapy group had been treated with three or four cycles of bleomycin, etoposide, and cisplatin	Battery of standardized neuropsychological tests	No differences in cognition were found when comparing groups A and B	Pedersen et al. (2009)
Breast cancer patients divided into 2 groups: 41 patients received chemotherapy followed by radiotherapy in 38 of these patients. The other group of 40 patients received radiotherapy without chemotherapy The study also included a control group of 45 healthy women	23 patients received DOX and CYA; 12 patients received taxotere, DOX and CYA; 6 patients received 5-FU, epirubicin and CYA	Battery of neuropsychological tests prior to the initiation of the treatment (T1), instantly after the first treatment (T2) and following the end of the last treatment (T3)	Both groups had difficulties for recruiting information in verbal memory, compared to T1. However, only patients treated with chemotherapy presented impaired verbal fluency	Quesnel et al. (2009)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
46 women receiving chemotherapy for stage II–III breast cancer	Patients received DOX and CYA with or without paclitaxel or taxotere	Neuropsychological tests to evaluate skills in five cognitive domains: immediate memory, delayed memory, attention, executive function and verbal fluency. Tests were performed 1 month following the end of chemotherapy	Patients had cognitive deficits in executive functioning and verbal fluency when comparing to calculated expected values	Reid-Arndt et al. (2009)
136 women with breast cancer scheduled to receive chemotherapy and 21 women also with breast cancer but not receiving chemotherapy	Several chemotherapeutic regimens that included 5-FU, epirubicin, CYA, DOX, MTX	Neuropsychological tests and self-report measures at baseline (before chemotherapy and after surgery) and 4 weeks after the last dose of chemotherapy	16.9% of the women treated with chemotherapy presented decline in verbal learning and memory abstract reasoning, and motor coordination. Moreover, the study also found that a decline in hemoglobin and increase in anxiety, over the course of chemotherapy, significantly predict impairment in multiple cognitive measures	Vearncombe et al. (2009)
Cancer patients scheduled to be treated with chemotherapy ($n=60$) and cancer patients not receiving chemotherapy ($n=72$) and 45 female healthy controls	Several chemotherapeutic regimens that included combinations of DOX, CYA, paclitaxel, taxotere, 5-FU, epirubicin and MTX	Standardized neuropsychological tests For chemotherapy-treated patients, the tests occurred after surgery but before the initiation of the treatment and 1, 6 and 18 months after treatment end. The other patients and healthy controls had the same testing scheduled	Older patients (60–70 years old) had lower cognitive reserves in the pre-treatment and performed worse after the treatment compared to non-chemotherapy patients and healthy controls (of the same age range)	Ahles et al. (2010)
122 breast cancer survivors free from chemotherapy at least 1 year but no more than 10 years and working actively 113 healthy control females also working actively	Not described	Perceived Cognitive Function: Cognitive Symptoms Checklist-Modified and Performance-Based Cognitive Function: Central Nervous System Vital Signs	Breast cancer survivors had more cognitive difficulties related to the work output	Calvio et al. (2010)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
Group 1: 22 women with invasive epithelial ovarian cancer disease-free and no relapses 5 to 10 years from the original diagnosis Group 2: 26 women with active invasive epithelial ovarian cancer who had at least one recurrence but were disease-free at 5 to 10 years from the original diagnosis Group 3: 8 women who had one or more relapses invasive epithelial ovarian cancer but were disease-free at 5 to 10 years from the original diagnosis	Most women had cycles of paclitaxel, carboplatin with or without cisplatin	Standardized neuropsychological tests	The average score on the tests did not differ significantly among groups, and the average score of the groups is within the range of healthy normative scores. Nevertheless, 28% of the women across the 3 groups had scores consistent with cognitive impairment	Correa et al. (2010)
124 breast cancer women aged < 60 years and 224 healthy controls aged-matched to the patients	Among the breast cancer patients, 75 received CYA and 5-FU every 3 weeks for seven cycles, 26 patients received tamoxifen and 19 received no adjuvant treatment	Neuropsychological testing was performed before any adjuvant treatment (T1) and approximately 6 months after the first test (T2)	No differences were found between groups in both time-points	Debess et al. (2010)
62,565 women who were diagnosed with stages I–IV breast cancer and were over 65 years old and cancer-free up to 16 years post-diagnosis and 9752 matched cohort that did not received chemotherapy	Not mentioned	Data base analysis of patients reports over the years	Chemotherapy-treated patients were 8% more likely to have cognitive impairments and dementia when compared to those without chemotherapy, but that was not statistically significant after adjusting for patient and tumor characteristic	Du et al. (2010)
101 cancer patients	Patients were randomized to receive one of two treatments: Treatment A (48 patients): epirubicin, CYA and paclitaxel Treatment B (53 patients): epirubicin and paclitaxel combined with CYA, MTX and 5-FU In each group, about half of the patients also received erythropoietin	The neuropsychological evaluation and self-perceived cognitive function took place before the start of preoperative chemotherapy (T1), before the last cycle of chemotherapy (T2, about 5 months later), and about 1 year after the first evaluation (T3)	No significant changes in self-reported cognitive changes nor in the neuropsychological tests were found	Hermelink et al. (2010)
27 women with confirmed diagnosis of stage III–IV epithelial ovarian or primary peritoneal cancer	Patients did at least six courses of platinum-based therapy	Battery of standardized neurocognitive tests before the initiation of chemotherapy, at the 3rd course and at the 6th course of chemotherapy	In the third course of chemotherapy, 92% of the women displayed symptoms of cognitive dysfunction when compared to baseline. The number declined to 86% from the 3rd to the 6th course of chemotherapy	Hess et al. (2010)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
61 women aged between 45–70 years old with breast cancer and post-menopausal. 30 of these women received chemotherapy and the remaining did not and were taken as controls	14 women received DOX, CYA and docetaxel/paclitaxel; 7 women received DOX, CYA and the other 9 received CYA, MTX and 5-FU	Battery of standardized neuropsychological tests and self-rated surveys at 3 times: before treatment (T1), 6 months after the first test (T2) and then 12 months after (T3)	The motor domain was the only one significantly worse in women treated with chemotherapy. In post-menopausal women, chemotherapy seems not to induce cognitive deficits	Tager et al. (2010)
42 patients with breast cancer	Patients were treated with 5-FU, DOX, CYA with or without paclitaxel	Battery of cognitive tests, mood measures and quality of life questionnaire. Tests were conducted before treatment, and then on average 2.9 months and 7.0 months (acute interval) and 13.1 months after the first assessment (late interval)	21% of the patients met the criteria of cognitive impairment before any treatment and the domains more affected were learning and memory, executive function, and processing speed. In the acute interval, the number increased to 65% being the same domains the most affected. In the late interval evaluation, 61% of the patients had cognitive decline, of whom, 30% presented new domains affected that were not affected before, mainly learning and memory impairments	Wefel et al. (2010)
30 women aged over 65 years and long-term survivors of breast cancer. They were at least 50 years old and at least 10 years post-chemotherapy. The patients had no cancer recurrence Each survivor was matched to a healthy control person	Patients were administered with different combinations of anticancer drugs that included CYA, MTX and 5-FU or DOX	Battery of neurocognitive tests selected to evaluate a broad spectrum of cognitive domains	The long-term survivor's group had impaired attention, working memory, psychomotor speed and aspects of executive functioning, performing 0.75 to 2.0 SD below their matched healthy control However, no impairment was detected in the domains of language, visuospatial and memory functioning	Yamada et al. (2010)
16 women in remission of breast cancer being previously treated with chemotherapy and 22 healthy women	Not described	Battery of standardized cognitive testing	Cancer patients displayed mild impairments in the immediate memory comparing to the controls	Ayala-Feliciano et al. (2011)
38 cancer survivors with the last treatment occurring from 4 months to 25 years earlier	Not described	Montreal Cognitive Assessment to detect mild cognitive impairment	37% of cancer survivors had scores consistent with mild cognitive impairment. There was no correlation between time since last treatment and tests scores	Baxter et al. (2011)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
42 patients with breast cancer	Patients did 4 cycles of paclitaxel or 4 cycles of 5-FU, DOX, and CYA	Neuropsychological testing that included measurement of cognition, mood and quality of life performed before treatment, and on average 3.7 and 13 months after the first evaluation	Before treatment, 21% of the patients had evidence of cognitive dysfunction, with the most common domains affected being: learning and memory, executive function, and processing speed Acute decline in cognitive function occurred in 65% of the patients. Late cognitive decline happened in 61% of the patients, of whom, 30% presented new domains affected that were not affected in the acute phase	Hermelink (2011)
71 women with breast cancer received chemotherapy	Group A: 22 patients had four cycles of DOX and CYA Group B: 49 patients had four cycles of DOX and CYA plus taxane	Neuropsychological testing was performed before the start of chemotherapy, 1 week after 4 cycles of treatment, 1 week after the taxane treatment and 6 months after finishing chemotherapy	At baseline, 23% of the women had scores consistent with cognitive dysfunction when comparing with pre-established scoring criteria. The main functions affected were language, visuospatial skills and motor function Throughout the study, in both groups, 52% of the women had a decreased of 1 or more SD for 2 or more tests, after chemotherapy, being the most impaired: visuospatial skills followed by motor function, attention, language, immediate memory and delayed memory At the 6-months evaluation, 20% of the women still presented a decreased of 1 or more SD from baseline scores	Jansen et al. (2011)
415 twin pairs over 65 years old (277 females and 138 male). One of the twins had cancer at least 3 years prior to the study and the other twin never had cancer	Not described	Telephonic cognitive screening	Female but not male cancer survivors were more likely to have cognitive impairment than their respective twin	Kurita et al. (2011)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
40 women with breast cancer that finished the treatment at least 18 months prior to evaluation	20 women did taxane and/or DOX-based chemotherapy and the other 20 women only took tamoxifen	Self-reported cognitive problems questionnaire and neuropsychological tests for assessment of long-term memory	Overall, the patients performed worse when comparing to the standard scores. When comparing the tamoxifen group versus chemotherapy group, there were no differences. In both groups, 40% of the women had scores consistent with mild cognitive impairment when compared to comparative values	Prokasheva et al. (2011)
122 testicular cancer patients post orchiectomy assorted into 3 groups: no-chemo; multiple-chemo and one-chemotherapy cycle	Most of the patients received cycles of bleomycin, etoposide, and cisplatin	Neuropsychological test battery after surgery but before any other chemotherapeutic treatment and repeated in average 12 months after	There were no differences between the three groups when multiple comparisons were performed	Skaali et al. (2011)
40 women undergoing chemotherapy for breast cancer	Patients received cycles of an anthracycline-based therapy	Cognitive testing and self-perception of cognition before (T1) and 6 months after chemotherapy (T2)	Comparing the pre-evaluation to the tests 6 months after treatment, there was a worsening in the overall cognition and in the visual selective attention Still, the patients did not had perception of cognitive impairment as the self-perception results did not correlate with objective evaluation	Biglia et al. (2012)
34 women with breast cancer and premenopausal treated with chemotherapy 16 women with breast cancer and not treated 19 healthy controls age matched	Patients did 6 cycles of 5-FU, epirubicin, and CYA or 3 cycles of 5-FU, epirubicin, and CYA plus 3 cycles of paclitaxel	Neuropsychological assessment before any treatment (T1) and 3–4 months after the last treatment (T2)	At T1, there were no differences among the 3 groups. In the T2 evaluation, the chemotherapy-treated group had significantly lower scores on attention and concentration tests, psychomotor speed and memory when comparing to the first assessment. Both groups not treated, and controls improved their scores overtime due to the learning ('practice') effect	Deprez et al. (2012)
196 female breast cancer survivors on average 21 years after treatment compared to 1509 females never diagnosed with cancer	6 cycles of CYA, MTX, 5-FU	Battery of neuropsychological tests	Women that received chemotherapy had considerably lower scores on cognitive tests of immediate and delayed verbal memory, processing speed, executive functioning and psychomotor speed when compared to the control population	Koppelmans et al. (2012a)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
47 colon cancer patients	Combination of oxaliplatin and 5-FU plus leucovorin	Neuropsychological tests before any treatment (T1), after end of the treatment (T2) and 6-months after the last treatment (T3)	No differences were found among time points on the conducted tests	Andreis et al. (2013)
40 breast cancer patients treated with chemotherapy and 40 age-matched healthy controls	Chemotherapy treatment with DOX, paclitaxel, CYA, and 5-FU	Neuropsychological tests that include event-based prospective memory task and time-based prospective memory task, 4 weeks after the end of the treatment	The chemotherapy group had lower scores in the MMSE, verbal fluency and digit span. This group also presented significantly lower scores in the event and time-based prospective memory task	Cheng et al. (2013)
60 women with early breast cancer 60 healthy control women	47 women received 5-FU, epirubicin, CYA with or without taxotere 7 women received CYA plus taxotere 5 women received DOX and CYA with or without paclitaxel	Battery of neuropsychological assessment after surgery but before chemotherapy and after each cycle of chemotherapy (7 time-points evaluated)	In the first assessment, both groups scored within the normal average scores and there were no differences among groups In the chemotherapy group, the cognition scores constantly changed from T1 to T7, reaching statistically difference in T5 (4th cycle of chemotherapy). The domain that declined the most was working memory on the chemotherapy group There were no difference in the scores of different anticancer treatments	Collins et al. (2013)
18 patients with breast cancer and 20 healthy controls	Cycles of DOX or epirubicin and CYA followed by paclitaxel/docetaxel	Battery of neuropsychological tests, before the treatment, 4 weeks after the last treatment and 6 months after the last treatment	Neither differences were found in cognition between the 3 time-points, nor between breast cancer patients and the healthy controls	Ando-Tanabe et al. (2014)
Breast cancer patients treated with chemotherapy ($n = 28$) or not ($n = 37$) and a control group ($n = 32$)	Chemotherapy regimen included several cycles of DOX and CYA with or without docetaxel or paclitaxel	Neuropsychological testing before chemotherapy and 1 month after the last treatment	No significant baseline differences were found regarding cognition among the groups. However, after treatment, the chemotherapy group had worse cognition scores than the other two groups	Askren et al. (2014)
Breast cancer patients that received chemotherapy ($n = 58$) or not ($n = 53$) and 55 matched healthy controls	Not described	Standardized neuropsychological testing that included evaluation of three distinct brain networks: alert, orientation, and executive networks. The tests were performed after the chemotherapy treatment	Chemotherapy treatment affected the alert and executive networks but not the orientation network. In addition, the chemotherapy group scored significantly lower in other cognitive domains (attention, memory and information processing tasks) compared to the other two groups	Chen et al. (2014)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
60 breast cancer patients that received chemotherapy 60 individually matched (age and education) controls	Most of patients received cycles of 5-FU, epirubicin and CYA	Standardized neuropsychological testing before the start of chemotherapy, at the end of each cycle (short-term evaluation) and 1 year after the last treatment (long-term evaluation)	At an individual level, there was a decline in score of 48% of the patients versus 9% of the control group, in the short-term evaluation. From the baseline evaluation to the long-term evaluation, the patients group presented a decline in 22% of the patients versus 6% in the control group	Collins et al. (2014)
81 patients scheduled to receive chemotherapy for colon cancer	Patients received oxaliplatin plus 5-FU/leucovorin	Standardized neuropsychological testing before chemotherapy, after the last cycle of chemotherapy and in a 6-month follow-up	Before treatment, 37% of the patients had cognitive impairment. At the end of the chemotherapy, the same percentage of patients were impaired. Six months after, 39% of the patients remained with cognitive dysfunction, thus no significant changes were seen	Cruzado et al. (2014)
32 women treated for breast cancer 32 healthy controls	41% of the patients received cycles of CYA, epirubicin and 5-FU. The remaining patients received DOX in combination CYA, docetaxel or paclitaxel	Standard cognitive testing done approximately 2 months after chemotherapy ended	Women treated with chemotherapy had higher percentage of cognitive impairment, being the domains of attention and working memory the most affected	Jung and Cimprich (2014)
1300 breast cancer survivors	93% of the patients received chemotherapy	Neuropsychological testing was done 18 months after diagnosis, while 1059 of these patients were evaluated again 36 months post-diagnosis	Despite the treatment, the women improved their cognitive function scores from the 18th month to the 36th month	Zheng et al. (2014)
72 men diagnosed testicular cancer 2–7 years prior to the evaluation	36 patients had an orchiectomy plus chemotherapy (3–4 cycles of bleomycin, etoposide, and cisplatin). The remaining patients only had surgery and 13 patients also had radiotherapy	Standardized neuropsychological testing	62.5% of the patients had cognitive impairment when compared to the normative data. Nevertheless, no differences were found in cognition when comparing the patients that received chemotherapy or not	Amidi et al. (2015)
99 women with breast cancer that received chemotherapy	Women received an anthracycline-based regimen consisting of DOX and CYA 600 mg/m ² or a taxane-based regimen consisting of docetaxel and CYA	Computerized neuropsychological testing and measurement of plasma cytokines levels before the start of the treatment (T1), after the 3rd cycle of chemotherapy (T2) and after the end of treatment (T3)	At T3, patients had dysfunction in memory (13.2%) and attention (7.3%) when comparing to T1. Considering the cytokines levels, only IL-6 was significantly increased from T1–T3	Cheung et al. (2015)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
231 women with ovarian cancer	Not described	Computerized cognitive function testing before chemotherapy (T1), before the 4 th cycle of chemotherapy (T2), after the end of the treatment (T3) and 6 months after the completion of the treatment (T4)	At T2, 25% of the patients had cognitive impairment when compared to baseline score. At T3, 21% of the patients had scores consistent with cognitive impairment and in the last evaluation (T4), 17% of the patients had cognitive dysfunction, being attention the domain most impaired	Hess et al. (2015)
76 cancer patients of whom 5 had not started chemotherapy, 20 had completed 1 cycle, 20 had completed 3 cycles, 19 had completed 7 cycles and 12 had completed the treatment at least 24 months prior to the study	Patients received cycles of 5-FU, epirubicin and CYA or CYA and DOX	Standard neuropsychological testing and measurement of blood levels of high-molecular-weight neurofilament subunit (pNH-F), a potential biomarker of neuronal or axonal damage	28.8% of all the patients that received chemotherapy had increased levels of pNH-F, but not the patients still waiting to receive treatment There was a positive correlation between the number of cycles and higher elevated levels of pNH-F No differences were observed in the neuropsychological tests among the groups of patients	Natori et al. (2015)
Early-stage breast cancer patients, over 65 years old, treated with chemotherapy ($n=58$) or radiotherapy ($n=61$). Sixty-two aged matched healthy controls were used	Chemotherapy treatment consisted of 5-FU, epirubicin and CYA with or without docetaxel	Neuropsychological testing after radiotherapy or before chemotherapy (T1) and after the end of the treatment (T2)	At T1, 46% of the patients that received chemotherapy had cognitive impairment. At T2, 64% of all patients had cognitive impairment. No differences in cognition were found between patients that received chemotherapy or radiotherapy	Lange et al. (2016)
357 cancer patients, over the age of 65 years	Not described	The Montreal cognitive assessment was applied during the chemotherapy treatment	Cognitive impairment was found in 46% of the patients. Patients that had cognitive impairment were up to 6 times more likely to die than patients with normal cognition	Libert et al. (2016)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
75 women with early-stage breast cancer	Most women received cycles of DOX, CYA and docetaxel	Cognitive testing and plasma concentration of several cytokines measured before the start of chemotherapy (T1), halfway through the treatment (T2), 6 months after the beginning of the treatment (T3), 1-year after the beginning (T4) and 2 years after the beginning (T5)	At T1, the granulocyte colony stimulating factor concentration was positively correlated with cognitive flexibility, executive functioning and psychomotor speed. On the other hand, IL-17 was negatively correlated with psychomotor speed. At T2, various cytokines levels were correlated with psychomotor speed, complex attentions, executive function, verbal memory and visual memory. At T3 and T4, there were also multiple relations between the levels of cytokines and cognition. At T5, less relations between levels of cytokines and cognition deficits were set when compared to earlier time-points	Lyon et al. (2016)
Breast cancer patients received chemotherapy, with or without endocrine therapy, ($n = 31$) or did not receive systemic chemotherapy ($n = 24$). The study also included 33 age-matched healthy controls	Anthracycline-based therapy	Standardized neuropsychological testing was performed before the start of the treatment (T1) and six months after chemotherapy (T2)	At T2, 16% of the patients treated with chemotherapy had impaired cognition, compared to 4% of the patients that did not receive chemotherapy and 6% in the healthy controls	Menning et al. (2016)
106 patients with non-Hodgkin's or Hodgkin's lymphoma and 53 healthy matched controls	Anthracycline-based chemotherapy	Neuropsychological testing	A subgroup of 16% of the patients had abnormal neuropsychological performance (1.5 SD below the norm in 3 of the 14 tests)	Wouters et al. (2016)
Women diagnosed with breast cancer scheduled to receive chemotherapy with taxane ($n = 25$) or without ($n = 26$)	Patients received cycles of 5-FU, epirubicin and CYA followed or not by taxane	Standard neuropsychological testing was performed before the start of chemotherapy (T1), after the end of the treatment (T2) and 75 weeks after T2 (T3)	At baseline, none of the patients had cognitive impairment. At T2, the taxane group performed worse than the group without taxane when compared to baseline. At T3, both groups still had cognitive impairment when compared to baseline	Cerulla et al. (2017)
136 cancer survivors (cancer-free for at least 5 years)	Not described	Standardized neuropsychological testing	15% of the cancer survivors had impaired performance in 2 or more tests and 32% had 1 or more impaired function, when comparing to published normative values	Muzzatti et al. (2017)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
480 breast cancer patients that received chemotherapy	Patients received several cycles of chemotherapeutic agents that included DOX, CYA, 5-FU, epirubicin or docetaxel	Montreal cognitive assessment and hospital anxiety and depression scale tests were performed before the start of the treatment and at the 1-year follow-up	At the 1-year follow-up, 81% of the patients had cognitive impairment. Moreover, there was an increasing risk of cognitive impairment in patients that did not presented anxiety in the first assessment	Ramalho et al. (2017)
403 cancer survivors, at least 4 years post diagnosis and treatment	Treatment data were available for 235 patients: 46.8% received exclusively chemotherapy, 32.3% received only surgery and 20.8% received only radiotherapy	Standard neuropsychological testing	The study found no association between cognitive function decline and chemotherapy treatment	Morin and Midlarsky (2018)
365 breast cancer patients over the age of 65 years that received chemotherapy	Not described	Neuropsychological testing to assess changes in attention and working memory. Tests were performed 6 times over two cycles of chemotherapy	The study identified 3 classes of attention functional profiles: low ($n = 134$), moderate (136) and high ($n = 95$) attentional function. Through the evaluation period, the scores of each group remained stable	Utne et al. (2018)
Breast cancer patients with different treatments: surgery ($n = 28$), radiotherapy ($n = 64$), chemotherapy ($n = 20$) or both ($n = 77$)	Not described	Neuropsychological testing	There were minimal differences in cognition among patients exposed to the different treatments	Van Dyk et al. (2018)
22 female breast patients that received chemotherapy	Anthracycline or non-anthracycline based chemotherapy	Neuropsychological testing and serum levels of inflammatory cytokines	Patients that had higher levels of soluble tumor necrosis factor receptor 1 had lower cognitive scores	Williams et al. (2018)
240 cancer survivors (mean age at the time of diagnosis was 68.41) of whom 34 only received chemotherapy were compared to 10,939 healthy controls	Not described	Neuropsychological testing	No significant cognitive decline was detected in the patients that only received chemotherapy	Zuniga and Bishop (2018)
94 women treated for breast cancer 3–6 years prior to the study	Not described	Neuropsychological testing and blood collection for biological markers of aging	Higher DNA damage and lower telomerase length in peripheral blood mononuclear cells were positively linked with lower executive function. In addition, decreased telomerase activity was correlated with worse attention and motor speed	Carroll et al. (2019)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
142 patients with breast cancer (51.4%) or colorectal cancer (48.6%) treated with chemotherapy	Breast cancer patients received a combination of 5-FU, epirubicin, CYA, DOX with or without docetaxel/paclitaxel Colorectal cancer patients received cycles of 5-FU, leucovorin, oxaliplatin, irinotecan and capecitabine	Psychomotor vigilance test and the trail-making test were performed prior to the start of the treatment (T1) and after the end of treatment (T2)	After chemotherapy, patients had decreased reaction time in the psychomotor vigilance test, indicating impaired orientation and attention. However, in the trail-making test, the patients did the test faster (6.7 s) and there were no significant alterations from T1–T2	Khan et al. (2019)
102 patients with testicular cancer that received surgery and chemotherapy ($n=61$) or only surgery ($n=41$)	Chemotherapy regimens consisted of cisplatin, etoposide and/or bleomycin or the single-agent carboplatin	Standardized cognitive testing before chemotherapy (T1) and follow-up assessment 12–18 months after (T2)	At T1, the group scheduled to receive chemotherapy had worse psychomotor function. However, at T2 no significant differences were found in cognition between the two groups	Whitford et al. (2019)
30 patients with hematologic cancer that received chemotherapy ($n=15$) or supportive care ($n=15$) were compared to 15 healthy controls	Chemotherapy consisted of one or more of the following agents: DOX, CYA, vincristine, rituximab, decitabine, and prednisone	Neuropsychological assessment of attention and executive function before chemotherapy (T1), after 1 cycle (T2) and after 3 cycles of treatment (T3)	Chemotherapy-treated patients displayed significant decline in the measures of attention and executive function when compared to healthy controls	Anderson et al. (2020)
356 breast cancer survivors (8 years post-treatment) were compared with 252 non-cancer female controls	79.5% of the cancer survivors received chemotherapy	Quality of Life in Adult Cancer Survivors scale and Hospital Anxiety and Depression scale	Breast cancer survivors with more advanced cancer and/or received chemotherapy presented significantly more cognitive problems than healthy controls	Carreira et al. (2020)
330 cancer survivors (62.7% women) who returned to work after diagnosis and treatment for cancer	59.9% of the patients received chemotherapeutic treatment	Cognitive functioning was assessed through the Cognitive Symptom Checklist-Work within 3 months of returning to work, and after 6-, 12-, and 18-months follow-up	Systemic chemotherapy-treated cancer survivors presented significant lower scores for executive function when comparing to the patients that received locoregional treatment	Ehrenstein et al. (2020)
51 patients with haematological cancer (66.7% male; mean age 68 years)	All patients received several cycles of multi-agent chemotherapy	Montreal Cognitive Assessment	Older chemotherapy-treated patients had substantial lower scores for language, learning and memory, visual spatial skills, and executive function when comparing to normative data	Koll et al. (2020)
182 cancer patients (25.8% breast, 24.2% colorectal, 24.7% prostate and 25.3% thyroid cancer)	First-line chemotherapy for the respective type of cancer	Battery of neuropsychological tests 1 day before the start of the treatment (T1), 1 day immediately after chemotherapy (T2) and 6 months after T2 (T3)	All patients had lower performance at T2 when comparing to T1, that remained stable at T3 in comparison to T2. Breast cancer patients displayed significantly worse performance on all cognitive domains assessed when comparing to other patients at T2 and T3	Megari (2020)

Table 1 (continued)

Number and conditions of patients	Anticancer treatment	Clinical and neuropsychological evaluations	Major findings	Study
Breast cancer patients that received chemotherapy ($n = 66$) or not ($n = 56$) were compared to 56 healthy controls	Chemotherapy regimen consisted of 5-FU, epirubicin, and CYA, followed by docetaxel	Battery of standardized cognitive testing before the start of chemotherapy (T1), and after the last cycle of treatment (T2)	Chemotherapy-treated women performed significantly poorer than the non-chemotherapy group and the healthy controls, particularly in the verbal fluency domain	Schrauwen et al. (2020)
145 patients with testicular cancer treated with chemotherapy ($n = 61$) or surgery alone ($n = 41$)	Cycles of etoposide and cisplatin with or without bleomycin or single agent carboplatin	Computerized battery of objective cognitive function tasks were applied before chemotherapeutic treatment (T1) and 12–18 months after (T2)	For both groups, no significant differences were observed in cognitive functioning from T1 to T2	Whitford et al. (2020)
150 patients with chronic lymphocytic leukaemia that received chemotherapy ($n = 50$) or not ($n = 100$)	Not described	Neuropsychological battery of tests of access various cognitive domains	Only patients categorized with higher risk of disease presented cognitive difficulties, regardless of treatment	Williams et al. (2020)
58 women diagnosed with breast cancer	43.1% of the women received docetaxel, DOX, CYA; 32.8% of the women had docetaxel and CYA; 19% of the patients were treated with docetaxel, carboplatin and trastuzumab; the remaining were treated with DOX and CYA	Computerized neurocognitive testing and evaluation of differential DNA methylation prior to receiving chemotherapy (T1) and one year the initiation of chemotherapy (T2)	DNA methylation in blood cells were enriched for genes involved in the modulation of cytokine concentrations and the memory domain, which was the only cognitive domain that did not improved from T1 to T2	Yang et al. (2020)

5-FU 5-fluorouracil, CYA Cyclophosphamide, DOX Doxorubicin, EORTC QLQ-C30 European Organization for Research and Treatment of Cancer Quality of Life Questionnaire C30, HSCS High Sensitivity Cognitive Screen, pNH-F high-molecular-weight neurofilament subunit, MTX Methotrexate, MMSE Mini mental state examination, SD standard deviation

domains more affected, but also how long were the patients still affected after the treatment ended. The first longitudinal trial was performed by Wefel and colleagues (Wefel et al. 2004b). They conducted neuropsychological evaluations on 18 breast cancer women, before they initiated the systemic chemotherapy, consisting of 5-fluorouracil (5-FU), DOX and cyclophosphamide (CYA), 3 weeks after the last cycle of chemotherapy and 1-year post-chemotherapy. The authors compared the results obtained by the patients with population normative data, adjusted for age and education, for each test. Before chemotherapy, 33% of the breast cancer patients had cognitive impairment, with the domains of verbal learning and memory being the most frequently impaired. Three weeks' post-chemotherapy, 61% of the patients exhibited a decline from the baseline evaluation. Furthermore, nonverbal memory, psychomotor processing speed and attention, confrontational naming, visuoconstruction and upper-extremity fine motor dexterity were impaired more frequently than expected after chemotherapy. One-year after the end of chemotherapy, only 50% of the patients presented improvements on their cognitive function (Wefel et al. 2004b). Regardless of the importance of its results, this longitudinal study compared the patients' scores to published normative values in healthy populations (Wefel et al. 2004b). Other authors also used the same method of comparison for patients scores (Amidi et al. 2015; Correa et al. 2010; Muzzatti et al. 2017). Even if the populations can be matched on age and education, cultural and socio-economic status can also influence cognitive scoring. Additionally, how the tests are undertaken differs among experimental groups, which may produce different results that may not be comparable (Von Ah et al. 2009). To ensure the most reliable results, the control group should have similar demographic characteristics to the patient group and tested in the same conditions. Moreover, and ideally, the testing should be applied by the same professionals to reduce confounding factors and a pre-treatment evaluation should be set.

As stated before, very few studies were made so far in non-breast cancer patients; moreover, many studies lack information of the chemotherapeutic regimen used (Table 1) and with the multitude of combinations possible of chemotherapeutic agents, it is hard to evaluate which specific chemotherapeutic agent or drug combination causes cognitive dysfunction (Vearncombe et al. 2009). Furthermore, small sample sizes, extremely heterogeneous groups and the lack of appropriate age and education matched control groups may influence the baseline cognitive outcome of each individual, leading to very scattered and confusing results regarding the existence or not of chemotherapy-induced cognitive impairment. Moreover, the lack of patient randomization regarding the inclusion in a drug treatment versus a non-drug treatment group can lead to potential bias. Notwithstanding, the neuropsychological testing performed

throughout the years in either cross-sectional studies or longitudinal studies provided an insight into the cognitive consequences of systemic chemotherapy and the cognitive domains more often impaired, namely on breast cancer treated women that underwent chemotherapy. Longitudinal clinical studies reported a prevalence of cognitive impairment between 28% (Schagen et al. 1999) and 75% (Wieneke and Dienst 1995) in women with breast cancer that received adjuvant chemotherapy. The domains reported more often to be impaired were verbal memory, psychomotor function (Ahles et al. 2002), visual memory, visuospatial and verbal learning (Castellon et al. 2004), memory function and attention (Wefel et al. 2004b).

After showing that chemobrain is, in fact, a real consequence of chemotherapy, one important question placed remains: is chemobrain a permanent condition? If not, how long does it persist? Two studies evaluated the long-term cognitive sequels of chemotherapy on 10-year (or more) breast cancer survivors over the age of 65 (Nguyen et al. 2013; Yamada et al. 2010). The chemotherapeutic treatment included CYA, methotrexate (MTX) and 5-FU or DOX. The two studies reported lower cognitive outcomes in chemotherapy-treated patients when comparing to the healthy controls. The domains more affected were attention, working memory, psychomotor speed and executive function (Nguyen et al. 2013; Yamada et al. 2010). Another study evaluated cancer survivors on average 20 years after chemotherapy (six cycles of CYA, MTX and 5-FU) and showed that they exhibited significantly lower cognitive capabilities when comparing to the healthy controls. The domains more frequently impaired were immediate and delayed verbal memory, processing speed, executive functioning and psychomotor speed (Koppelmans et al. 2012a, b). These reports seem to show that cognitive deficits are seen in multiple fields and may be more prolonged in time, although some studies referred in Table 1 show other results.

Another important aspect is that the large bulk of clinical studies carried out so far were performed in female breast cancer patients who received chemotherapy (Schagen et al. 2006a). Thus, less information on the cognitive consequences of cytostatic agents used in men and/or other cancer types is available (Sadighi et al. 2006). Besides, most of the studies done in men were performed on testicular cancer patients with similar drug regimens. Male cancer patients with testicular cancer treated with BEP regimen (bleomycin, etoposide and cisplatin), 2–7 years post-treatment, had their neuropsychological evaluation compared to normative data. The authors found that 62.5% of the survivors had impaired cognition across multiple domains such as verbal learning and memory, visual learning and memory, processing speed, executive functioning and attention and working memory (Amidi et al. 2015). On the other hand, a 3-year post-chemotherapy neuropsychological evaluation

performed on 70 testicular cancer patients, also receiving BEP, showed no significant differences in mean cognitive test performance when compared with 57 patients with the same pathology but that only underwent surgery (Schagen et al. 2008). Therefore, it is not clear whether chemotherapy does or does not affect men's cognitive functions or if the evaluation of only one cancer type and one regimen blurred any accurately taken and solid conclusions. In fact, some studies were performed in men but did not made analysis taking sex into consideration (Table 1).

Moreover, the research on chemotherapy-induced cognitive dysfunction has been mainly focused on middle-aged women and neglected other groups or cancer types. In older cancer patients, the existence of cognitive impairment prior to chemotherapy is associated with poor prognosis for cancer, mainly to the fact that recommendations from health care professionals or family are harder to remember, therefore jeopardizing the anticancer treatment. In fact, patients with impaired cognition have six times more risk to die than patients without cognitive impairment (Libert et al. 2016). Moreover, older cancer survivors are at higher risk for cognitive impairment due to comorbidities (poor cardiovascular function, diabetes, anemia, or hypertension) and lower cognitive reserves. In addition, older age is an increased risk for neurodegenerative diseases and higher incidence of cancer (Edwards et al. 2018), which can be a bias to the studies. Older patients (60–70 years old) treated with chemotherapy had lower cognitive reserves at baseline and performed significantly lower than healthy controls and non-chemotherapy treated patients of the same age range at the 1-month follow-up (Ahles et al. 2010). Although age at diagnosis seems a contributing factor to chemotherapy-induced cognitive dysfunction, it has been hitherto poorly studied.

In summary, chemobrain is now a well-recognized side effect of chemotherapy (Weiss 2008), but there is still a lot of uncertainty into what extend it affects treated patients, as described in this work. Hence, the research on the neurotoxicity of chemotherapy is crucial to the following: (1) access the real incidence of cognitive dysfunction on chemotherapy-treated patients and individual susceptibility factors; (2) clarify the putative mechanisms of chemobrain; and (3) establish standard care and management options to ameliorate the quality of life of cancer survivors (Parsons and Dietrich 2019). In addition, what are the true mechanisms and the possible measures to be undertaken to ameliorate its impact on cancer survivors' quality of live are still unknown (Pendergrass et al. 2018).

Electrophysiological and neuroimaging studies

The brain optimal function is highly dependent on the maintenance of its integrity and structure. The grey matter compromises the regions of the CNS enriched in cell bodies, unmyelinated axons, dendrites and glial cells (Snell 2010). Beneath the grey matter cortex, lies the white matter, which is composed by oligodendrocytes that involve the axons in lipid myelin sheets. Myelin is essential for the fast transmission of electrical impulses and the loss of integrity and axonal demyelination of the white matter occur in some neurodegenerative diseases, such as Alzheimer's and multiple sclerosis (Kassmann et al. 2007). The performance of cognitive tasks depends on the highly organized and synchronized brain networks and the integrity of brain white matter pathways to transmit the neural signals effectively (Deprez et al. 2013).

Multiple structural and neuroimaging techniques have been used to understand the changes in the neural substrate of chemotherapy-treated patients that may lead to poor cognition. Voxel-based morphometry (VBM) and Diffusion-Tensor imaging (DTI) are structural neuroimaging techniques that can detect alterations in the grey matter and white matter tissue, respectively (Simo et al. 2013). On the other hand, functional Magnetic Resonance Imaging (fMRI), Positron Emission Tomography (PET) and electroencephalogram (EEG) can find changes in brain function even when structural changes are not evident (Simo et al. 2013). Scans that use magnetic resonance imaging (MRI) such as VBM, DTI and fMRI are not invasive and provide the best image quality, both in 2 or 3 dimensions, preventing exposure of patients to ionizing radiation or radioactive probes (Scherling and Smith 2013). These techniques provide an insight on the brain of chemotherapy-treated patients attempting to explain why they have declines in cognitive functions, focusing in what areas are affected and what are some inherent mechanisms. Summary data and results of the studies using electrophysiological and neuroimaging techniques are presented in Table 2.

Electrophysiology studies or determinations focusing on brain function can be diverse and expensive; however, they represent an objective way, without interferences of practice (or education factor), to assess neurotoxicity. Nevertheless, earlier studies on chemobrain, using these techniques, suffered the same problems regarding grouping, namely controls, or lower number of subjects when compared to cognitive testing mentioned in the previous section. In 1992, Siddiqui and colleagues made the first EEG study in cancer-treated patients to assess the baseline cognition before the treatment. They evaluated patients neither having brain metastasis nor presenting signs of dementia or other

Table 2 Clinical studies with neuroimaging and neurophysiological testing and major findings

Number and conditions of patients	Anticancer treatment	Clinical neuroimaging and neurophysiological evaluation	Major findings	Study
6 patients with breast cancer treated with 5-FU, whom developed leukoencephalopathy	4 patients received 5-FU, CYA, epirubicin and carmofofur, one received the same regime except the 5-FU and the last patient only received carmofofur	Mini mental state examination (MMSE) Routine laboratory tests, brain biopsies and electroencephalography	5-FU and carmofofur seem to be the cause of leukoencephalopathy in these patients	Choi et al. (2001)
33 post-surgery high-risk cancer patients that received chemotherapy 14 stage I breast cancer control patients	16 patients received 5 cycles of chemotherapy (5-FU, CYA and epirubicin) radiotherapy and tamoxifen 17 patients received high-dose chemotherapy where the 5th course of chemotherapy was replaced by CYA, thiotepa, carboplatin and peripheral blood progenitor cell transplantation The control group, after surgery was treated with radiotherapy only	Event-related potential (ERP) quantitative electroencephalography performed 2 years post-treatment	Cognitive dysfunction was established in 32% of the patients treated with high-dose chemotherapy, in 17% of the standard-dose patients and in 9% of the stage I breast cancer control group. Regarding the electroencephalograph results, 7 patients of the high-dose regimen and 2 patients of standard dose regimen had asymmetric alpha rhythms versus no alteration in the control group	Schagen et al. (2001)
26 breast-cancer patients underwent surgery, received locoregional radiotherapy and chemotherapy 23 stage I breast-cancer patients underwent surgery, received locoregional radiotherapy and had no chemotherapy	6 cycles of CYA, MTX, and 5-FU	Electroencephalogram (EEG), ERP and battery of neuropsychologic tests. The chemotherapy group was tested on average 5.1 years after the end of treatment and the surgery group was tested 3.6 after radiotherapy	The chemotherapy-treated group had an earlier and reduced P3 amplitude	Kreukels et al. (2005)
12 patients with breast cancer treated with high-dose chemotherapy 17 patients with breast cancer treated with standard-dose chemotherapy 23 control patients with breast cancer that did not receive chemotherapy	The standard-dose group received 5 courses of 5-FU, epirubicin, and CYA every 3 weeks. In the high-dose group, the fifth course was replaced by a course of CYA, thiotepa and carboplatin with autologous peripheral blood hematopoietic progenitor cell transplantation. All patients received locoregional radiation therapy after the end of chemotherapy and were treated with tamoxifen for 2–5 years if they had hormone receptor-positive cancer	EEG and neuropsychological testing. All patients were evaluated approximately 4 years after treatment	The group treated with high-dose chemotherapy had significantly lower P3 component compared with control patients. However, there were no significant differences between the standard-dose group and the control group	Kreukels et al. (2006)

Table 2 (continued)

Number and conditions of patients	Anticancer treatment	Clinical neuroimaging and neurophysiological evaluation	Major findings	Study
16 women who had breast cancer and received chemotherapy 5–10 years prior 8 control women, who did not had chemotherapy and had breast cancer ($n=5$) or not ($n=3$)	Not described	Positron emission tomography (PET) scan	Chemotherapy-treated women had significantly altered blood-flow in specific regions of frontal cortex and cerebellum while performing short-term recall tasks. Moreover, these patients' cerebral activation differed from untreated patients in inferior frontal gyrus. Metabolism of the basal glia was considerably decreased in chemotherapy patients that also received tamoxifen when compared to the other groups, even when compared with other women that underwent chemotherapy	Silverman et al. (2007)
17 post-chemotherapy women with breast cancer 18 healthy controls 10 non-chemotherapy treated women with breast cancer (radiotherapy or hormonal therapy)	Chemotherapy cycles included 5-FU, epirubicin, and CYA	Diffuser tensor imaging (DTI) Neuropsychological testing	White matter integrity was diminished in chemotherapy treated patients when comparing to healthy controls and non-chemotherapy treated patients. Moreover, there were significant correlations between white matter integrity and neuropsychological tests covering the domains of attention and processing/psychomotor speed in the temporal and parietal areas	Deprez et al. (2011)
32 patients with Hodgkin's lymphoma or non-Hodgkin's lymphoma treated with chemotherapy 18 healthy controls	Cancer patients were treated according to the ABVD (Bleomycin-Dacarbazine-DOX-Vinblastine) protocol and the CHOP (CYA-DOX-Vincristine-Prednisolone) protocol	Neuropsychological testing and PET scanning in average 7 months after the last treatment	The chemotherapy-treated group had lower glucose metabolism in multiple regions of the brain being the prefrontal cortex the main area affected	Baudino et al. (2012)
34 women with history of breast cancer and treated with chemotherapy 27 healthy controls matched for age, education, and overall intelligence	Most women received cycles of DOX, CYA and taxol	Resting state functional magnetic resonance imaging (fMRI)	Breast cancer patients presented altered global brain network when comparing to the healthy controls	Bruno et al. (2012)
Chemotherapy group: 17 breast cancer patients treated almost 10 years prior to evaluation No chemotherapy group: 15 patients that had stage I breast cancer and were currently disease-free	4 cycles of 5-FU, CYA, followed by one cycle of high-dose CYA, thiopeta, carboplatin, and autologous peripheral blood hematopoietic progenitor-cell transplantation. Patients also took 40 mg of tamoxifen daily on average for 3.8 years	Neuropsychological testing Magnetic resonance imaging (MRI) scanning	Almost 10 years after chemotherapy, patients had significant reductions in grey matter volume and signs of axonal injury when comparing to the non-chemotherapy group. The chemotherapy-treated group also performed worse on the neuropsychological testing	de Ruiter et al. (2012)

Table 2 (continued)

Number and conditions of patients	Anticancer treatment	Clinical neuroimaging and neurological evaluation	Major findings	Study
37 female breast cancer survivors (in average 4 years after chemotherapy) 38 age-matched healthy controls	Most women did several cycles of DOX and CYA	MRI scanning	Women treated with chemotherapy for breast cancer had alterations in the properties of grey matter, particularly in the frontotemporal regions when comparing to the control group	Hosseini et al. (2012)
184 invasive breast cancer survivors on average 21 years after chemotherapy 368 age-matched healthy controls	6 cycles of CYA, MTX, and 5-FU	MRI and voxel-based morphometry (VBM) for analysis of different brain regions and measurement of total brain volume and hippocampal volume	The chemotherapy-treated group had smaller total brain volume, and smaller grey matter volume than the control group. No differences were found in the volume of the hippocampal region, white matter and cerebral-spine fluid volume compared to the control group	Koppelmans et al. (2012b)
Female breast cancer patients treated with chemotherapy ($n = 16$) or without ($n = 12$) Healthy controls ($n = 15$)	Most patients that received chemotherapy did cycles of DOX, CYA and paclitaxel or docetaxel, DOX and CYA or DOX and CYA	MRI scan performed before any treatment, 1 month after chemotherapy and 1 year later	At baseline, both groups of patients had increased bifrontal and decreased left parietal activation and at the 1-month follow up, those groups had reduced frontal hyperactivation with partial return to the baseline patterns after one year. Comparing chemotherapy-treated patients with non-treated, they differ in the pattern of activation in the frontal lobes	McDonald et al. (2012)
9 women with breast cancer	Not described	Functional MRI session before the start of the chemotherapy (T1), 1 month after (T2) and 1 year after the last treatment (T3)	At T2 exam, there was decreased functional connectivity in the dorsal attention network that partially returned to baseline values after 1-year. Regarding the default mode network, there was decreased connectivity that did not return to normal after 1-year	Dumas et al. (2013)

Table 2 (continued)

Number and conditions of patients	Anticancer treatment	Clinical neuroimaging and neurophysiological evaluation	Major findings	Study
44 women survivors of breast cancer, on average 4.8 years post-chemotherapy 38 healthy female controls	36 women received DOX, CYA or paclitaxel 6 women received CYA, 5-FU and paclitaxel or MTX	MRI exam, hippocampal volume measurement and plasma cytokines measurement	The left hippocampal volume was significantly reduced after chemotherapy when comparing with the control group. The right hippocampal volume was slightly reduced ($p=0.07$) compared to the controls Concerning the cytokine levels, only TNF- α and IL-6 levels were significantly higher in the chemotherapy group, comparing to the control group	Kesler et al. (2013a)
30 women with breast cancer that received chemotherapy 27 women with breast cancer that did not receive chemotherapy 24 healthy controls	Not mentioned	Neuroimaging study with multivariate pattern analyses	Breast cancer patients treated with chemotherapy had abnormal default mode network connectivity compared with the non-chemotherapy breast cancer patients and with the healthy controls	Kesler et al. (2013b)
21 early breast cancer patients 21 matching controls	15 women received 5-FU, epirubicin, CYA and docetaxel or with epirubicin 6 women received CYA with docetaxel or DOX	Neuropsychological assessment and self-report mood state, MRI scanning and verbal recall task before and after chemotherapy	Before chemotherapy, cancer patients activated the anterior cingulate fewer times than controls during memory recall. In the patients' group, there were also changes in brain activation after the treatment. On contrary, the control group had no changes in activation in either time-points assessed	Lopez Zunini et al. (2013)
Female with breast cancer treated with chemotherapy ($n=27$) or without ($n=28$) 24 healthy controls	Standard dose chemotherapy that included taxane	Apolipoprotein (ApoE) genotyping and MRI scanning before any treatment and a month after the end of chemotherapy	No significant changes were observed among groups in the first evaluation Chemotherapy-treated patients, in the second evaluation, had reduced grey matter density, in the left middle frontal gyrus, when compared to the controls. This change was not observed in the patients that did not receive chemotherapy No correlation was found between changes in the grey matter size and the ApoE gene	McDonald et al. (2013)

Table 2 (continued)

Number and conditions of patients	Anticancer treatment	Clinical neuroimaging and neurophysiological evaluation	Major findings	Study
Cancer patients scheduled to receive chemotherapy ($n = 18$) or not ($n = 16$) Age matched healthy controls ($n = 17$)	Patients received cycles of 5-FU, epirubicin, CYA	fMRI scanning pre-treatment and 4–6 months after the end of treatment	Before chemotherapy, there were no significant differences among the groups. At the 6 months follow-up, the chemotherapy group had decreased brain activity in the multitasking network	Deprez et al. (2014)
19 female breast cancer patients 19 individually matched healthy controls	Most patients received 5-FU, epirubicin and CYA plus docetaxel	Neuropsychological testing and MRI scanning before chemotherapy (T1), one month after the last treatment (T2) and one year (T3) after the last treatment	In the treated group, at the one-year evaluation (T3), the MRI scan showed a decrease in the grey matter volume in the frontal, temporal, parietal, and occipital regions, comparing to the last evaluation. The control had no significant alterations. Regarding the neuropsychological testing, scores decreased over time in the chemotherapy-treated group	Lepage et al. (2014)
8 breast cancer patients 8 matched controls	Not described	EEG after physical and cognitive testing, performed before chemotherapy (T1), after 3/4 cycles of chemotherapy (T2) and one year (T3) after completion of the treatment	Before treatment, there were no significant differences between patients and controls. However, during the treatment (T2), the EEG power value was higher in patients compared with the controls. At the 1-year assessment (T3), there were no differences among the two groups	Moore et al. (2014)
Breast cancer patients that received chemotherapy ($n = 27$) or not ($n = 26$) Healthy controls ($n = 26$)	Standard dose chemotherapy that included DOX and CYA	MRI scanning before treatment and one month after the last cycle of chemotherapy	At baseline, there were no differences among groups. After treatment, the chemotherapy group had an increased perfusion comparing to baseline assessment and significant differences in the precentral gyrus when compared to the control group	Nudelman et al. (2014)
21 patients with non-Hodgkin's lymphoma	Not described	PET imaging before and after chemotherapy	After the treatment, the patients had overall reduced cerebral cortical metabolism, reduced partial volume-corrected average glucose uptake. However, the overall brain volume was not altered	Sorokin et al. (2014)

Table 2 (continued)

Number and conditions of patients	Anticancer treatment	Clinical neuroimaging and neurophysiological evaluation	Major findings	Study
Breast cancer patients treated with chemotherapy within 2 years of the study and that had cognitive complaints ($n = 15$) or not ($n = 13$)	Not described	MRI scanning	Women who reported cognitive complaints had disrupted functional connectivity between brain networks implicated in cognitive control	Piccirillo et al. (2015)
10 breast cancer survivors over the age of 65 years that had cancer at least 10 years prior to the study	Multi-agent regimens that included CYA, MTX and 5-FU	PET scanning	The patients that had cognitive complaints had hypometabolism in the bilateral orbital gyri, in the <i>substantia nigra</i> and brainstem when comparing to the healthy controls	Ponto et al. (2015)
10 healthy controls age matched				
Patients with testicular cancer treated with chemotherapy ($n = 38$) or surgery only ($n = 49$) were evaluated on average 14 years post-treatment	Patients had cycles of bleomycin, etoposide and cisplatin	Battery of standardized neuropsychological testing MRI scanning	The chemotherapy-group had lower cognitive performance than the surgery group. The chemotherapy also affected the white matter microstructure, but that change did not correlate with cognitive performance	Stouten-Kemperman et al. (2015)
Cancer patients more than 2 years post-treatment, that received an anthracycline based chemotherapy ($n = 20$) or non-anthracycline based chemotherapy ($n = 19$) or did not received chemotherapy at all ($n = 23$)	Anthracycline based therapy consisted of DOX, CYA and paclitaxel. Non-anthracycline based therapy consisted of CYA, MTX and 5-FU	Neuropsychological testing MRI scanning	The anthracycline group had considerably lower verbal memory and left precuneus connectivity than those who received non-anthracycline regimens and the non-chemotherapy group	(Kessler and Blayney 2016)
22 female breast cancer patients 22 healthy controls	Patients received CYA, DOX and taxotere	Neuropsychological testing MRI scanning	Breast cancer patients had lower connectivity in the medial temporal lobe and dorsal medial prefrontal cortex when compared to the healthy controls	Miao et al. (2016)
15 cancer patients receiving chemotherapy 14 age-matched healthy controls	Not described	MRI scanning after 3 cycles have been completed	Patients had lower prefrontal cortex activation when compared to the control group	Wang et al. (2016)
Patients with testicular cancer that received chemotherapy ($n = 22$) or just surgery ($n = 43$) 25 healthy controls	Chemotherapeutic cycles consisted of bleomycin, etoposide and platinum	Neuropsychological testing and MRI scanning before the start of chemotherapy (T1) and 6 months after the end of the treatment (T2) ApoE genotyping was also performed	Both cancer patient groups had decline in cognition when comparing with healthy controls, regardless of therapy undertaken Chemotherapy-treated patients had grey matter reductions in the prefrontal and parieto-frontal regions when compared with the other two groups	Amidi et al. (2017)

Table 2 (continued)

Number and conditions of patients	Anticancer treatment	Clinical neuroimaging and neurophysiological evaluation	Major findings	Study
34 breast cancer patients that were treated with chemotherapy 31 healthy controls	Not described	fMRI scanning	The chemotherapy-group had lower hippocampal connectivity compared to the control group	Cheng et al. (2017)
18 ovarian cancer patients that finished chemotherapy 1–4 months prior to the study 18 matched healthy controls	Taxane and cisplatin-based therapy	MRI scanning and neuropsychological testing	Patients had significantly reduced grey matter volume when comparing to healthy controls. No significant differences were seen in the neuropsychological tests amount the two groups	Correa et al. (2017)
Breast cancer patients treated with chemotherapy ($n = 28$) or without ($n = 34$) Healthy controls ($n = 30$)	Most patients received a combination of DOX, CYA, and paclitaxel	fMRI scanning and verbal working memory task were performed before the start of the treatment (T1), 5 months after (T2) and approximately 1 year after the first test (T3)	The chemotherapy-treated group showed impaired verbal working memory task aggravated from T1 to T3 and the fMRI showed changes in the network connectivity during the task when comparing to the non-chemotherapy group and healthy controls	Jung et al. (2017)
14 gastric cancer patients 11 healthy controls	Most patients received a combination of capecitabine and cisplatin	fMRI scanning and neuropsychological testing before the start of chemotherapy and after the last treatment	After the treatment, patients displayed diminished attention, working memory and executive function when comparing to the healthy controls	Kim et al. (2017)
Breast cancer patients received chemotherapy ($n = 28$) or not ($n = 24$) Healthy controls ($n = 31$)	Patients that received chemotherapy had an anthracycline-based therapy	fMRI scanning while performing executive functioning task and episodic memory task. The scans were done before and 6 months after the end of the treatments	No changes were seen in the execution of the tasks among the groups at baseline and after the treatment. However, the chemotherapy-group had increased parietal activation compared to baseline results	Menning et al. (2017)
19 patients received chemotherapy 11 healthy controls	Patients received combinations of 5-FU, CYA, epirubicin and paclitaxel	Resting-State fMRI, diffusion tensor imaging and neuropsychological testing before the treatment (T1) and after the treatment (T2)	Chemotherapy-treated group performed poorly in the memory test when comparing T1 time-point with controls. Regarding the neuroimaging scans, at T1 there were no differences between the two groups. After treatment, the patients had significant alterations in some brain areas when comparing to T1 but after normalizing for anxiety and depression scores, the changes were no longer meaningful	Mo et al. (2017)

Table 2 (continued)

Number and conditions of patients	Anticancer treatment	Clinical neuroimaging and neurophysiological evaluation	Major findings	Study
33 breast cancer patients 31 healthy controls	Patients received cycles of DOX, paclitaxel, and CYA without any hormonal treatment	Resting-state fMRI	The chemotherapy-group had lower functional connectivity	Tao et al. (2017)
16 breast cancer women, over the age of 60 years that received chemotherapy 15 age matched healthy controls	Most patients received a combination of docetaxel and CYA or paclitaxel and trastuzumab	MRI scanning and cognitive testing before the start of the treatment (T1) and after the end of chemotherapy (T2)	At T1, there were no differences in grey matter density between both groups. In the chemotherapy group, from T1 to T2 there was a considerable reduction in grey matter density in several brain areas. In opposition, the healthy controls had no significant changes from T1 to T2. Considering the cognitive tests, there were no differences among groups	Chen et al. (2018)
19 women that had breast cancer and received chemotherapy less than 6 months before assessment started 20 healthy control women	Patients received cycles of taxotere and epirubicin	Diffusion MRI scanning	The chemotherapy group had lower white matter integrity. Moreover, the brain network integration also deteriorated in the patients group	Li et al. (2018a)
28 breast cancer patients 29 matched healthy controls	Patients received cycles of DOX and paclitaxel	MRI scanning was performed after each cycle of chemotherapy	The grey matter density decreased in the chemotherapy group comparing with the control group	Li et al. (2018b)
Breast cancer patients received chemotherapy ($n = 26$) or not ($n = 23$) Healthy controls ($n = 30$)	Patients received combinations of DOX, CYA, epirubicin, docetaxel, paclitaxel, and 5-FU	MRI scanning before the start of chemotherapy (T1) and at 6 months follow-up (T2)	At T2, the chemotherapy group had subtle but significant deterioration in white matter integrity when comparing to the patients that did not receive chemotherapy and to healthy controls	Menning et al. (2018)
Patients with small cell lung cancer after receiving chemotherapy ($n = 17$) or prior to chemotherapy ($n = 15$) 18 healthy controls	Not described	Neuropsychological testing. Event-related potentials recording	The chemotherapy-group performed worse in speed-processing, executive function and visuospatial abilities when compared to healthy controls Both patient groups had abnormal event-related potentials when comparing to the healthy controls	Simo et al. (2018)

Table 2 (continued)

Number and conditions of patients	Anticancer treatment	Clinical neuroimaging and neurophysiological evaluation	Major findings	Study
122 women with early-stage breast cancer that received chemotherapy ($n=72$) or not ($n=49$) 59 healthy control	Women received cycles of 5-FU, epirubicin and CYA with or without docetaxel	Neuropsychological testing and assessment of volume changes through VBM before the start of treatment (T1) and after the end of the treatment (T2)	At T1 assessment, there were no differences among groups. In T2, chemotherapy-treated patients performed worse on attention/concentration, memory and processing speed. In opposition, the other groups improved their scores in the testing done. In the younger patients (<52 years old) treated with chemotherapy, the VBM results showed widespread volume alterations when comparing to the other groups	Blommaert et al. (2019)
19 women with breast cancer and over the age of 60 14 age-matched healthy controls	Most chemotherapy-treated patients received docetaxel in combination with DOX or CYA	DTI before treatment (T1) and one month after chemotherapy (T2)	In T1, both groups performed similarly in the neuropsychological testing. At T2, the chemotherapy group performed worst only in the oral recognition test. However, from T1 to T2, the chemotherapy group had significant white matter alterations when compared to the control group	Chen et al. (2019)
Women with breast cancer treated with chemotherapy ($n=18$) or not ($n=22$) 22 healthy controls	Not described	Neuropsychological testing and fMRI scanning post-diagnosis and before treatment (T1) and at least 1 month (T2) and 7 months (T3) after the end of the treatment	The chemotherapy-group at T2 was considerably affected by the treatment and had lower performance on the digit span test compared to T1. However, they recovered in T3. Similarly, in T2 that same group had abnormal functional connectivity in parietal and frontal regions of the brain compared to T1 and a partial return to baseline values in T3. These patterns were not so significantly altered neither in the non-chemotherapy group nor in the healthy controls	Kardan et al. (2019)
13 breast cancer patients received chemotherapy 13 matched healthy controls	Patients received cycles of DOX, CYA and taxol or CYA, MTX, and 5-FU	fMRI scanning during memory encoding and retrieval before and after chemotherapy	After chemotherapy, patients had diminished activity in posterior brain regions, especially those involved in visual processing when compared to the control group	Pergolizzi et al. (2019)

Table 2 (continued)

Number and conditions of patients	Anticancer treatment	Clinical neuroimaging and neurophysiological evaluation	Major findings	Study
85 patients with colorectal cancer that received chemotherapy ($n=49$) or not ($n=26$)	Chemotherapy regimens consisted of cycles of 5-FU with or without oxaliplatin	Standardized neuropsychological and MRI assessment before the start of the treatment (T1) and after 12 months (T2)	On T2, the chemotherapy-treated group performed worse on executive function when comparing to the non-treated group at the same time-point	Sales et al. (2019)
19 breast cancer patients that received chemotherapy within 6 months of the study	Patients received docetaxel and epirubicin	MRI scanning and neuropsychological testing	Patients had alterations in the frontoparietal lobe and occipital lobe when compared to the healthy controls	Shen et al. (2019)
20 matched healthy controls				
28 female patients that received chemotherapy for breast cancer	Not described	Neuroimaging techniques to evaluate the striatal dopamine transporter binding ratio	Chemotherapy-treated patients had significantly dopaminergic decrease in all the dorsal striatum when comparing with the healthy controls	(Vitor et al. 2019)
22 healthy women				
20 patients with gynecological cancer	Chemotherapy regimens included paclitaxel, carboplatin, cisplatin, DOX, bleomycin, and MTX	Neuropsychological testing and MRI scanning before the start of the treatment (T1) and after treatment end (T2)	In T1, besides psychomotor speed, the groups did not present significant differences. However, at T2, the chemotherapy group performed significantly worst in particularly in the domains of verbal memory, immediate and delayed recall and information processing speed	Zeng et al. (2019)
20 healthy controls				
Patients with lung cancer treated with chemotherapy ($n=21$) or not ($n=27$)	Cisplatin-based therapy	fMRI scanning	The group treated with chemotherapy had considerably lower functional connectivity between the posterior cingulate cortex and the right anterior cingulate cortex, when compared to the non-chemotherapy treated group	Zhang et al. (2019)
30 healthy controls				
35 premenopausal women with breast cancer who received chemotherapy were compared with 32 age- and sex- matched healthy controls	Chemotherapeutic treatment included mainly DOX, paclitaxel and CYA	Resting-state fMRI scanning and cognitive assessments of attention, memory and executive functioning, on average 25 days after the end of the treatment	Breast cancer patients had lower interhemispheric functional connectivity in the cingulated posterior, insular and postcentral regions. Furthermore, the scan values of the cingulated posterior and in the insular region correlated with decreased executive function and memory, respectively	Tao et al. (2020)

Table 2 (continued)

Number and conditions of patients	Anticancer treatment	Clinical neuroimaging and neurophysiological evaluation	Major findings	Study
19 older breast cancer patients (> 60 years old) treated with chemotherapy were compared with 14 age-matched healthy controls	Regimens consisted of combinations of docetaxel, CYA, carboplatin, trastuzumab, DOX, and paclitaxel	DTI and neuropsychological assessment before (T1) and one month (T2) after the treatment	Chemotherapy-treated patients had increased mean diffusivity and radial diffusivity in the genu of the corpus callosum from T1 to T2. No meaningful changes were observed in the control group	(Chen et al. 2020a)
102 breast cancer patients who received chemotherapy ($n=57$) or not ($n=45$), and 70 healthy controls	Not described	Resting-state fMRI after patients finished chemotherapy	Functional network disruptions occurred regardless of chemotherapeutic treatment when comparing to the control group	Chen et al. (2020b)
17 breast cancer patients that received chemotherapy	Patients received combinations of DOX, CYA, docetaxel, and epirubicin	Resting-state fMRI before chemotherapy (T1), 1 week after completing chemotherapy (T2), and 6 months after completing chemotherapy (T3)	Patients had significantly changes within-network functional connectivity at T2 and T3	Feng et al. (2020)
Breast cancer patients treated with chemotherapy ($n=46$) were compared with matched healthy controls ($n=50$)	Not described	MRI scanning before any treatment (T1), 1 month after chemotherapy (T2), and 1-year post chemotherapy (T3)	Chemotherapy-treated patients had decreased cortical thickness along with accelerated predicted brain age from T1 to T2 when comparing to controls	Henneghan et al. (2020)
28 patients with acute lymphoblastic leukemia compared with 20 matched healthy controls	Not described	MRI scanning after patients completed chemotherapy	Chemotherapy-treated patients had fewer highly interactive nodes in temporal regions compared to the healthy controls	Wang et al. (2020)
529 primary breast cancer patients evaluated of which 178 cases before treatment (Group 1), 167 during chemotherapy (Group 2), and 184 after chemotherapy (Group 3)	Not described	Electrophysiological measure of P300 component of ERP	Group 2 and group 3 had significant differences in P300 latency in comparison with Group 1	Xu et al. (2020)

ApoE Apolipoprotein E, *DTI* Diffusion tensor imaging, *EEG* Electroencephalogram, *ERP* Event Related Potentials, *fMRI* Functional magnetic resonance imaging, *MMSE* Mini mental state examination, *MRI* Magnetic resonance imaging, *PET* Positron emission tomography, *VBM* Voxel-based morphometry

neurologic problems and used an electroencephalographer to measure the event-related potential (ERP) P300 (Siddiqui et al. 1992). The P300 is a positive component of the event-related potential that peaks at 300 ms or more after a stimulus and integrates cognitive domains such as attention and working memory (Siddiqui et al. 1992). A change on these cognitive domains leads to a latency in the potential registered (Linden 2005). The study aimed to detect subclinical cognitive defects in cancer patients. In the 16 patients analyzed, 9 presented abnormalities in the P300, while the 28 healthy controls had no abnormalities (Siddiqui et al. 1992). In a study carried out by Silverman et al., they used a positron-emitting glucose analogue (18F-fluorodeoxyglucose) in combination with PET and the scans revealed that the chemotherapy-treated group had hypometabolism in the inferior frontal cortex during the cognitive test. Thus, chemotherapy exposure could alter cortical activation while performing a short-term memory recall (Silverman et al. 2007). Metabolic differences due to chemotherapy exposure were also observed in breast cancer survivors even 10 years after treatment and when comparing with healthy controls. Cancer survivors had hypometabolism in the bilateral orbital frontal gyri, in the basal ganglia (right substantia nigra), brainstem, left postcentral gyrus of the parietal lobe and corpus callosum (Ponto et al. 2015).

The large portion of clinical neuroimaging studies regarding chemobrain has used MRI and fMRI-based techniques. De Ruiter and colleagues used fMRI and investigated the effects of chemotherapy on the brain of breast cancer survivors more than 10 years post-treatment. Cancer survivors that had high-dose chemotherapy had hyporesponsiveness of the dorsolateral prefrontal cortex and parahippocampal gyrus while performing the tower of London test and the test of episodic memory, respectively, when comparing with the non-chemotherapy group (de Ruiter et al. 2011).

Studies have gone further and determined white matter integrity after chemotherapy. Deprez et al. performed DTI in cancer patients, approximately 3 months after treatment (3 to 6 cycles of 5-FU, epirubicin and CYA with or without taxol). The DTI revealed that chemotherapy-treated patients had decreased frontal and temporal white matter integrity when comparing with healthy controls and non-chemotherapy treated patients (Deprez et al. 2011). As previously mentioned, the integrity of the white matter is essential for proper cognitive function. In fact, the authors verified that the loss white matter integrity in the temporal and parietal areas correlated with poor outcomes in neuropsychological tests done in these patients, in specific regarding domains of attention and processing/ psychomotor speed (Deprez et al. 2011). Moreover, white matter microstructure alterations were also detected in older women (more than 60 years old) that underwent chemotherapy, since the MRI scan performed 1 month after chemotherapy revealed significant alterations

in the genu of corpus callosum and left superior longitudinal fasciculus (Chen et al. 2019).

Regarding grey matter volume, Lepage et al. reported reductions in grey matter of frontal region, also in the parietal and occipital regions in cancer patients 1 month after treatment (5 or 6 cycles of 5-FU, epirubicin and CYA) in comparison to the baseline scans. At one-year examination, partial recovery happened in some areas, but the frontal and temporal regions were still altered when compared to baseline (Lepage et al. 2014). These changes had positive correlations with the poor cognitive performance in the domains of processing speed, working memory and visual memory seen in that study (Lepage et al. 2014). Hippocampus volume has also been evaluated in breast cancer patients in average 3.4 years post-chemotherapy, mostly treated with DOX and CYA. It was shown that those patients had significantly reduced left hippocampal volume and that the right hippocampal volume was slightly decreased when comparing to healthy controls (Kesler et al. 2013a, b).

Chemotherapy can also affect core processes of the brain and the default network mode (DNM) connectivity has been reported to be decreased in chemotherapy-treated patients. DNM of the brain comprises areas that are usually more active during rest than during active task performance and that are thought to support important processes such as implicit learning, memory retrieval, prospection and other internally focused thought processes (Mars et al. 2012). In a cohort of women with breast cancer and treated with CYA and/or an anthracycline, fMRI scanning was performed and showed that from the baseline evaluation to the 1-year following chemotherapy time-point, there was a decrease connectivity in the DMN of chemotherapy treated women (Dumas et al. 2013). Furthermore, the intrinsic functional connectivity pattern in the DMN was analyzed in lung cancer patients treated with cisplatin-based therapy, and 6 months after treatment, chemotherapy-treated patients had significantly reduced functional connectivity between the posterior cingulate cortex and the right anterior cingulate cortex, left inferior parietal lobe and left medial prefrontal cortex, when comparing to the non-chemotherapy group (Zhang et al. 2019).

In summary, electrophysiological and neuroimaging studies provided evidence for the functional and structural damages evoked by chemotherapy, being even more sensitive and by far more objective than cognitive studies. In addition, very often they were done after longer periods post-treatment and drugs other than anthracyclines were analyzed. Nevertheless, fewer patients *per* study were evaluated because of costs, and breast cancer as well as women are still more represented in the studies. Conversely, it has become clear that chemotherapy causes brain abnormalities in the values of P300 (Kreukels et al. 2005, 2006), reduced metabolism in certain area (Ponto et al. 2015), alterations

in the white matter microstructure and integrity (Chen et al. 2019), reduction of the grey matter volume (Lepage et al. 2014), especially in the frontal region, and alters the brain networks (Dumas et al. 2013). Those changes are reported to remain years after chemotherapy (Kreukels et al. 2005).

Animal studies

All experimental in vivo studies found in this review were performed on rodent models since as it happens in cancer patients, the rodents display cognitive impairment following systematic chemotherapy treatment. Moreover, there is a wide variety of behavioral tests that can evaluate the effects of chemotherapy in rodents and determine what cognitive domains are impaired. Besides behavioral testing, the rodent brain can be removed and used to study the underlying cell-biological and molecular mechanisms affected by chemotherapy (Seigers et al. 2013). The use of laboratory animals allows the evaluation of the effects commonly used drugs, either as chemotherapeutic combination regimens or as single chemotherapeutic agents and to understand potential differential effects among male and female brains. All of these aspects are difficult or even impossible to assess in a cohort of cancer patients because of lower number of enrolled patients (Matsos and Johnston 2019) or even sex specific or gender prone cancers for obvious ethic reasons.

For systematic reasons, a table with animal species and strain used in the studies, drug dosage and main results regarding the putative neurotoxicity of a single anticancer drug regimen will be presented next in Table 3.

Regarding combination regimens, male rats were administered intraperitoneal (i.p.) with DOX (2 mg/kg) or CYA (50 mg/kg) or both once *per* week for 4 weeks. The authors observed that co-administration and CYA administration alone reduced cell proliferation in the subgranular zone of the dentate gyrus, while the co-administration of the drugs significantly reduced cell survival of newly divided cells. Plasma levels of brain-derived neurotrophic factor (BDNF) in the co-administration group were significantly decreased when compared to saline controls; however, brain levels of BDNF presented no differences among groups (Kitamura et al. 2015).

In animal studies, the brain structure most frequently investigated has been the hippocampal formation due to its role in cognition and its inherent active neurogenesis on adult animals. Furthermore, DOX has been the chemotherapeutic agent more studied in animal models, and oxidative stress and increased levels of cytokines seem to be involved in the chemobrain observed. MTX seems to influence hippocampal neurogenesis, causing depressive-like behaviors and memory deficits, accompanied with increased activity of inducible nitric oxide synthase (iNOS) and

cyclooxygenase-2 (COX-2), while 5-FU also impacted on neurogenesis and cellular viability on hippocampal neurons (Yang et al. 2012). Cisplatin has also been seen to cause high impact on neuronal population on cortex as in hippocampus (Huo et al. 2018; Lomeli et al. 2017; Mu et al. 2015; Zhou et al. 2016). CYA and thiotepa significantly impaired the memory (Alexander et al. 2018; Janelins et al. 2016).

In vitro studies

In vitro neuronal models have been extensively used in the field of neurotoxicology since they allow a better understanding of the molecular and cellular mechanisms by which xenobiotics affect the different cell types that compose the brain. There are several models available with different complexities, ranging from subcellular systems, single cell types to systems that try to maintain the tissue structure and function (e.g. brain layers). The most suited in vitro model depends on the hypothesis being tested and the previous knowledge of the pharmacokinetics and pharmacodynamics of the compound being studied (Barbosa et al. 2015).

One of the first studies using CNS cells to study the biological basis of the chemotherapy-induced cognitive dysfunction was performed by Dietrich and colleagues. They evaluated the effects of carmustine (5–200 μ M), cisplatin (0.1–100 μ M) and cytosine arabinoside (0.01–2 μ M) in multiple tumor cell lines and pluripotent neuroepithelial stem cells that can differentiate into neurons, astrocytes and oligodendrocytes. The results obtained demonstrated that these cytostatic drugs were more toxic for the progenitor cells of the CNS and to non-dividing oligodendrocytes than they were for multiple cancer cell lines (Dietrich et al. 2006). The loss of oligodendrocytes and, consequently, myelination may result on the loss of neuronal connectivity and cognitive deficits (Guest 2020). In another study, Andres and colleagues used neural stem cells and differentiated hippocampal neurons to evaluate the neurotoxic effects of cisplatin. The authors observed that exposure to low concentration of cisplatin (0.1 μ M) caused loss of dendritic spines and synapses within 30 min and longer exposures injured dendritic branches and decreased dendritic complexity. However, at this concentration there was no reduction in viability or increase in apoptosis in the neural stem cells. For higher concentrations of cisplatin (1 μ M) a rapid loss of synapses, dendritic disintegration and neuronal apoptosis occurred in differentiated hippocampal neurons, but not in the neural stem cells. The authors concluded that the synaptic and dendritic damage observed could contribute to the cognitive impairment seen in chemotherapy-treated patients and that differentiated neurons were more prone to injury of cisplatin (Andres et al. 2014).

Table 3 Animal studies evaluating neurotoxicity induced by cytostatic drugs, animal species and strain, and major results

Cytostatic agent	Experimental setting	Animals	Major results found in the drug-treated animals	Study
5-FLUOROURACIL	5 injections [20 mg/kg, intravenous (i.v.)] over the course of 2 weeks	Adult male Lister-hooded rats	Altered spatial working memory; decrease in hippocampal brain-derived neurotrophic factor (BDNF) and doublecortin (DCX)	Mustafa et al. (2008)
	5 injections [25 mg/kg, intraperitoneal (i.p.)] over 2 weeks	Male Lister-hooded rats	Impaired spatial working memory; reduced cell proliferation and survival in the sub granular zone of the dentate gyrus	Lyons et al. (2012)
	6 injections (20 mg/kg i.p.) over 2 weeks. Animals were sacrificed 1 day, 2 weeks or 6 weeks after the end of the treatment	Male Lister-hooded rats	Decreased expression of Ki67 protein at the 2- and 6-weeks evaluation; decreased survival of newly divided neurons at day 1 persisted at 6 weeks	El et al. (2012)
	2 injections (25 mg/kg i.v.) over 1 week	Adult male Wistar rats	Decreased dopamine release in the striatum; impaired attentional shifting	Jarmolowicz et al. (2019)
METHOTREXATE	37.5, 75, 150 or 300 mg/kg, i.v. Animals were sacrificed 1 or 7 days after the last administration	Adult male Wistar rats	Dose-dependent negative effect on hippocampal cell proliferation	Seigers et al. (2008)
	250 mg/kg i.v. Animals were sacrificed 1 or 3 weeks after treatment	Adult male Wistar rats	Decreased hippocampal cell proliferation; decreased white matter density in the lateral corpus callosum; impaired memory in all time points evaluated	Seigers et al. (2009)
	40 mg/kg i.p. Animals were tested 1 day after treatment	Female C3H/HeN mice	Depression-like behavior and cognitive impairment; decreased cell proliferation on the dentate gyrus; upregulation of pro-inflammatory enzymes such as inducible nitric oxide synthase (iNOS) and cyclooxygenase type 2 (COX-2)	Yang et al. (2012)
	100 mg/kg i.p. Animals were tested 4 weeks after treatment	Wild-type CD57BL/6 mice breed with CD1 mice	Significant decrease in myelin sheath thickness in the corpus callosum; increased microglia activation in the white matter but not in the grey matter	Gibson et al. (2019)
DOXORUBICIN	0.5, 2 and 8 mg/kg i.p. Animals were tested 3 h or 1 or 7 days after administration	Adult male Wistar rats	Impaired learning capacity, long-term memory and sparing short-term memory, 1 and 7 days after administration	Liedtke et al. (2009)
	1.5 and 3 mg/kg i.p. Animals were tested 3 or 24 h after administration	Adult male Wistar rats	Inhibition of the dopamine system uptake in the frontal cortex, hippocampus, and striatum	Antkiewicz-Michaluk et al. (2016)
	25 mg/kg i.p. Animals were sacrificed within 24 h after treatment	Male B6C3F1J mice	Slower uptake of glutamate in the frontal cortex and dentate gyrus	Thomas et al. (2017)

Table 3 (continued)

Cytostatic agent	Experimental setting	Animals	Major results found in the drug-treated animals	Study
	4 mg/kg i.p. weekly for 5 weeks. Animals performed behavioral after the last administration	Male rats	Impaired learning and memory	Alharbi et al. (2020)
	2 mg/kg i.p. weekly for 4 weeks	Sprague–Dawley male rats	Significant learning and memory impairment. Reduction of glutathione (GSH) levels and increased lipid peroxidation. Increase in nuclear factor kappa B (NF- κ B) and pro-inflammatory mediators [COX-2/tumor necrosis factor (TNF- α), and glial fibrillary acidic protein (GFAP). Decrease in acetylcholine levels and increased neuronal cell apoptosis	Ali et al. (2020)
	2 mg/kg i.p. weekly for 4 weeks. Animals were sacrificed 28 days after the last administration	6-week-old male Sprague–Dawley rats	Decreased spatial memory and short-term memory Nuclear pyknosis and degeneration of neuronal cells in the hippocampus Increased hippocampal oxidative stress and pro-inflammatory response	Tong et al. (2020)
	2 mg/kg i.p. weekly for 4 weeks. Animals were sacrificed 9 days after the last administration	Male albino rats	Memory impairment Increase in pro-inflammatory mediators IL-17 and TNF- α and transforming growth factor beta (TGF- β) Decreased mitochondrial antioxidants Autophagic and apoptotic neuronal death in the hippocampus	Wahdan et al. (2020)
	2.5 mg/kg DOX i.p. every 2 days for a total of 7 injections over a 2-week period	Male C57BL/6 mice	Impaired spatial learning and memory Increased hippocampal cytokines and GFAP levels Redox imbalance in the hippocampus Mitochondrial dysfunction	Wang et al. (2021)
CYCLOPHOSPHAMIDE	7 injections (30 mg/kg i.v.) 2 days apart. Animals were sacrificed 5 days after the last administration	Male Lister-hooded rats	No significant alterations on spatial working memory or hippocampal proliferation	Lyons et al. (2011)
	4 injections (200 mg/kg i.v.) 1-week interval. Animals were behaviorally tested at the beginning of the treatment and for the following 3 months	Male C57BL/6 mice	Gradual memory decline from the 3rd injection until the 3 month follow up	Janelins et al. (2016)
	4 injections (100 mg/kg i.p.) every 2 days. Animals were tested 24 h after administration and the following 3 days	Male mice	Significant decline in memory	Alhowail et al. (2019)

Table 3 (continued)

Cytostatic agent	Experimental setting	Animals	Major results found in the drug-treated animals	Study
THIOTEPA	3 injections (10 mg/kg i.p.) weekly for 3 weeks. Animals were tested 5 weeks after treatment	Male mice C57BL/6 J	Decreased memory retention; decreased hippocampal dendritic arborization and length; decreased cytokine production in the hippocampal formation	Alexander et al. (2018)
CISPLATIN	6–8 injections (4 mg/kg i.p.) once every 7 days. Animal behavioral testing started after the 3rd administration	Male Sprague–Dawley rats	Anxiety-like behavior and impaired spatial reference memory	Mu et al. (2015)
	3 cycles of 5 daily administrations (34.5 mg/kg i.p.). Animals were tested 5 days after the last administration and sacrificed 4 weeks after the last administration	Female C57/BL6J mice	Deficits in spatial orientation and memory; decreased white matter integrity and reduced number of dendritic spines and neuronal arborizations in the cingulate cortex	Zhou et al. (2016)
	3 mg/kg/day for 2 days given i.p. Animals were sacrificed 5 weeks after the end of administrations	Male Sprague Dawley rats	Reduced hippocampal dendritic branching and spine density of CA3 pyramidal neurons; mitochondrial degradation and vacuolization in the CA3 hippocampal neurons	Lomeli et al. (2017)
	2 cycles of 5 daily administrations (2.3 mg/kg/day i.p.). Animals were behavioral tested 3 days after treatment and sacrificed 4 weeks after the last administration	Male C57BL/6 J mice	Attention deficits; abnormal synaptic integrity in the prefrontal cortex	Huo et al. (2018)

BDNF Brain-derived neurotrophic factor, *COX-2* Cyclooxygenase type 2, *DCX* Doublecortin, *GFAP* glial fibrillary acidic protein, *GSH* glutathione, *iNOS* Inducible nitric oxide synthase, *i.p.* Intraperitoneal, *i.v.* Intravenous, *NF-κB* Nuclear factor kappa B, *TNF-α* Tumor necrosis factor-α, *TGF-β* Transforming growth factor β

In the vast majority of the studies, the neurotoxicity of chemotherapeutic drugs has been evaluated as either one drug alone (Gopal et al. 2012) or with the most used combinations of chemotherapeutic drugs (Long et al. 2011). However, few studies have compared the neurotoxic effects of two chemotherapeutic drugs applied together. To try to fill this gap, Almeida and co-workers compared the neurotoxicity of DOX and mitoxantrone (0.13, 0.2 and 0.5 μM) in the SH-SY5Y human neuronal cell model. The authors observed that, despite both chemotherapeutics agents causing significant cytotoxicity, mitoxantrone was more neurotoxic than DOX in differentiated SH-SY5Y cells, while in undifferentiated (and therefore more proliferative) SH-SY5Y cells, DOX was more neurotoxic than mitoxantrone, in the same experimental conditions. However, both DOX and mitoxantrone caused a decrease in the transmembrane mitochondrial potential following a 48-h exposure at the lowest concentration. The authors also sought to evaluate the neuroprotective effect of some compounds and observed that tiron (100 μM ; an antioxidant) was the only drug among those tested capable of partially avoiding the toxicity of mitoxantrone. On the other hand, cycloheximide (10 nM; a protein synthesis inhibitor) and Ac-DEVD-CHO (100 μM ; an inhibitor of caspase 3/7) were able to partially protect against DOX-induced neurotoxicity and unexpectedly, the antioxidant tiron and the GSH precursor N-acetylcysteine did not provide any neuroprotection. The higher neurotoxicity of mitoxantrone in differentiated SH-SY5Y could be due to its greater lipophilicity, as it most easily accumulates inside the cells (Almeida et al. 2018). Mitoxantrone is used in the treatment of several types of cancer and was also approved for the treatment of multiple sclerosis (Hartung et al. 2002), thus its higher neurotoxicity is of great concern as its neurotoxic effects have been poorly studied over the years and the BBB might be impaired in multiple sclerosis (Minagar and Alexander 2003).

Regarding DOX, a study by Lopes and colleagues evaluated its neurotoxicity in rat cortical neurons exposed to concentrations ranging from 0.01 to 20 μM , for 24 or 48 h. The obtained results showed that DOX induced neuronal cell death in differential ways, depending on the concentration, being that concentrations up to 0.5 μM of DOX, neuronal death was due to apoptosis while at higher concentrations, necrosis became prevalent (Lopes et al. 2008). The exposure of rat cortical primary cell cultures to DOX resulted in a significant generation of reactive oxygen and nitrogen species (RONS), activation of nuclear factor kappa B (NF- κB), depletion of glutathione (GSH) and ATP and cell death, in a concentration biphasic manner. Death at the lower concentration (up to 0.5 μM) was prevented using a nitric oxide synthase (NOS) pharmacological inhibitor. DOX also significantly increased protein-bound quinone products in neurons

in a concentration-dependent manner (Lopes et al. 2011), demonstrating local oxidative stress.

The emerging high-throughput technologies have also been used to study the phenomena of chemotherapy-induced cognitive dysfunction. Seo and colleagues evaluated the changes on transcriptome of human T98G neuroglial cells exposed to the combination of 5-FU (50 $\mu\text{g}/\text{ml}$), epirubicin (0.5 $\mu\text{g}/\text{ml}$) and CYA (2 $\mu\text{g}/\text{ml}$), commonly used in the treatment of breast cancer (Schagen et al. 2006b). The results obtained showed that this anticancer combination deregulated 67 genes involved in neuronal development, 37 genes involved in development of the sensory system, 12 genes in extension of axons and 3 genes in migration of neurons. Moreover, this combination downregulated the expression of integrins subunits $\beta 1$ and αV and the biosynthesis of cell adhesion molecule 1, which are essential for development and maintenance of synaptic structure plasticity and synaptic long-term potential (Seo et al. 2018).

In vitro methods can be a more simplistic approach when compared to animal models, but they allow a better understanding of the biological pathways impaired by chemotherapy exposure. In addition, they allow a rapid screening for testing drugs and their combinations.

Possible molecular mechanisms of chemobrain

Direct passage of chemotherapeutic agents through the blood–brain barrier

The BBB represents the first line of defense of the brain. The endothelial cells linked by tight junctions, the lack of fenestrations and low pinocytotic capacity prevent the intercellular passage of molecules (Seigers and Fardell 2011). The transport of compounds and nutrients essential for proper brain function is assured by selective uptake and efflux transport mechanisms. Nevertheless, compounds without endogenous selective transporter can only enter the brain by passive diffusion, therefore, the ability of a compound to cross the BBB is established by its lipid solubility, molecular weight (less than 400–500 daltons), degree of ionization, plasma protein and tissue binding and local cerebral flow (de Vries et al. 2006). Regardless, many compounds are substrates of the efflux ATP-binding cassette transporters of the multidrug resistance protein family in the BBB, in which the P-glycoprotein and the breast cancer resistance protein (BCRP) are included and most prevalent (Löscher and Potschka 2005). Altogether, the entrance of anticancer drugs into the brain and its accumulation is difficult (de Vries et al. 2006; Seigers and Fardell 2011) and it has been considered that the physical and functional role of the BBB protects the brain from the entrance of chemotherapeutic

agents. While systemic administration of anthracyclines (da Ros et al. 2015), platinum-based compounds (Jacobs et al. 2009) and taxanes (Rice et al. 2003) does not seem to be able to reach the CNS, carmustine, temozolomide and procarbazine (Bhowmik et al. 2015) can pass through the BBB. Moreover, some chemotherapeutic agents can cross the BBB, when its permeability is changed, as mentioned before. Some specific examples will be described next.

In the literature, there are several case-reports of cancer patients treated with 5-FU that rapidly developed leukoencephalopathy and recovered partially when 5-FU regimens were halted (Choi et al. 2001; Kim et al. 2006; Kwon et al. 2010). This molecule has a molecular weight of 130.08 daltons (Jansen et al. 2005) and has a short half-life in serum (10–25 min) both in humans and rodents and can readily cross the BBB by passive diffusion (Wigmore et al. 2010). A study in primates that received an i.v. infusion of 5-FU demonstrated a rapid diffusion of the compound to all brain regions and cerebrospinal fluid (Bourke et al. 1973). In another study, the CNS permeability of oral administration of 400 mg/kg of capecitabine (that it is converted to 5-FU) in mice was determined. After 1 h, 5-FU was detected in the brain and in the cerebrospinal fluid (Zhang et al. 2015).

CYA has the molecular weight of 261.08 daltons and it is a lipophilic alkylating agent capable of crossing the BBB (Jansen et al. 2005). It is an inactive prodrug that needs enzymatic activation and is metabolized to both active (phosphoramidate mustard and acrolein) and inactive metabolites (de Jonge et al. 2005). A study performed on rats that were treated with 30 mg/kg CYA (i.p.) detected its presence on the brain (Graul et al. 1967). Furthermore, in a clinical study, 10 patients that had CNS neoplasms received cycles of CYA (500–1250 mg/m²) and the levels of CYA were measured in cerebrospinal fluid. One-hour after the administration, the levels of the drug were around 8 µg/mL. However, CNS neoplasms can cause weakening of the blood-cerebrospinal fluid barrier and of the BBB, making the entrance of CYA on the brain even easier (Egorin et al. 1982). Despite CYA's chemical characteristics make it prone to cross the BBB, there are no studies in cancer patients without CNS tumor or metastasis that evaluated its entry after systemic administration.

Although the entrance of cisplatin in the brain is considered poor, it was analyzed in nonhuman primates (adult male rhesus monkeys). The animals received 10 mg/kg of cisplatin i.v. over 1 h (equivalent to 200 mg/m² in humans), and cisplatin was detected in the brain (the area under the concentration–time curve over the 4-h sampling was 1.1 µM h and 2.8 µM h) and in the cerebral spine fluid (4 µM h and 4.8 µM h). Therefore, even in relatively small amounts, cisplatin can reach the brain (Jacobs et al. 2010).

DOX is a water-soluble molecule with the molecular weight of 580 daltons (Jansen et al. 2005). Moreover, DOX

is a substrate of the efflux pump P-glycoprotein, that significantly restricts its entrance to the brain (Ohnishi et al. 1995). It has been largely assumed that DOX cannot cross the BBB, as it was not detected after systemic administration in areas protected by the BBB such as the cortex and hippocampus; meanwhile, DOX was detected in the choroid plexus that is protected by the more permeable blood-cerebrospinal fluid barrier (Tangpong et al. 2006). However, *post-mortem* analysis of human brains detected the presence of DOX (Stewart et al. 1993). *Post-mortem* distribution has not been described in the deep tissues where DOX accumulates nor even in studies on cancer patients to assess the BBB integrity on treatment. Despite this, in an intact BBB, DOX neurotoxicity caused by direct entrance after systemic administration seems highly unlikely at pharmacological relevant concentrations due to its physical properties and because it is a substrate of P-glycoprotein. Nevertheless, when morphine was administered prior to the drug administration, DOX has been detected in the brain (Sardi et al. 2011). And also, DOX has been found in the brain of brain-tumor bearing mice, though at residual levels (Gaillard et al. 2014).

The BBB integrity might also be compromised after chemotherapy. Branca and colleagues investigated the effects of oxaliplatin in the properties of the BBB using a rat brain endothelial cell line. The incubation of oxaliplatin (10 µM) dysregulated the expression of tight junction proteins zonula occludens-1 and F-actin, causing the disruption of the monolayer and pore-formation, making that barrier no longer impermeable to the entrance of oxaliplatin. The authors decided to evaluate the role of reactive oxygen species (ROS) formation in tight junction protein alterations and observed that oxaliplatin exposure caused significant increase of intracellular ROS, which on the other hand, increased endoplasmic reticulum stress. Since endoplasmic reticulum stress can induce caspase 3 activation, the activity of this enzyme was measured, and the authors found that caspase 3 was significantly activated in the oxaliplatin-treated cells when compared to the controls. Furthermore, the oxaliplatin-exposed cells also had higher levels of extracellular ATP and the extrusion of cellular ATP is dependent upon the activation of pannexin-1 channels. The authors concluded that ROS formation leads to the loosening of BBB and could cause further damage on the brain of chemotherapy-treated patients (Branca et al. 2018).

BBB has a crucial role on protecting the brain from most chemotherapeutic agents used in a systemic administration, although its permeability can be disrupted in several pathological conditions, namely pro-inflammatory cytokines surges, chemotherapeutic agents or even illness like multiple sclerosis, all situations that may make cancer patients more prone to neurotoxicity. In brain cancer, attempts using nanodelivery systems and others have been done to increase the

levels of chemotherapeutics on the brain, but in this review, brain tumors will not be mentioned.

Oxidative stress

Oxidative stress has been linked to the progression of many neurodegenerative diseases, such as Parkinson's, Alzheimer's and amyotrophic lateral sclerosis (Andersen 2004). The brain is highly susceptible to oxidative stress due to its high demand for oxygen, high content in polyunsaturated fatty acids and relative low antioxidant capacity (Joshi et al. 2007). An imbalance in the production and elimination of reactive species leads to cell damage since RONS can cause protein, DNA and RNA oxidation or lipid peroxidation and compromise essential cellular functions (Gandhi and Abramov 2012). Although it is known that ROS formation in cancer cells by some chemotherapeutic agents is responsible for tumor regression, more than half of the approved chemotherapeutic agents by the US Food and Drug Administration have been reported to induce oxidative stress in non-targeted tissues (Chen et al. 2007). Among those, anthracyclines such as DOX or epirubicin, platinum-based compounds such as cisplatin, carboplatin or oxaliplatin, alkylating agents and topoisomerase inhibitors produce the highest levels of ROS. On the other hand, taxanes (e.g., paclitaxel and docetaxel), vinca alkaloids (e.g. vinblastine) and antimetabolites (e.g. 5-FU) produce lower levels of ROS peripherally (Conklin 2004; Yang et al. 2018). Chemotherapeutic agents can cause oxidative stress by undergoing redox cycle and disrupt essential cellular functions or by depleting antioxidants that inhibit cellular detoxification, or even causing mitochondria dysfunction (Yang et al. 2018). The generation of oxidative stress in the brain may cause a downstream of events leading to mitochondrial dysfunction, glial cell activation, programmed cell death triggering and proteasomal malfunction (Andersen 2004). Effectively, oxidative stress caused by ROS after exposure to chemotherapeutic agents can be a trigger for chemobrain (El-Agamy et al. 2019).

Protein oxidation, one of the consequences and a marker of oxidative stress, occurs by oxidative protein cleavage, which results in carbonyl groups (Czerska et al. 2015; El-Agamy et al. 2018; Yokoyama et al. 2017). Also, 3-nitrotyrosine (3-NT) is one of the main products of protein oxidation and nitration, more specifically, of tyrosine oxidation (Ruiz-Ojeda et al. 2018). This marker is formed by replacing hydrogen with a nitro crane in the phenolic ring of the tyrosine residues (El-Agamy et al. 2018; Ruiz-Ojeda et al. 2018; Yokoyama et al. 2017). On the other hand, malondialdehyde (MDA) and 4-hydroxynonenal (HNE) levels are indexes of lipid peroxidation, consequently of oxidative stress (Czerska et al. 2015; El-Agamy et al. 2018; Yokoyama et al. 2017). Indirectly, the evaluation of thiobarbituric acid reacting

substances (TBARS) is also used as a marker of lipid peroxidation (Cardoso et al. 2020).

To investigate if DOX causes oxidative stress in the brain, Joshi et al. divided male mice (2–3 months old) into 4 groups. Two groups received saline and 2 other groups received 150 mg/kg of γ -glutamyl cysteine ethyl ester (GCEE). One mouse from each of the previous group received a single dose of DOX (20 mg/kg, i.p.) 4 h after saline or GCEE injection. GCEE is a precursor in the synthesis of GSH, which is an important non-protein thiol antioxidant. Seventy-two hours after DOX administration, the brain levels of GSH, protein carbonyls, 3-NT and HNE were determined. The mice that only received DOX had significantly lower levels of GSH when compared to the saline controls or the mice that only received GCEE. The administration of DOX per se caused the elevation of protein carbonyls, 3-NT and HNE when compared with the saline control and GCEE-administrated mice. Importantly, GCEE partially reverted the effects of DOX on the brain levels of GSH, of protein carbonyl, 3-NT and protein-bound HNE (Joshi et al. 2007). Cardoso et al. also investigated whether DOX administration could affect the brain oxidation status of male Wistar rats (3 months of age). They injected DOX (2.5 mg/kg/week, i.p., 4 weeks) or saline. DOX injected rats presented increased levels of TBARS, and glutathione reductase (GR) activity, but decreased oxidized glutathione (GSSG) and ferritin levels in the brain homogenate. No differences were found related to catalase (CAT), superoxide dismutase, glutathione peroxidase (GPx) activities, GSH, or iron contents (Cardoso et al. 2020). On another study, Tangpong et al. injected a single dose of DOX (20 mg/kg i.p.) into 8-week-old male B6C3 mice, and the results showed elevation of protein carbonyls, 3-NT, and HNE levels in the brain (Tangpong et al. 2011). Kwatra et al. showed increased MDA levels, decreased levels of GSH, SOD, and CAT activities in the hippocampus of male Wistar rats injected with DOX (15 mg/kg, i.p.) (Kwatra et al. 2016). A study by Aluisse et al. upon the administration of a single injection of DOX (20 mg/kg, i.p.) in B6C3 mice (2–3 months of age) showed a significant increase in plasma levels of protein carbonyls, 3-NT, and HNE, when compared with mice treated with saline alone (Aluisse et al. 2009). Therefore, the authors found that DOX induces oxidative stress on the brain. These claims were also supported by the study of Keeney et al. that have searched for brain and plasma biomarkers of oxidative stress in male mice (2–3 months old). The mice were injected with a single dose of DOX (25 mg/kg, i.p.) and upon 72 h, DOX-treated mice had increased protein–carbonyl and HNE levels in the brain and plasma (Keeney et al. 2018).

More studies have been done to demonstrate the increase in oxidative stress in the brain, some of which are described in Table 4.

Table 4 Summary of studies that demonstrated oxidative stress induced by anticancer drugs in laboratory

Drug	Experimental setting	Animal	Major findings	References
Doxorubicin	2 mg/kg, i.p. injected once a week for 4 weeks. Animals were tested 28 days after treatment	Six-week-old male Sprague–Dawley rats	Increase in malondialdehyde (MDA) levels and decrease in GSH dipoles in the hippocampus	Tong et al. (2020)
	10 injections (2.5 mg/kg, i.p.) once in 5 days over a period of 50 days. Animals were tested 11 days after treatment	Female Wistar rats (10–12 weeks of age)	The decrease in catalase (CAT), superoxide dismutase (SOD), GSH, and total thiols levels in the frontal cortex and hippocampal regions	Nandakumar et al. (2018)
	2 mg/kg, i.p. injected weekly for 3 weeks. Animals were tested 1 week after treatment	Male albino rats	Non-adverse effects in locomotive function, but pyknosis and degeneration were observed in most of the neuronal cells in the hippocampus	El-Agamy et al. (2018)
Combination of Cyclophosphamide and doxorubicin	2 mg/kg, i.p. injected weekly for 4 weeks. Animals were tested 1 week after treatment	Male albino rats	Decrease in GSH levels by 65% and in CAT activity by 72%, increase in MDA levels by 53% in the hippocampus	Wahdan et al. (2020)
	injection of the combination of CYA (40 mg/kg, i.v.) and DOX (4 mg/kg, i.v.) weekly for 3 weeks. Animals were tested 1 week after treatment	Ovariectomized female Sprague–Dawley rats (8–10 weeks of age)	Decrease in MnSOD levels by 80% in the hippocampal homogenate	Bagnall-Moreau et al. (2019)
			Increase in 8-hydroxy 2 deoxyguanosine (8-oxodG) immunoreactivity and GPx1 levels in the hippocampus	

DOX Doxorubicin, *CYA* cyclophosphamide, *3-NT* 3-nitrotyrosine, *HNE* 4-hydroxynonenal, *GST* glutathione-S-transferase, *GSH* reduced glutathione, *MDA* malondialdehyde, *CAT* catalase, *SOD* superoxide dismutase, *MnSOD* manganese superoxide dismutase, 8-oxodG 8-hydroxy 2 deoxyguanosine, *GPx1* glutathione peroxidase 1

In this sense, studies have emerged with the aim of linking the increase in oxidative stress to other events, which would induce chemobrain as a result. Ren et al. dedicated to test the hypothesis that tumor necrosis factor alpha (TNF- α) may be related to oxidative stress, hippocampal neurochemical profiles and mitochondrial function in wild type (WT), but not in TNF- α null (TNFKO) mice following DOX administration. A dose of DOX (25 mg/kg, i.p.) was administered and caused an increase in oxidative stress markers in the brain and plasma, as seen by an increase in brain protein carbonyl and HNE levels of WT mice, but not in the brain of TNFKO mice. According to these results, it is possible to conclude that TNF- α may be a key player in the elevation of DOX-induced oxidative stress in the brain (Ren et al. 2019b).

Although oxidative stress has been the major link for cognitive dysfunction induced by chemotherapy, these previous studies also show that oxidative stress may be linked to other phenomena, namely neuro-inflammation, which will be discussed next.

Pro-inflammatory cytokines and neuroinflammation

Pro-inflammatory cytokines are produced mostly by activated macrophages, but can also be produced by monocytes or by nonimmune cells such as fibroblasts and endothelial cells (Zhang and An 2007) and prompt the organism inflammatory response in interlinked complex pathways. In fact, pro-inflammatory cytokines control the induction, production, stimulation, and inhibition of other cytokines. Some pro-inflammatory cytokines are IL-1 β , IL-6 and TNF- α (Khan et al. 2018). These molecules can cross easily the BBB by simple diffusion (e.g. TNF- α), saturable influx transport or retrograde axonal transport system (e.g. IL-1 α , IL-6 and TNF- α) or gain access to the brain through the circumventricular organs, where the barrier is more permeable (Yarlagadda et al. 2009). In the brain, the pro-inflammatory cytokines can activate microglia that will further exacerbate the production of pro-inflammatory cytokines *in situ*. The microglia functions as brain macrophages, a primary source of pro-inflammatory cytokines and pivotal in the mediation of neuro-inflammation (Colonna and Butovsky 2017). Nevertheless, to some extent, astrocytes and neurons can also produce some pro-inflammatory cytokines (Harry and Kraft 2008). Moreover, even without entering the brain, the pro-inflammatory cytokines can damage the BBB by destroying the tight junctions of microvascular endothelial cells of the BBB and increasing its permeability (Yarlagadda et al. 2009).

Cytokines also have significant impact on brain function. Interferon- α can increase the levels of IL-1 α , IL-6 and

TNF- α , and high levels of these pro-inflammatory cytokines are associated with impairments of memory, fine motor skills, executive functions and mood that indicate frontal-subcortical defects (Meyers et al. 2005). In addition, microglia activation and release of pro-inflammatory cytokines have been linked to several neurodegenerative diseases (Smith et al. 2012). Considering this, the increase in pro-inflammatory cytokines due to chemotherapeutic agents has been pointed as a possible mechanism for the cognitive changes seen in chemotherapy-treated patients (Tables 1 and 2). In a study of 35 breast cancer patients, with an average of 4 years after the end of chemotherapy (DOX, CYA and paclitaxel), with age- and educational-matched healthy controls, the serum levels of pro-inflammatory cytokines and hippocampal volume were measured, and cognitive tests were performed. The results revealed that chemotherapy-treated patients had significantly higher serum levels of IL-6 and TNF- α when compared with the control group. Moreover, in the chemotherapy group, higher levels of TNF- α were associated with reduced left hippocampal volume and the levels of IL-6 and TNF- α correlated with reduced performance on verbal memory (Kesler et al. 2013a, b). Pomykala and co-workers also sought to evaluate the relationship between chemotherapy-induced cognitive dysfunction and pro-inflammatory cytokines. In breast cancer patients, the levels of serum pro-inflammatory cytokines were measured, and PET scans were performed to evaluate regional cerebral metabolism. The evaluations were performed immediately after treatment and one-year later. In the chemotherapy group, positive correlation between left medial frontal and right inferior lateral anterior temporal cortices metabolism and the pro-inflammatory cytokines IL-1 α , IL-6 and TNF- α and the acute-phase protein C-reactive protein (whose synthesis is induced by IL-6 and TNF- α) were seen. No correlations were observed in the patients that did not receive chemotherapy. At the 1-year assessment, the correlations between cytokines and metabolism continued in the medial frontal cortex and temporal cortex in the chemotherapy group. The placed hypothesis states that the initial inflammatory response causes a downstream of events that has long-term impact on brain metabolism (Pomykala et al. 2013).

Following the reasoning of the study by Ren et al. mentioned earlier, several studies demonstrated changes in inflammatory processes that may be involved in the mechanism of cognitive dysfunction caused by anticancer drugs (Ren et al. 2019b). Aluise et al. investigated the effects of DOX on plasma markers of oxidative stress and inflammation from patients undergoing treatment for various malignancies and identified a possible mechanism by which oxidative stress may be the link to the elevation of TNF- α in the periphery. The initial part of the study was done on patients who received DOX as a part of their prescribed

multiagent chemotherapy regimen and had vascular access devices for blood sample acquisition. Blood samples were obtained immediately before and 6 h after DOX administration. The results regarding inflammatory processes demonstrated a significant increase in TNF- α in plasma after DOX administration, while there were no significant differences in the levels of other cytokines (Aluise et al. 2011). Regarding oxidative stress, Aluise et al. observed an increase in protein carbonyls, and the abundant plasma protein apolipoprotein A1 (Apo-1) was oxidized in plasma after DOX exposure. In the other part of the study, they administered a single dose of DOX (25 mg/kg) into male B6C3 mice (2–3 months of age), from which the highly abundant plasma protein, Apo-1, was oxidized also in plasma. Additionally, there was an increase in peripheral levels of TNF- α , which may migrate across the BBB and cause oxidative stress and inflammation in the brain parenchyma (Aluise et al. 2011).

In a study by El-Agamy et al. mentioned already, the generation of a robust inflammatory response was reported, with increased levels of TNF- α by 110%, prostaglandin E2 (PGE2) by 211% and COX-2 by 85%, in brain homogenates of rats after intraperitoneal injection of DOX, when compared to the control group, emulating heightened neuro-inflammatory response (El-Agamy et al. 2018). Tangpong and colleagues also investigated the role of the pro-inflammatory cytokine TNF- α in DOX-induced neurotoxicity. TNF- α levels, both circulating and brain levels were significantly higher in the drug-treated group of mice when compared to the saline control group. In addition, in DOX-treated mice, TNF- α was especially found in the cortex and hippocampus when compared with saline controls (Tangpong et al. 2006).

Regarding other chemotherapeutic agents, female rats administrated with a combination of CYA, MTX and 5-FU, presented significant cognitive impairment, with increase in brain pro-inflammatory mediators IL-1, TNF- α and COX-2, a decrease in the anti-inflammatory IL-10 and decreased myelination and thickness of axons. However, when COX-2 was inhibited to block the inflammatory response, there were significant improvements in cognition and integrity and structure of axons (Briones and Woods 2014). Other studies focusing on inflammatory responses after treatment are summarized in Table 5.

Thus, inflammatory cytokines seem to play a role on chemobrain, indicating that inflammation may be part of the mechanism that induces cognitive dysfunction from chemotherapeutic agents. The first key event may be ROS, produced by anticancer drugs as previously described, which can trigger neurotoxicity as it can indirectly elicit cytokine production, such as TNF- α (Barkoukis 2012; El-Agamy et al. 2019; Wang et al. 2015). In fact, the oxidation of Apo-1 by anticancer drugs triggers an inflammatory process (Ren et al. 2017). Apo-1 is part of the

high-density lipoprotein (HDL) complex for transporting cholesterol and phospholipids to the liver for degradation. Among its function, Apo-1 is also responsible for regulation the inflammatory response. This regulation is done by the interaction of Apo-1 with the cholesterol transport protein, ATP-binding cassette transporter (ABCA1), which consequently activates the Janus kinase 2 (JAK2)/signal transducer and activator of transcription 3 (STAT3) pathway, a signaling cascade that, in this case, signals for the decrease of the release of pro-inflammatory cytokine from the macrophage. Additionally, Apo-1 also inhibits the secretion of pro-inflammatory cytokines from stimulated T cells and monocytes. Therefore, its action prevents the overproduction of inflammatory cytokines, such as interleukin 1 beta (IL-1 β) and TNF- α (Aluise et al. 2011; El-Agamy et al. 2018; Keeney et al. 2018; Ren et al. 2019a). However, when Apo-1 is oxidized by anticancer drugs, it leads to an increase in pro-inflammatory cytokines in the periphery because its inhibitory action on TNF- α production is not possible (El-Agamy et al. 2019, 2018; Ren et al. 2019a). As Aluise et al. described, proteomics analysis revealed a high abundance of oxidized Apo-1 in mice plasma after DOX administration (Aluise et al. 2011). Therefore, oxidative modification of Apo-1 inhibits its anti-inflammatory activity, giving rise to an exacerbate release of cytokines (El-Agamy et al. 2019; Ren et al. 2019a). Peripheral pro-inflammatory cytokines can cross the BBB, either by passive diffusion through leaky regions or by receptor-mediated endocytosis, resulting in an uninterrupted feedback loop of inflammation between the periphery and the CNS (Ren et al. 2019a).

Once in the brain, pro-inflammatory cytokines can act directly on the CNS and activate and induce microglia and astrocytes to secrete even more cytokines (El-Agamy et al. 2018; Ren et al. 2017). Together, all these cytokines, namely TNF- α , trigger an immune response and induce the increase of oxidative stress and mitochondrial damage (Nguyen and Ehrlich 2020; Ren et al. 2019a). Then, pro-inflammatory cytokines also activate NF- κ B, which is the redox-sensitive transcription factor capable of coordinating the regulation of immune and inflammatory responses, as well as cell development, proliferation and survival, and it is involved in the regulation of cytokine production (Barkoukis 2012; Keeney et al. 2018; Kwatra et al. 2016; Ren et al. 2019a; Ren et al. 2017; Trachootham et al. 2008). TNF- α activates NF- κ B, that, in turn, activates the transcription of multiple target genes, including inflammatory cytokines, increasing the expression of IL-1 β and TNF- α (Barkoukis 2012; Trachootham et al. 2008). The resulting pro-inflammatory state induces the expression of iNOS. Then, the high expression of iNOS results in excess •NO, and, therefore, there is an increase in nitrosative stress in the brain (El-Agamy et al. 2018). The oxidative and nitrosative stress can affect the

pro-inflammatory

Table 5 Summary of studies focused on inflammatory processes induced by anticancer drugs in laboratory animals

Drug	Experimental setting	Animals	Major findings	Study
Doxorubicin	Single dose (20 mg/kg, i.p.). Animals were tested during and after treatment	Eight-week-old male B6C3 mice	Increase in levels of TNF- α in peripheral circulation and brain homogenates	Tangpong et al. (2006)
	Single dose (20 mg/kg, i.p.). Animals were tested 72 h after treatment	Eight-weeks-old male B6C3 mice	Increase in levels of TNF- α in brain homogenates	Tangpong et al. (2011)
	4 injections (2 mg/kg, i.p.) weekly for 4 weeks. Animals were tested 1 week after treatment	Male albino rats	Increase in IL-17 and TNF- α by 290% and 590% respectively, a decrease of IL-10 level by 57% in the hippocampal homogenate	Wahdan et al. (2020)
	4 injections (2 mg/kg, i.p.) weekly for 4 weeks. Animals were tested 4 weeks after treatment	Six-month-old male wild type mice (C57BL/6 J, Jackson)	Increased IL-1 β , IL-3, IL-5, IL-12 levels in hippocampal tissues	Allen et al. (2019)
	4 injections (2 mg/kg, i.p.) weekly for 4 weeks. Animals were tested 28 days after treatment	Six-week-old male Sprague–Dawley rats	Increased TNF- α , PGE2, and COX-2 levels in the hippocampus	Tong et al. (2020)
Combination of cyclophosphamide and doxorubicin	Single dose of DOX (15 mg/kg, i.p.), on the 10 th day. Animals were tested 30 min after treatment	Male Wistar rats	Increased IL-1 β and TNF- α in the hippocampus	Kwatra et al. (2016)
	3 injections of the combination of CYA (40 mg/kg, i.v.) and DOX (4 mg/kg, i.v.) weekly for 3 weeks. Animals were tested 1 week after treatment	Ovariectomized female Sprague–Dawley rats (8–10 weeks old)	Increased nuclear NF- κ B levels (p65 subunit), and TNF- α , IL-2, IL-3, IL-4, IL-6, IL-10 in the hippocampal lysates	Bagnall-Moreau et al. (2019)

DOX Doxorubicin, CYA cyclophosphamide, TNF- α tumor necrosis factor alpha, IL-17 interleukin 17, IL-10 interleukin 10, IL-1 β interleukin 1 beta, IL-3 interleukin 3, IL-5 interleukin 5, IL-12 interleukin 12, NF- κ B nuclear factor kappa B, IL-2 interleukin 2, IL-4 interleukin 4, IL-6 interleukin 6, PGE2 prostaglandin E2, COX-2 cyclooxygenase-2

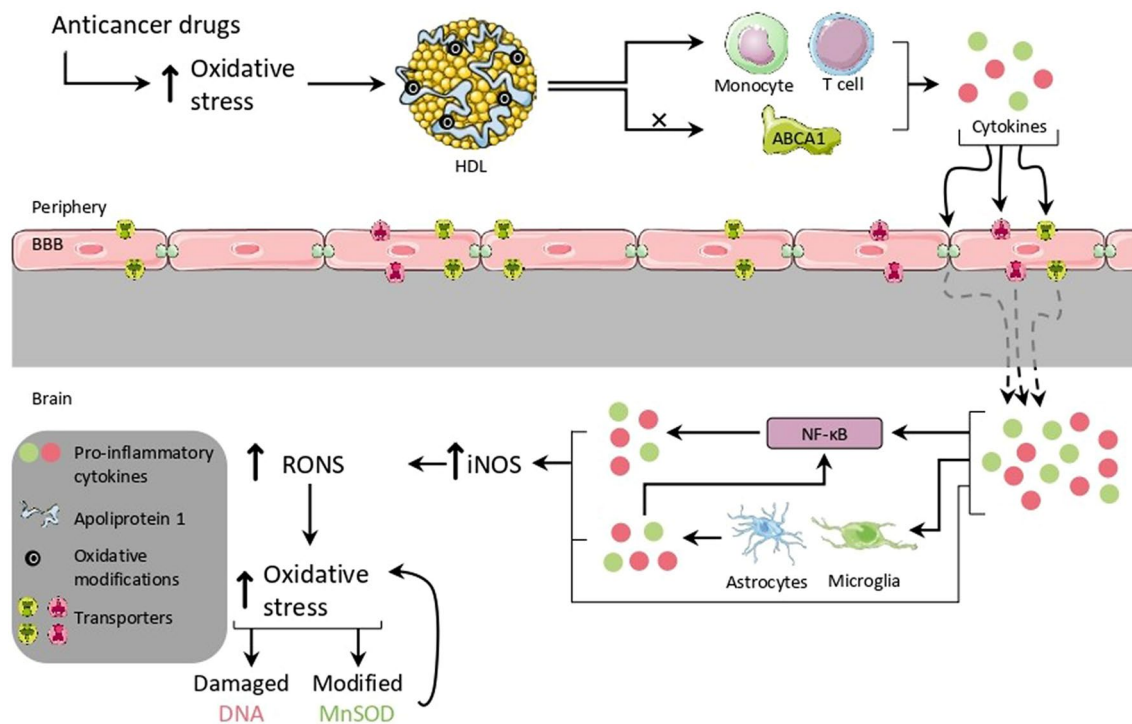


Fig. 4 Schematic representation of the proposed mechanism of chemotherapy-induced cognitive dysfunction mediated by oxidative stress generated from anticancer drugs and the associated inflammatory process. Anti-cancer drugs in the periphery trigger oxidative stress that results in protein oxidation and lipid peroxidation, namely in apolipoprotein 1 (Apo-1) oxidation. That oxidation leads to high production of pro-inflammatory cytokines, namely tumor necrosis factor alpha (TNF- α). Pro-inflammatory cytokines can cross the BBB by passive diffusion through leaky regions or by receptor-mediated endocytosis. Once in the brain, pro-inflammatory cytokines trigger an immune

response, promoting the production of more pro-inflammatory cytokines (represented by pink and green balls) by astrocytes, microglia cells, and nuclear factor kappa B (NF- κ B), in the hippocampus. Consequently, there is an increased expression of inducible nitric oxide synthase (iNOS) which triggers the increase of reactive oxygen species/reactive nitrogen species (RONS) and oxidative stress in the brain. This causes damage to deoxyribonucleic acid (DNA) and damage and modification of several proteins, namely manganese superoxide dismutase (MnSOD), which is unable to protect neuronal cells from oxidative stress, promoting their continuous increase

DNA repair systems causing its damage and consequently cell death (Bagnall-Moreau et al. 2019; Ren et al. 2017; Tacar et al. 2013). At the same time, excess $\bullet$ NO causes modification and damage to MnSOD in the hippocampus, rendering neurons unable to protect themselves from overproduced ROS (Ren et al. 2019a). The proposed mechanism for the inflammatory process involved in chemobrain is summarized in Fig. 4.

Cell death

The CNS is primarily formed by neurons and glia that can undergo cell death after noxious stimuli. Apoptosis, or programmed cell death, can be triggered by an external signal that acts on a plasma membrane receptor, or internal events such as DNA damage, infection, oxidative stress, among others (Nelson and Cox 2013) or via intrinsic pathways involving the mitochondria. Some studies, described below, report cell death in the brain, associated with increase of apoptotic

and autophagic markers after anticancer drug exposure. Ren et al. in a study previously mentioned demonstrated a decline in mitochondrial respiration, after DOX administration in mice (Ren et al. 2019b). Kwatra et al. verified in animals treated with DOX, significant impairment of the mitochondrial complexes I, II, IV and mitochondrial redox activity, in comparison with the control group (Kwatra et al. 2016). Tangpong et al. used pyruvate plus malate and succinate as the substrates to assess the effect of DOX on brain mitochondrial function and metabolism. The results revealed a decrease in respiration ratios after treatment with DOX, and the level of succinate dehydrogenase was not altered. Furthermore, mitochondrial dysfunction has also been observed after treatment with cisplatin (Chiu et al. 2017; Lomeli et al. 2017; Wang et al. 2019). These changes in mitochondrial function can trigger apoptosis. Pro-apoptotic protein Bax, and an anti-apoptotic protein, B-cell lymphoma-extra-large (Bcl-xL), have been shown to be increased in mitochondria after DOX (Tangpong et al. 2006). These results are in line with the increase in TNF- α , seen previously since

this cytokine causes a decline in mitochondrial respiration and subsequent increase in oxidative stress markers. Accordingly, oxidative stress causes mitochondrial dysfunction and initiates the mitochondrial apoptosis pathway by the translocation of pro-apoptotic proteins, p53 and Bax, to mitochondria. Finally increases in the release of cytochrome c, as well as its concentration in cytosol, and also an increase in caspase 3 activity have been shown in the brains of mice treated with DOX (Tangpong et al. 2006). Tong et al. in a study mentioned before, detected in the hippocampus of rats, after DOX administration, an increase in the levels of cleaved caspase 3 and cleaved caspase 9 (Tong et al. 2020). A study by El-Agamy et al. also found an increase in the expression of hippocampal cytochrome c by 57% and the activity of caspase 3 by 642%, after DOX administration (El-Agamy et al. 2018). Tangpong et al. detected an increase in the levels of p53, Bax, Bcl-xL and also caspase 3 activity in DOX-treated mice (Tangpong et al. 2011). Wahdan et al., in a study previously mentioned, determined that the administration of DOX triggered hippocampal apoptotic activation, which caused the increase in the expression of the active form of the apoptotic marker, caspase 3, by 363% (Wahdan et al. 2020).

Taking all these data into account, it is plausible to hypothesize that DOX causes cognitive dysfunction and triggers cell death in the brain. Mitochondrial dysfunction can be responsible for mitochondrial apoptosis pathway activation (Tangpong et al. 2006). Furthermore, oxidative stress and/or inflammation cause the activation of pro-apoptotic proteins, p53 and Bax (Ren et al. 2017; Tangpong et al. 2006). In turn, there is an increase in the susceptibility of mitochondria to the opening of the calcium-mediated permeability transition pores, leading to mitochondrial swelling, membrane degradation, and release of apoptotic proteins, which decrease the regular mitochondrial function followed by the formation of mitochondrial permeability transition pore. This opening leads to the release of cytochrome c and the subsequent activation of caspase signaling with caspase 3 cleavage (El-Agamy et al. 2018; Ren et al. 2019a; Tangpong et al. 2006). Ultimately, these series of reactions are responsible for apoptotic death of brain cells and may be responsible, at least partially, for the structural and functional alteration of several cognition-related brain areas, causing cognitive deficits of chemotherapy-treated cancer survivors (Ren et al. 2019a; Ren et al. 2017). The proposed mechanisms related to cell death are summarized in Fig. 5.

Autophagy is a process of continuous renewal and defense of cells against environmental/external aggressions and allows adaptation to unfavorable environments. It is defined as catabolic lysosomal degradation of cellular proteins and other components and has a protective role in neuronal cells (Glick et al. 2010; Nelson and Cox 2013). Several studies, reported below, have also shown alterations in autophagy markers after

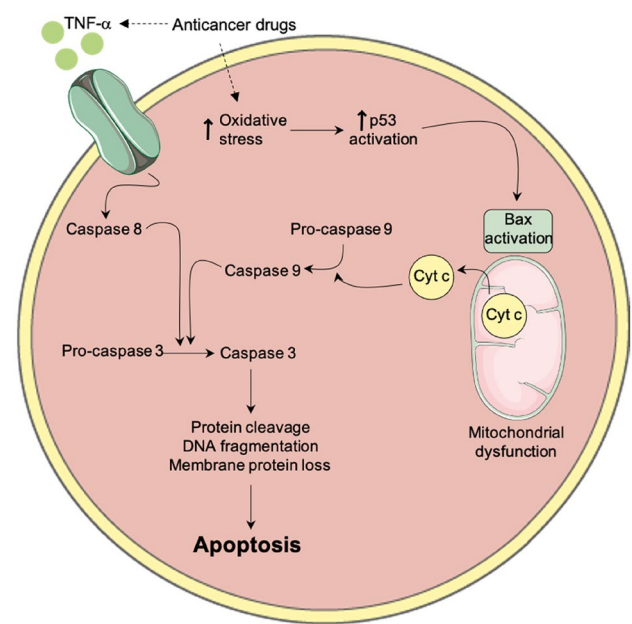


Fig. 5 Schematic representation of the possible intracellular mechanisms of brain cells death involved in chemotherapy-induced cognitive dysfunction. Apoptosis can be induced by an intrinsic or an extrinsic route. In this case, the extrinsic pathway starts with tumor necrosis factor alpha (TNF- α), whose formation is indirectly induced by anti-cancer drugs, which binds to its receptor and transduces the signal into the activation of caspase 8, leading to initiation of the extrinsic pathway. Likewise, oxidative stress generated indirectly in the brain by anticancer drugs induces the activation of p53 and Bax, which causes increased susceptibility to mitochondria to the opening of the calcium-mediated permeability transition pores, leading to mitochondrial swelling, membrane degradation, and release of apoptotic proteins, which lead to mitochondrial dysfunction. Cytochrome c (Cyt c) is released from mitochondria to the cytosol, followed by activation of caspase 9 through the formation of apoptosome. Both caspase 9 (intrinsic pathway) and caspase 8 (extrinsic pathway) can cleave and activate caspase 3, which cleaves its protein substrates, resulting in apoptosis

DOX administration. Wahdan et al., in a study mentioned before, observed the effects of DOX treatment on hippocampal autophagy status by measuring the protein expression levels of p62, beclin1, light chain 3 I (LC3-I), light chain 3 II (LC3-II), total protein kinase B (t-AKT) and phosphorylated AKT (p-AKT) among the treated groups. In DOX-treated animals, there was a significant increase in autophagic dysfunction effect, due to the increase in p62 levels and a reduction of beclin1 levels, and also a decrease in LC3-I/LC3-II and p-AKT/t-AKT ratios (Wahdan et al. 2020). Manchon et al. aimed to determine whether DOX induced impairment on primary neurons obtained from the cortex of rat embryos. Their study showed that the levels of LC3-II were lower in neurons incubated only with DOX than in neurons incubated with the combination of DOX and ammonium chloride (NH₄Cl). Since NH₄Cl can neutralize the lysosomal pH and inhibit the last step of autophagy-mediated degradation, these results mean that

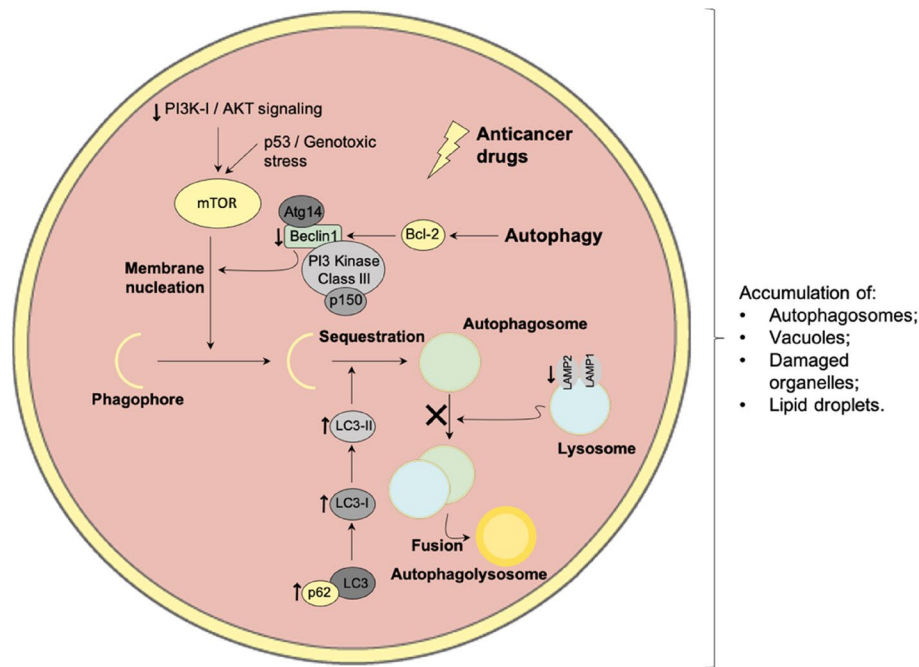


Fig. 6 Schematic representation of the autophagy process and the altered autophagy markers reported by different studies after administration of anticancer drugs in neuronal cells. The changes reported in the studies include a decrease in the p-protein kinase B (AKT) / t-AKT ratio [phosphoinositide 3-kinase I (PI3K-I) / AKT signaling], which will consequently decrease the phosphorylation of the mTOR (mTOR) protein. In addition, there was a decrease in beclin1 [which forms a complex with p150, PI3K-III and autophagy-related pro-

tein 14 (Atg14)], light chain 3 I (LC3-I), and light chain 3 II (LC3-II), represents defective autophagosome formation. The observed increase and accumulation of p62 prevent the autophagy's damage and promotes apoptosis. Additionally, a decrease in the expression of lysosome-associated membrane protein 2 (LAMP2) is also described, suggestive of the decrease in the total number of lysosomes and impaired degradative capacity. Thus, the fusion of autophagosome and lysosome does not occur, causing the accumulation of the former

autophagy would be upregulated. Then, the authors evaluated beclin1-GFP, a marker of autophagic morphology since beclin1 is part of the pre-autophagosomal complex. As expected, they obtained the formation of beclin-GFP-positive puncta, like what neuronal autophagy inducers do. From the analysis of p62, a selective autophagy substrate that binds to LC3, its high protein levels were observed in neurons exposed to DOX, indicating impeding autophagy. Additionally, it was also found that DOX caused the accumulation of vacuoles, damaged organelles, autophagosomes and lipid droplets, proving that this anticancer drug may impair the degradative autophagic pathway (Moruno-Manchon et al. 2016). Finally, Zhou et al. set to explore the autophagic response of the hippocampus towards DOX treatment in the male Sprague–Dawley rats. Their results showed impaired autophagic flow and dysfunctional lysosomes in the brain after treatment with DOX (2.5 mg/kg i.p. injected every day for a total of seven injections), by increasing the abundance of autophagosomes, increasing the expression of LC3-II and p62, probably derived from the accumulation of the autophagosome and preventing the elimination of autophagosome. Additionally, they also verified the lysosomal impairment induced by DOX, by the suppression of lysosome-associated membrane protein 2 (LAMP2) expression

and inhibition of cathepsin D (CTSD) expression and activity, suggestive of the decrease in the total number of lysosomes and impaired degradative capacity (Zhou et al. 2018).

From the results of the reported studies, it was found that DOX induces dysfunctional and inefficient autophagy, which causes the accumulation of autophagosomes, vacuoles, damaged organelles and lipid droplets. This accumulation is related to the neurodegeneration seen in many cognitive diseases and can be partially responsible for the chemotherapy-induced cognitive dysfunction (Glick et al. 2010; Moruno-Manchon et al. 2016). The proposed mechanisms are summarized in Fig. 6. Other drugs are in urgent need of attention regarding this intracellular pathway.

Neurotransmitters

Neurotransmitter dysregulation, often a reduction in their availability, is observed in most neurological disorders. However, neurotransmitter alterations induced by anticancer drugs remain poorly known (El-Agamy et al. 2019; Purves 2004). Some studies, presented next, have described changes in neurotransmitters due to the administration of anticancer drugs.

First, Thomas et al. studied the hypothesis that a dose of DOX (single i.p. of 25 mg/kg DOX) in male B6C3F1 J mice caused glutamate dynamics alteration in the brain. It was found that DOX-induced slower clearance of glutamate, which may result from changes in glutamate transporter trafficking, transport capacity, transporter affinity, transcription and translation and abnormal transporter post-translational modifications (Thomas et al. 2017). Interestingly, in peripheral blood mononuclear cells isolated from lung cancer patients displaying cognitive dysfunction and previously treated with platinum-based compounds, a significantly altered glutamate metabolism was seen and correlated with the cognitive deficits observed (Michalak et al. 2016).

read In a study by Keeney et al., it was demonstrated that the administration of DOX caused a decrease in the compounds containing choline, as evaluated by the choline-containing compounds/creatine ratios in the hippocampus in rats, probably related to myelin damage and which also indicates decreased cell density and necrosis. Additionally, the activities of phosphatidylcholine-specific phospholipase C and phospholipase D were severely decreased after DOX administration, consequently leading to a decrease in acetylcholine availability (Keeney et al. 2018). The authors also found that treatment with DOX induced a significant reduction in the levels of the monoamines dopamine, noradrenaline and 5-hydroxytryptamine (serotonin) in the brain, compared to the control group (Kwatra et al. 2016). The changes in all these monoamines can interact with the results of oxidative processes, already verified, during DOX-induced neurotoxicity (Kwatra et al. 2016). On the other hand, administration of 2 injections of 5-FU (25 mg/kg i.v.) in rats over the course of 2 weeks revealed that 5-FU caused significant impairment in dopamine release when compared with saline controls (Jarmolowicz et al. 2019).

Despite the results found, the changes in the levels of neurotransmitters may or may not be related to the induction of cognitive dysfunction or to other related and already described pathways. The disturbances in neurotransmitters can be due to several factors, such as changes in their production, release, reception, breakage and reuptake, or even in the number and affinity of their receptors. Thus, it is possible to conclude that anticancer drugs can affect neurotransmitters (Maiese 2019). However, further studies are needed to clarify both the triggering events for this, but also the consequences of those changes.

Neurogenesis

Adult neurogenesis is the process that leads to the formation of new neurons from stem cells in specific adult brain areas. Two brain areas have been focused, namely the subventricular zone and the subgranular zone of the dentate gyrus of the hippocampal formation, where new neurons are formed from

a neuronal precursor, a neuroblast, in the adult (Garthe and Kempermann 2013; Purves 2004). The hippocampal formation is one of the most important structure involved in memory formation and spatial processing, and adult-born neuron integration into the circuitry has a pivotal role in hippocampal neurogenesis and, hence, proper maintenance of cognitive functions (El-Agamy et al. 2019). These facts make post-natal neurogenesis disruption a possible mechanism for explaining cognitive deficits induced by anticancer drugs (Janelins et al. 2010).

Christie et al. set out to address the alterations in cognition induced by exposure to CYA or DOX. For this, they evaluated the effects of chronic treatment with CYA or DOX in the absence of disease, by assessing cognitive changes using paradigms known to engage the hippocampus. They also evaluated the effects of CYA and DOX on hippocampal neurogenesis. In their results, they report that rats treated with CYA (50 mg/kg/week) and DOX (2 mg/kg/week) for 4 weeks, showed significantly impaired performance in the new task of place recognition and the task of conditioning contextual fear compared to untreated controls, suggesting memory function changes in the hippocampus. Furthermore, chemotherapy-treated animals showed a significant decline in neurogenesis (Christie et al. 2012). Moreover, Janelins et al. tested the hypothesis that anticancer drugs that easily cross the BBB (namely 5-FU or CYA), compared to those that do not (DOX and paclitaxel), are more likely to affect neurogenesis. To do so, the authors investigated the effect of the four drugs frequently used in chemotherapy on neural mitotic activity in the subgranular zone of the dentate gyrus in mice. The results showed a reduction in BrdU positive cells following 5-FU (60 mg/kg) and a reduction following CYA (50 mg/kg). Similarly, non-BBB crossing agents DOX (5 mg/kg) and paclitaxel (5 mg/kg) led to a reduction in newly divided cells (Janelins et al. 2010). On the other hand, the study by Seigers et al. aimed to explore the effects of different cytostatic agents CYA (i.p. injection of 150 mg/kg), docetaxel (33 mg/kg i.v.), DOX (5 or 10 mg/kg i.v.), 5-FU (75 mg/kg i.p.) and MTX (250 or 500 mg/kg i.v.) on the brain in male C57Bl/6j mice. For that, they analyzed the effect of these agents on neurogenesis with markers for cell proliferation (Ki-67) and cell survival (doublecortin), but they did not observe any significant changes in neurogenesis for any cytotoxic agent. In addition, the treatment with different cytotoxic agents did not affect the maturation of the newly formed cells in adult animals (Seigers et al. 2016). Nevertheless, other studies found that CYA decreased hippocampal cell proliferation (Lyons et al. 2011), 5-FU decreased hippocampal plasticity and cell survival (Mustafa et al. 2008) and also 5-FU reduced cell proliferation and survival in the sub granular zone of the dentate gyrus (Lyons et al. 2012).

Decreased neurogenesis after chemotherapy may involve broader implications (Janelins et al. 2010). The presented studies have opposite results, depending on the dose or drug used. In addition, post-natal neurogenesis can be affected by anxiety, sleep disorders and depression, which are common symptoms in cancer patients, making future studies needed to understand the neurobiological effects of anticancer drugs in clinically relevant doses in post-natal neurogenesis (Janelins et al. 2010).

Other factors

Many researchers have also focused their attention on the screening for potential genetic risk factors that could be important to identify patients more susceptible to chemotherapy-induced cognitive dysfunction and allow a closer monitoring. Apolipoprotein E (ApoE) plays a role not only in plasmatic protein involved in lipid and lipoprotein metabolism, but also in stress-response pending on the isoform involved. The allele ϵA is associated with an increased risk for Alzheimer's disease and mild cognitive impairment (Dose et al. 2016). Ahles and colleagues compared the cognitive outcome of cancer survivors (on average 4 years after chemotherapy) who carried the ϵA allele with those absent for at least one ϵA allele. The patients that underwent chemotherapy, and that had both ϵA alleles, presented significantly impaired visual memory and spatial ability and tended to score lower in executive functioning when compared to patients who did not carried the ϵA allele. Therefore, patients homozygous for the ϵA alleles could be at higher risk for cognitive dysfunction after receiving chemotherapeutic treatment and this could be due to differential capacity to repair the microvascular and neuronal damage caused by chemotherapy (Ahles et al. 2003).

Catechol-*O*-methyltransferase (COMT) is an enzyme involved in the metabolism of catecholamines such as dopamine, adrenaline and noradrenaline. The activity of this enzyme is particularly important in the breakdown of dopamine in the prefrontal cortex, since in this brain area the dopamine transporter has low expression within synapses; thus, COMT is essential to control the dopamine flux and degradation (Akil et al. 2003). The COMT gene has two isoforms, the less active form being the result of a substitution of a valine by a methionine. The homozygous for the valine isoform has a higher rate of dopamine metabolism leading to less availability of this neurotransmitter. Small and co-workers performed cognitive and genotype testing in breast cancer survivors treated with chemotherapy or/and radiotherapy and compared their scores with healthy controls. The chemotherapy regimen consisted of an average of 5 anthracycline-based cycles. The patients with COMT homozygous for valine had significantly impaired attention,

verbal fluency and motor speed when comparing to patients with COMT methionine homozygous. Moreover, the chemotherapy-treated group COMT homozygous for valine performed worst on tests of attention than the COMT homozygous for valine healthy controls (Small et al. 2011), placing the hypothesis that cancer patients COMT homozygous for valine could be at higher risk for chemotherapy-induced cognitive impairment.

BDNF is a neurotrophic factor that is highly expressed in the brain, particularly in the prefrontal cortex and hippocampus, and it is essential for the survival and maintenance of mature neurons. Moreover, BDNF impairment has been associated with some neurodegenerative diseases (Zuccato and Cattaneo 2009). The single nucleotide substitution at nucleotide 196 results in an amino acid exchange (methionine to a valine) and the protein formed has decreased activity on BDNF secretion. In a prospective cohort study in breast cancer patients scheduled to receive anthracycline- or taxane-based chemotherapy, BDNF genotyping was performed. Furthermore, neuropsychological testing was done prior to treatment, 6 weeks after the start and at the end of treatment. The homozygous patients for methionine were associated with significantly lower chances of developing cognitive impairment. Moreover, the valine homozygotes were more likely to have impaired verbal fluency and multi-tasking ability when compared to the methionine homozygotes (Ng et al. 2016).

Epigenetic alterations have also been reported to occur due to chemotherapy exposure, with the most pronounced deregulations in miRNA associated with altered BDNF levels in the prefrontal cortex (Kovalchuk et al. 2017), suggesting another important contributor to chemotherapy-induced cognitive dysfunction.

Conclusions and future directions

The main goal of this review was to assess the increasing interest of chemobrain on the scientific and clinical community. Chemotherapy-induced cognitive impairment or chemobrain has been established as a long-term secondary effect of chemotherapy treatment that may persist for years following the treatment, which significantly diminishes the life quality of cancer survivors. The neuropsychological testing performed reported prevalence of cognitive impairment between 28% (Schagen et al. 2002) and 75% (Wieneke and Dienst 1995). The cognitive domains reported more often to be impaired are verbal memory, psychomotor function, visual memory, visuospatial and verbal learning, memory function and attention. Despite some methodological difficulties and confounding factors inherent to neuropsychological tests, the electrophysiological and neuroimaging studies provided more objective evidence of brain changes in

chemotherapy-treated patients. Chemotherapy causes functional alterations as seen by alterations in P300 value (Kreukels et al. 2005), reduction of metabolism in some brain areas (Silverman et al. 2007), altered brain networks (Miao et al. 2016) and also structural changes in both the white (Chen et al. 2019) and grey matter (Lepage et al. 2014).

Despite the growing number of studies in chemotherapy-treated patients, these have been mainly performed on female breast cancer survivors, and, therefore, data on men or other cancers in women are scarce. Moreover, the older cancer survivors that potentially are at higher risk for cognitive impairment have also been poorly studied. In retrospective, the anticancer treatments are widening and so is the survival rate rendering hard not only to determine both the neurotoxic potential of new drugs used in combination with classical chemotherapy, but also to see the impact of chemotherapy on brain in the coming decades following treatment.

Regarding the chemotherapeutic agents that might cause higher incidence of neurotoxic effects, conclusions are hard to take from the clinical studies, since the patients are frequently treated with combinations of chemotherapeutic agents and not with a single agent. Nevertheless, anthracyclines are the most studied drugs. To bridge some of the gaps from clinical studies, animal studies arose as a valid model to study the neurotoxicity of chemotherapeutic agents. The administration of commonly used chemotherapeutic agents also caused cognitive dysfunction in rodents. Furthermore, these studies revealed that most chemotherapeutic agents significantly decrease hippocampal post-natal neurogenesis and even modification of neurotransmitter levels or glial activation. In addition to these mechanisms, there are pre-existing susceptible factors such as genetic, hormonal changes due to chemotherapy and brain ageing that might increase the risk for chemotherapy-induced cognitive impairment. Nevertheless, regarding the underlying mechanisms of chemobrain, they are still poorly studied. Regardless, DOX has been the chemotherapeutic agent most studied in animal models, and oxidative stress and increased cytokine levels seem to be involved in the observed chemobrain, as stated above. In addition, carmustine, an alkylating agent used in several cancers, seems to activate similar routes, although in this case, it is also able to cross the BBB. Furthermore, MTX seems to influence hippocampal post-natal neurogenesis, causing depressive-like behaviors and memory deficits, accompanied with increased activity of iNOS and COX-2, while 5-FU also impacted on post-natal neurogenesis and cellular viability on hippocampal neurons. Cisplatin has also been seen to cause high impact on neuronal population on cortex and in hippocampus.

Therefore, oxidative stress and neuronal damage have been the most studied aspects *in vivo*, being that neuroinflammation can be the common link between several anticancer drugs. In fact, several clinical studies showed

increased cytokine levels, thus exposing the clinical relevance and translation of animal studies. Nevertheless, the anticancer therapies usually use combination regimens, and those are still under evaluation in animal models, and importantly the combination of several cytostatic agents might exacerbate the neurotoxic actions of a single drug.

Finally, translational research is essential to understand the complexity of chemobrain and find solutions that can ensure the post-treatment life quality of patients. Despite the great progress made in understanding the neurotoxicity of chemotherapy, there is still much work to be done. Moreover, there are no established guidelines for identifying cancer patients more susceptible to chemobrain to help and rehabilitate these patients. Future studies are needed to better understand predisposition factors and biomarkers of chemobrain, as well as underlying mechanisms, to allow a more precise guidance of the chemotherapy-treated patients.

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Declarations

Conflict of interest The authors declare no competing financial interest.

References

- Adams-Price CE, Morse LW, Cross GW, Williams M, Wells-Parker E (2009) The effects of chemotherapy on Useful Field of View (UFOV) in younger and older breast cancer patients. *Exp Aging Res* 35(2):220–234. <https://doi.org/10.1080/03610730902720497>
- Ahles TA, Saykin AJ (2007) Candidate mechanisms for chemotherapy-induced cognitive changes. *Nat Rev Cancer* 7(3):192. <https://doi.org/10.1038/nrc2073>
- Ahles TA, Saykin AJ, Furstenberg CT et al (2002) Neuropsychologic impact of standard-dose systemic chemotherapy in long-term survivors of breast cancer and lymphoma. *J Clin Oncol* 20(2):485–493. <https://doi.org/10.1200/jco.2002.20.2.485>
- Ahles TA, Saykin AJ, McDonald BC et al (2010) Longitudinal assessment of cognitive changes associated with adjuvant treatment for breast cancer: impact of age and cognitive reserve. *J Clin Oncol* 28(29):4434–4440. <https://doi.org/10.1200/jco.2009.27.0827>

- Ahles TA, Saykin AJ, Noll WW et al (2003) The relationship of APOE genotype to neuropsychological performance in long-term cancer survivors treated with standard dose chemotherapy. *Psychooncology* 12(6):612–619. <https://doi.org/10.1002/pon.742>
- Akil M, Kolachana BS, Rothmond DA, Hyde TM, Weinberger DR, Kleinman JE (2003) Catechol-O-methyltransferase genotype and dopamine regulation in the human brain. *J Neurosci* 23(6):2008–2013. <https://doi.org/10.1523/JNEUROSCI.23-06-02008.2003>
- Alexander TC, Kiffer F, Groves T et al (2018) Effects of thioTEPA chemotherapy on cognition and motor coordination. *Synapse*. <https://doi.org/10.1002/syn.22085>
- Alharbi I, Alharbi H, Almogbel Y, Alalwan A, Alhowail A (2020) Effect of Metformin on Doxorubicin-Induced Memory Dysfunction. *Brain Sci* 10(3). <https://doi.org/10.3390/brainsci10030152>
- Alhowail AH, Chigurupati S, Sajid S, Mani V (2019) Ameliorative effect of metformin on cyclophosphamide-induced memory impairment in mice. *Eur Rev Med Pharmacol Sci* 23(21):9660–9666. https://doi.org/10.26355/eurrev_201911_19460
- Ali MA, Menze ET, Tadros MG, Tolba MF (2020) Caffeic acid phenethyl ester counteracts doxorubicin-induced chemobrain in Sprague-Dawley rats: Emphasis on the modulation of oxidative stress and neuroinflammation. *Neuropharmacology* 181:108334. <https://doi.org/10.1016/j.neuropharm.2020.108334>
- Allen BD, Apodaca LA, Syage AR et al (2019) Attenuation of neuroinflammation reverses Adriamycin-induced cognitive impairments. *Acta Neuropathol Commun* 7(1):186. <https://doi.org/10.1186/s40478-019-0838-8>
- Almeida D, Pinho R, Correia V, et al. (2018) Mitoxantrone is More Toxic than Doxorubicin in SH-SY5Y Human Cells: A 'Chemobrain' In Vitro Study. *Pharmaceuticals (Basel)* 11(2) doi:<https://doi.org/10.3390/ph11020041>
- Aluise CD, Miriyala S, Noel T et al (2011) 2-Mercaptoethane sulfonate prevents doxorubicin-induced plasma protein oxidation and TNF- α release: implications for the reactive oxygen species-mediated mechanisms of chemobrain. *Free Radic Biol Med* 50(11):1630–1638. <https://doi.org/10.1016/j.freeradbiomed.2011.03.009>
- Aluise CD, St Clair D, Vore M, Butterfield DA (2009) In vivo amelioration of adriamycin induced oxidative stress in plasma by gamma-glutamylcysteine ethyl ester (GCEE). *Cancer Lett* 282(1):25–29. <https://doi.org/10.1016/j.canlet.2009.02.047>
- Amidi A, Agerbaek M, Wu LM et al (2017) Changes in cognitive functions and cerebral grey matter and their associations with inflammatory markers, endocrine markers, and APOE genotypes in testicular cancer patients undergoing treatment. *Brain Imaging Behav* 11(3):769–783. <https://doi.org/10.1007/s11682-016-9552-3>
- Amidi A, Wu LM, Pedersen AD et al (2015) Cognitive impairment in testicular cancer survivors 2–7 years after treatment. *Support Care Cancer* 23(10):2973–2979. <https://doi.org/10.1007/s00520-015-2663-3>
- Andersen JK (2004) Oxidative stress in neurodegeneration: cause or consequence? *Nat Med* 10(7):S18–S25. <https://doi.org/10.1038/nrn1434>
- Anderson DE, Kedar S, Bhatt VR, Schmid K, Holstein SA, Rizzo M (2020) Neurophysiologic and ophthalmic markers of chemotherapy-related cognitive impairment in patients diagnosed with hematologic cancer: a feasibility study. *J Neurol Sci* 410:116644. <https://doi.org/10.1016/j.jns.2019.116644>
- Ando-Tanabe N, Iwamitsu Y, Kuranami M et al (2014) Cognitive function in women with breast cancer receiving adjuvant chemotherapy and healthy controls. *Breast Cancer* 21(4):453–462. <https://doi.org/10.1007/s12282-012-0405-7>
- Andreis F, Ferri M, Mazzocchi M et al (2013) Lack of a chemobrain effect for adjuvant FOLFOX chemotherapy in colon cancer patients. A Pilot Study. *Support Care Cancer* 21(2):583–590. <https://doi.org/10.1007/s00520-012-1560-2>
- Andres AL, Gong X, Di K, Bota DA (2014) Low-doses of cisplatin injure hippocampal synapses: a mechanism for “chemo” brain? *Exp Neurol* 255:137–144. <https://doi.org/10.1016/j.expneurol.2014.02.020>
- Antkiewicz-Michaluk L, Krzemieniecki K, Romanska I, Michaluk J, Krygowska-Wajs A (2016) Acute treatment with doxorubicin induced neurochemical impairment of the function of dopamine system in rat brain structures. *Pharmacol Rep* 68(3):627–630. <https://doi.org/10.1016/j.pharep.2016.01.009>
- Arndt V, Koch-Gallenkamp L, Jansen L et al (2017) Quality of life in long-term and very long-term cancer survivors versus population controls in Germany. *Acta Oncol* 56(2):190–197
- Askren MK, Jung M, Berman MG et al (2014) Neuromarkers of fatigue and cognitive complaints following chemotherapy for breast cancer: a prospective fMRI investigation. *Breast Cancer Res Treat* 147(2):445–455. <https://doi.org/10.1007/s10549-014-3092-6>
- Ayala-Feliciano M, Pons-Valerio JJ, Pons-Madera J, Acevedo SF (2011) The relationship between visuospatial memory and coping strategies in breast cancer survivors. *Breast Cancer (auckl)* 5:117–130. <https://doi.org/10.4137/bcbr.s6957>
- Aziz NM (2002) Cancer survivorship research: challenge and opportunity. *J Nutr* 132(11):3494S–3503S. <https://doi.org/10.1093/jn/132.11.3494S>
- Bagnall-Moreau C, Chaudhry S, Salas-Ramirez K, Ahles T, Hubbard K (2019) Chemotherapy-induced cognitive impairment is associated with increased inflammation and oxidative damage in the hippocampus. *Mol Neurobiol* 56(10):7159–7172. <https://doi.org/10.1007/s12035-019-1589-z>
- Barbosa DJ, Capela JP, de Lourdes BM, Carvalho F (2015) In vitro models for neurotoxicology research. *Toxicology Research* 4(4):801–842. <https://doi.org/10.1039/C4TX00043A>
- Barkoukis TJ (2012) Sleep-Related Medical Disorders. In: Blumer J (ed) *Therapy in Sleep Medicine*. 1st edn. Elsevier/Saunders, p 607–615
- Baudino B, D'Agata F, Caroppo P et al (2012) The chemotherapy long-term effect on cognitive functions and brain metabolism in lymphoma patients. *Q J Nucl Med Mol Imaging* 56(6):559–568
- Baxter MF, Dulworth AN, Smith TM (2011) Identification of mild cognitive impairments in cancer survivors. *Occup Ther Health Care* 25(1):26–37. <https://doi.org/10.3109/07380577.2010.533251>
- Becker H, Henneghan A, Mikan SQ (2015) When do I get my brain back? Breast cancer survivors' experiences of cognitive problems. *Clin J Oncol Nurs* 19(2):180–184. <https://doi.org/10.1188/15.cjon.180-184>
- Bender CM, Sereika SM, Berga SL et al (2006) Cognitive impairment associated with adjuvant therapy in breast cancer. *Psychooncology* 15(5):422–430. <https://doi.org/10.1002/pon.964>
- Bhowmik A, Khan R, Ghosh MK (2015) Blood brain barrier: a challenge for effectual therapy of brain tumors. *Biomed Res Int* 2015:320941. <https://doi.org/10.1155/2015/320941>
- Biglia N, Bounous VE, Malabaila A et al (2012) Objective and self-reported cognitive dysfunction in breast cancer women treated with chemotherapy: a prospective study. *Eur J Cancer Care (engl)* 21(4):485–492. <https://doi.org/10.1111/j.1365-2354.2011.01320.x>
- Blommaert J, Schroyen G, Vandenbulcke M et al (2019) Age-dependent brain volume and neuropsychological changes after chemotherapy in breast cancer patients. *Hum Brain Mapp* 40(17):4994–5010. <https://doi.org/10.1002/hbm.24753>
- Bourke RS, West CR, Chheda G, Tower DB (1973) Kinetics of entry and distribution of 5-fluorouracil in cerebrospinal fluid and brain following intravenous injection in a primate. *Cancer Res* 33(7):1735–1746

- Branca JJV, Maresca M, Morucci G et al (2018) Oxaliplatin-induced blood brain barrier loosening: a new point of view on chemotherapy-induced neurotoxicity. *Oncotarget* 9(34):23426–23438. <https://doi.org/10.18632/oncotarget.25193>
- Bray F, Ferlay J, Soerjomataram I, Siegel RL, Torre LA, Jemal A (2018) Global cancer statistics 2018: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J Clin* 68(6):394–424. <https://doi.org/10.3322/caac.21492>
- Brezden CB, Phillips KA, Abdoell M, Bunston T, Tannock IF (2000) Cognitive function in breast cancer patients receiving adjuvant chemotherapy. *J Clin Oncol* 18(14):2695–2701. <https://doi.org/10.1200/jco.2000.18.14.2695>
- Briones TL, Woods J (2014) Dysregulation in myelination mediated by persistent neuroinflammation: possible mechanisms in chemotherapy-related cognitive impairment. *Brain Behav Immun* 35:23–32. <https://doi.org/10.1016/j.bbi.2013.07.175>
- Bruno J, Hosseini SM, Kesler S (2012) Altered resting state functional brain network topology in chemotherapy-treated breast cancer survivors. *Neurobiol Dis* 48(3):329–338. <https://doi.org/10.1016/j.nbd.2012.07.009>
- Calvio L, Peugeot M, Bruns GL, Todd BL, Feuerstein M (2010) Measures of cognitive function and work in occupationally active breast cancer survivors. *J Occup Environ Med* 52(2):219–227. <https://doi.org/10.1097/JOM.0b013e3181d0bef7>
- Cardoso CV, de Barros MP, Bachi ALL et al (2020) Chemobrain in rats: Behavioral, morphological, oxidative and inflammatory effects of doxorubicin administration. *Behavioural Brain Res* 378:112233. <https://doi.org/10.1016/j.bbr.2019.112233>
- Carreira H, Williams R, Dempsey H, Stanway S, Smeeth L, Bhaskaran K (2020) Quality of life and mental health in breast cancer survivors compared with non-cancer controls: a study of patient-reported outcomes in the United Kingdom. *J Cancer Surviv*. <https://doi.org/10.1007/s11764-020-00950-3>
- Carroll JE, Van Dyk K, Bower JE et al (2019) Cognitive performance in survivors of breast cancer and markers of biological aging. *Cancer* 125(2):298–306. <https://doi.org/10.1002/cncr.31777>
- Castellon S, Ganz PA (2009) Neuropsychological studies in breast cancer: in search of chemobrain. *Breast Cancer Res Treat* 116(1):125–127. <https://doi.org/10.1007/s10549-008-0211-2>
- Castellon SA, Ganz PA, Bower JE, Petersen L, Abraham L, Green-dale GA (2004) Neurocognitive performance in breast cancer survivors exposed to adjuvant chemotherapy and tamoxifen. *J Clin Exp Neuropsychol* 26(7):955–969. <https://doi.org/10.1080/13803390490510905>
- Cerulla N, Arcusa A, Navarro JB et al (2019) Cognitive impairment following chemotherapy for breast cancer: The impact of practice effect on results. *J Clin Exp Neuropsychol* 41(3):290–299. <https://doi.org/10.1080/13803395.2018.1546381>
- Cerulla N, Arcusa A, Navarro JB et al (2017) Role of taxanes in chemotherapy-related cognitive impairment: a prospective longitudinal study. *Breast Cancer Res Treat* 164(1):179–187. <https://doi.org/10.1007/s10549-017-4240-6>
- Chen BT, Jin T, Patel SK et al (2018) Gray matter density reduction associated with adjuvant chemotherapy in older women with breast cancer. *Breast Cancer Res Treat* 172(2):363–370. <https://doi.org/10.1007/s10549-018-4911-y>
- Chen BT, Ye N, Wong CW et al (2019) Effects of chemotherapy on aging white matter microstructure: a longitudinal diffusion tensor imaging study. *J Geriatr Oncol*. <https://doi.org/10.1016/j.jgo.2019.09.016>
- Chen BT, Ye N, Wong CW et al (2020a) Effects of chemotherapy on aging white matter microstructure: a longitudinal diffusion tensor imaging study. *J Geriatr Oncol* 11(2):290–296. <https://doi.org/10.1016/j.jgo.2019.09.016>
- Chen VC, Lin KY, Tsai YH, Weng JC (2020b) Connectome analysis of brain functional network alterations in breast cancer survivors with and without chemotherapy. *PLoS ONE* 15(5):e0232548. <https://doi.org/10.1371/journal.pone.0232548>
- Chen X, Li J, Ren J et al (2014) Selective impairment of attention networks in breast cancer patients receiving chemotherapy treatment. *Psychooncology* 23(10):1165–1171. <https://doi.org/10.1002/pon.3539>
- Chen Y, Jungsuwadee P, Vore M, Butterfield DA, St Clair DK (2007) Collateral damage in cancer chemotherapy: oxidative stress in nontargeted tissues. *Mol Interventions* 7(3):147. <https://doi.org/10.1124/mi.7.3.6>
- Cheng H, Li W, Gong L et al (2017) Altered resting-state hippocampal functional networks associated with chemotherapy-induced prospective memory impairment in breast cancer survivors. *Sci Rep* 7:45135. <https://doi.org/10.1038/srep45135>
- Cheng H, Yang Z, Dong B et al (2013) Chemotherapy-induced prospective memory impairment in patients with breast cancer. *Psychooncology* 22(10):2391–2395. <https://doi.org/10.1002/pon.3291>
- Cheung YT, Ng T, Shwe M et al (2015) Association of proinflammatory cytokines and chemotherapy-associated cognitive impairment in breast cancer patients: a multi-centered, prospective, cohort study. *Ann Oncol* 26(7):1446–1451. <https://doi.org/10.1093/annonc/mdv206>
- Chiu GS, Maj MA, Rizvi S et al (2017) Pifithrin-mu Prevents cisplatin-induced chemobrain by preserving neuronal mitochondrial function. *Cancer Res* 77(3):742–752. <https://doi.org/10.1158/0008-5472.can-16-1817>
- Choi SM, Lee SH, Yang YS, Kim BC, Kim MK, Cho KH (2001) 5-fluorouracil-induced leukoencephalopathy in patients with breast cancer. *J Korean Med Sci* 16(3):328–334. <https://doi.org/10.3346/jkms.2001.16.3.328>
- Christie LA, Acharya MM, Parihar VK, Nguyen A, Martirosian V, Limoli CL (2012) Impaired cognitive function and hippocampal neurogenesis following cancer chemotherapy. *Clin Cancer Res* 18(7):1954–1965. <https://doi.org/10.1158/1078-0432.ccr-11-2000>
- Collins B, MacKenzie J, Tasca GA, Scherling C, Smith A (2013) Cognitive effects of chemotherapy in breast cancer patients: a dose-response study. *Psychooncology* 22(7):1517–1527. <https://doi.org/10.1002/pon.3163>
- Collins B, Mackenzie J, Tasca GA, Scherling C, Smith A (2014) Persistent cognitive changes in breast cancer patients 1 year following completion of chemotherapy. *J Int Neuropsychol Soc* 20(4):370–379. <https://doi.org/10.1017/s1355617713001215>
- Colonna M, Butovsky O (2017) Microglia function in the central nervous system during health and neurodegeneration. *Annu Rev Immunol* 35:441–468. <https://doi.org/10.1146/annurev-immunol-051116-052358>
- Conklin KA (2004) Chemotherapy-associated oxidative stress: impact on chemotherapeutic effectiveness. *Integr Cancer Ther* 3(4):294–300. <https://doi.org/10.1177/1534735404270335>
- Conroy T, Paillot B, Francois E et al (2005) Irinotecan plus oxaliplatin and leucovorin-modulated fluorouracil in advanced pancreatic cancer—a Groupe Tumeurs Digestives of the Federation Nationale des Centres de Lutte Contre le Cancer study. *J Clin Oncol* 23(6):1228–1236. <https://doi.org/10.1200/jco.2005.06.050>
- Correa DD, Root JC, Kryza-Lacombe M et al (2017) Brain structure and function in patients with ovarian cancer treated with first-line chemotherapy: a pilot study. *Brain Imaging Behav* 11(6):1652–1663. <https://doi.org/10.1007/s11682-016-9608-4>
- Correa DD, Zhou Q, Thaler HT, Maziarz M, Hurley K, Hensley ML (2010) Cognitive functions in long-term survivors of ovarian cancer. *Gynecol Oncol* 119(2):366–369. <https://doi.org/10.1016/j.ygyno.2010.06.023>

- Cruzado JA, Lopez-Santiago S, Martinez-Marin V, Jose-Moreno G, Custodio AB, Feliu J (2014) Longitudinal study of cognitive dysfunctions induced by adjuvant chemotherapy in colon cancer patients. *Support Care Cancer* 22(7):1815–1823. <https://doi.org/10.1007/s00520-014-2147-x>
- Cull A, Hay C, Love SB, Mackie M, Smets E, Stewart M (1996) What do cancer patients mean when they complain of concentration and memory problems? *Br J Cancer* 74(10):1674–1679
- Czerska M, Mikolajewska K, Zielinski M, Gromadzinska J, Wasowicz W (2015) Today's oxidative stress markers. *Med Pr* 66(3):393–405. <https://doi.org/10.13075/mp.5893.00137>
- da Ros M, Iorio AL, Lucchesi M, Stival A, de Martino M, Sardi I (2015) The Use of Anthracyclines for Therapy of CNS Tumors. *Anticancer Agents Med Chem* 15(6):721–727. <https://doi.org/10.2174/1871520615666150407155319>
- Davis BD, Fernandez F, Adams F, et al. (1987) Diagnosis of dementia in cancer patients. Cognitive impairment in these patients can go unrecognized. *Psychosomatics* 28(4):175–9. [https://doi.org/10.1016/s0033-3182\(87\)72542-0](https://doi.org/10.1016/s0033-3182(87)72542-0)
- de Boer AG (2014) The European cancer and work network: CAN-WON. *J Occup Rehabil* 24(3):393–398. <https://doi.org/10.1007/s10926-013-9474-5>
- de Jonge ME, Huitema AD, Rodenhuis S, Beijnen JH (2005) Clinical pharmacokinetics of cyclophosphamide. *Clin Pharmacokinetics* 44(11):1135–1164. <https://doi.org/10.2165/00003088-200544110-00003>
- de Ruiter MB, Reneman L, Boogerd W et al (2012) Late effects of high-dose adjuvant chemotherapy on white and gray matter in breast cancer survivors: converging results from multimodal magnetic resonance imaging. *Hum Brain Mapp* 33(12):2971–2983. <https://doi.org/10.1002/hbm.21422>
- de Ruiter MB, Reneman L, Boogerd W et al (2011) Cerebral hyporesponsiveness and cognitive impairment 10 years after chemotherapy for breast cancer. *Hum Brain Mapp* 32(8):1206–1219. <https://doi.org/10.1002/hbm.21102>
- de Vries NA, Beijnen JH, Boogerd W, van Tellingen O (2006) Blood-brain barrier and chemotherapeutic treatment of brain tumors. *Expert Rev Neurother* 6(8):1199–1209. <https://doi.org/10.1586/14737175.6.8.1199>
- Debess J, Riis JO, Engebjerg MC, Ewertz M (2010) Cognitive function after adjuvant treatment for early breast cancer: a population-based longitudinal study. *Breast Cancer Res Treat* 121(1):91–100. <https://doi.org/10.1007/s10549-010-0756-8>
- Deprez S, Amant F, Smeets A et al (2012) Longitudinal assessment of chemotherapy-induced structural changes in cerebral white matter and its correlation with impaired cognitive functioning. *J Clin Oncol* 30(3):274–281. <https://doi.org/10.1200/jco.2011.36.8571>
- Deprez S, Amant F, Yigit R et al (2011) Chemotherapy-induced structural changes in cerebral white matter and its correlation with impaired cognitive functioning in breast cancer patients. *Hum Brain Mapp* 32(3):480–493. <https://doi.org/10.1002/hbm.21033>
- Deprez S, Billiet T, Sunaert S, Leemans A (2013) Diffusion tensor MRI of chemotherapy-induced cognitive impairment in non-CNS cancer patients: a review. *Brain Imaging Behav* 7(4):409–435. <https://doi.org/10.1007/s11682-012-9220-1>
- Deprez S, Vandenbulcke M, Peeters R et al (2014) Longitudinal assessment of chemotherapy-induced alterations in brain activation during multitasking and its relation with cognitive complaints. *J Clin Oncol* 32(19):2031–2038. <https://doi.org/10.1200/jco.2013.53.6219>
- Dietrich J, Han R, Yang Y, Mayer-Proschel M, Noble M (2006) CNS progenitor cells and oligodendrocytes are targets of chemotherapeutic agents in vitro and in vivo. *J Biol* 5(7):22. <https://doi.org/10.1186/jbiol50>
- Donovan KA, Small BJ, Andrykowski MA, Schmitt FA, Munster P, Jacobsen PB (2005) Cognitive functioning after adjuvant chemotherapy and/or radiotherapy for early-stage breast carcinoma. *Cancer* 104(11):2499–2507. <https://doi.org/10.1002/cncr.21482>
- Dose J, Huebbe P, Nebel A, Rimbach G (2016) APOE genotype and stress response—a mini review. *Lipids Health Dis* 15(1):121. <https://doi.org/10.1186/s12944-016-0288-2>
- Du XL, Xia R, Hardy D (2010) Relationship between chemotherapy use and cognitive impairments in older women with breast cancer: findings from a large population-based cohort. *Am J Clin Oncol* 33(6):533–543. <https://doi.org/10.1097/COC.0b013e3181b9cflb>
- Duijts SF, van der Beek AJ, Boelhouwer IG, Schagen SB (2017) Cancer-related cognitive impairment and patients' ability to work: a current perspective. *Curr Opin Support Palliat Care* 11(1):19–23. <https://doi.org/10.1097/spc.0000000000000248>
- Dumas JA, Makarewicz J, Schaubhut GJ et al (2013) Chemotherapy altered brain functional connectivity in women with breast cancer: a pilot study. *Brain Imaging Behav* 7(4):524–532. <https://doi.org/10.1007/s11682-013-9244-1>
- Eberhardt B, Dilger S, Musial F, Wedding U, Weiss T, Miltner WH (2006) Medium-term effects of chemotherapy in older cancer patients. *Support Care Cancer* 14(3):216–222. <https://doi.org/10.1007/s00520-005-0894-4>
- Eberling JL, Wu C, Tong-Turnbeaugh R, Jagust WJ (2004) Estrogen and tamoxifen-associated effects on brain structure and function. *Neuroimage* 21(1):364–371. <https://doi.org/10.1016/j.neuroimage.2003.08.037>
- Edwards BJ, Zhang X, Sun M et al (2018) Neurocognitive deficits in older patients with cancer. *J Geriatr Oncol* 9(5):482–487. <https://doi.org/10.1016/j.jgo.2018.02.010>
- Egorin MJ, Kaplan RS, Salzman M et al (1982) Cyclophosphamide plasma and cerebrospinal fluid kinetics with and without dimethyl sulfoxide. *Clin Pharmacol Therapeutics* 32(1):122–128
- Ehrenstein JK, van Zon SKR, Duijts SFA et al (2020) Type of cancer treatment and cognitive symptoms in working cancer survivors: an 18-month follow-up study. *J Cancer Surviv* 14(2):158–167. <https://doi.org/10.1007/s11764-019-00839-w>
- El-Agamy SE, Abdel-Aziz AK, Esmat A, Azab SS (2019) Chemotherapy and cognition: comprehensive review on doxorubicin-induced chemobrain. *Cancer Chemother Pharmacol* 84(1):1–14. <https://doi.org/10.1007/s00280-019-03827-0>
- El-Agamy SE, Abdel-Aziz AK, Wahdan S, Esmat A, Azab SS (2018) Astaxanthin ameliorates doxorubicin-induced cognitive impairment (Chemobrain) in experimental rat model: impact on oxidative, inflammatory, and apoptotic machineries. *Mol Neurobiol* 55(7):5727–5740. <https://doi.org/10.1007/s12035-017-0797-7>
- El M, Mustafa S, Umka J et al (2012) The effect of 5-fluorouracil on the long term survival and proliferation of cells in the rat hippocampus. *Brain Res Bull* 88(5):514–518. <https://doi.org/10.1016/j.brainresbull.2012.05.005>
- Fan HG, Houede-Tchen N, Yi QL et al (2005) Fatigue, menopausal symptoms, and cognitive function in women after adjuvant chemotherapy for breast cancer: 1- and 2-year follow-up of a prospective controlled study. *J Clin Oncol* 23(31):8025–8032. <https://doi.org/10.1200/jco.2005.01.6550>
- Fan HG, Park A, Xu W et al (2009) The influence of erythropoietin on cognitive function in women following chemotherapy for breast cancer. *Psychooncology* 18(2):156–161. <https://doi.org/10.1002/pon.1372>
- Feng Y, Wang YF, Zheng LJ, Shi Z, Huang W, Zhang LJ (2020) Network-level functional connectivity alterations in chemotherapy treated breast cancer patients: a longitudinal resting state functional MRI study. *Cancer Imaging* 20(1):73. <https://doi.org/10.1186/s40644-020-00355-6>

- Fitzmaurice C, Abate D, Abbasi N et al (2019) Global, regional, and national cancer incidence, mortality, years of life lost, years lived with disability, and disability-adjusted life-years for 29 cancer groups, 1990 to 2017: a systematic analysis for the global burden of disease study. *JAMA Oncol* 5(12):1749–1768. <https://doi.org/10.1001/jamaoncol.2019.2996>
- Folstein MF, Fetting JH, Lobo A, Niaz U, Capozzoli KD (1984) Cognitive assessment of cancer patients. *Cancer* 53(10 Suppl):2250–2257
- Freeman JR, Broshek DK (2002) Assessing cognitive dysfunction in breast cancer: what are the tools? *Clin Breast Cancer* 3(Suppl 3):S91–S99
- Gaillard PJ, Appeldoorn CC, Dorland R et al (2014) Pharmacokinetics, brain delivery, and efficacy in brain tumor-bearing mice of glutathione pegylated liposomal doxorubicin (2B3–101). *PLoS one* 9(1):e82331. <https://doi.org/10.1371/journal.pone.0082331>
- Gallego Perez-Larraya J, Hildebrand J (2014) Brain metastases. *Handbook Clin Neurol* 121:1143–1157. <https://doi.org/10.1016/b978-0-7020-4088-7.00077-8>
- Gandhi S, Abramov AY (2012) Mechanism of oxidative stress in neurodegeneration. *Oxid Med Cell Longev* 2012: <https://doi.org/10.1155/2012/428010>
- Garthe A, Kempermann G (2013) An old test for new neurons: refining the Morris water maze to study the functional relevance of adult hippocampal neurogenesis. *Front Neurosci* 7:63. <https://doi.org/10.3389/fnins.2013.00063>
- Gerstenecker A, Nabors LB, Meneses K et al (2014) Cognition in patients with newly diagnosed brain metastasis: profiles and implications. *J Neuro-Oncol* 120(1):179–185. <https://doi.org/10.1007/s11060-014-1543-x>
- Gibson EM, Nagaraja S, Ocampo A et al (2019) Methotrexate Chemotherapy Induces Persistent Tri-glial Dysregulation that Underlies Chemotherapy-Related Cognitive Impairment. *Cell* 176(1–2):43–55.e13. <https://doi.org/10.1016/j.cell.2018.10.049>
- Glick D, Barth S, Macleod KF (2010) Autophagy: cellular and molecular mechanisms. *J Pathol* 221(1):3–12. <https://doi.org/10.1002/path.2697>
- Gopal KV, Wu C, Shrestha B, Campbell KC, Moore EJ, Gross GW (2012) d-Methionine protects against cisplatin-induced neurotoxicity in cortical networks. *Neurotoxicol Teratol* 34(5):495–504. <https://doi.org/10.1016/j.ntt.2012.06.002>
- Graul EH, Schaumloffel E, Hundeshagen H, Wilmanns H, Simon G (1967) Metabolism of radioactive cyclophosphamide Animal tests and clinical studies. *Cancer* 20(5):896–899. [https://doi.org/10.1002/1097-0142\(1967\)20:5%3c896::aid-cnrcr2820200551%3e3.0.co;2-d](https://doi.org/10.1002/1097-0142(1967)20:5%3c896::aid-cnrcr2820200551%3e3.0.co;2-d)
- Greimel ER, Bjelic-Radisic V, Pfisterer J, Hilpert F, Daghofer F, du Bois A (2006) Randomized study of the arbeitgemeinschaft gynaekologische onkologie ovarian cancer study group comparing quality of life in patients with ovarian cancer treated with cisplatin/paclitaxel versus carboplatin/paclitaxel. *J Clin Oncol* 24(4):579–586. <https://doi.org/10.1200/jco.2005.02.4067>
- Grisold W, Grisold A, Löscher W (2017) Cancer therapy and neuromuscular complications: a mini review. *Neurology (ECRicon)* 9(1):20–26
- Guest PC (2020) MK-801 treatment of oligodendrocytes as a cellular model of aging. *Methods Mol Biol (clifton, NJ)* 2138:431–447. https://doi.org/10.1007/978-1-0716-0471-7_32
- Gutmann DH (2019) Clearing the Fog surrounding Chemobrain. *Cell* 176(1–2):2–4. <https://doi.org/10.1016/j.cell.2018.12.027>
- Harry GJ, Kraft AD (2008) Neuroinflammation and microglia: considerations and approaches for neurotoxicity assessment. *Expert Opin Drug Metab Toxicol* 4(10):1265–1277. <https://doi.org/10.1517/17425255.4.10.1265>
- Hartung H-P, Gonsette R, König N et al (2002) Mitoxantrone in progressive multiple sclerosis: a placebo-controlled, double-blind, randomised, multicentre trial. *Lancet* 360(9350):2018–2025. [https://doi.org/10.1016/S0140-6736\(02\)12023-X](https://doi.org/10.1016/S0140-6736(02)12023-X)
- Heck JE, Albert SM, Franco R, Gorin SS (2008) Patterns of dementia diagnosis in surveillance, epidemiology, and end results breast cancer survivors who use chemotherapy. *J Am Geriatr Soc* 56(9):1687–1692. <https://doi.org/10.1111/j.1532-5415.2008.01848.x>
- Heflin LH, Meyerowitz BE, Hall P et al (2005) Cancer as a risk factor for long-term cognitive deficits and dementia. *J Natl Cancer Inst* 97(11):854–856. <https://doi.org/10.1093/jnci/dji137>
- Henneghan A, Rao V, Harrison RA et al (2020) Cortical brain age from pre-treatment to post-chemotherapy in patients with breast cancer. *Neurotox Res* 37(4):788–799. <https://doi.org/10.1007/s12640-019-00158-z>
- Hermelink K (2011) Acute and late onset cognitive dysfunction associated with chemotherapy in women with breast cancer. *Cancer* 117(5):1103; author reply 1103–4 doi:<https://doi.org/10.1002/cncr.25708>
- Hermelink K, Kuchenhoff H, Untch M et al (2010) Two different sides of “chemobrain”: determinants and nondeterminants of self-perceived cognitive dysfunction in a prospective, randomized, multicenter study. *Psychooncology* 19(12):1321–1328. <https://doi.org/10.1002/pon.1695>
- Hess LM, Chambers SK, Hatch K et al (2010) Pilot study of the prospective identification of changes in cognitive function during chemotherapy treatment for advanced ovarian cancer. *J Support Oncol* 8(6):252–258. <https://doi.org/10.3816/cbc.2002.s.019>
- Hess LM, Huang HQ, Hanlon AL et al (2015) Cognitive function during and six months following chemotherapy for front-line treatment of ovarian, primary peritoneal or fallopian tube cancer: An NRG oncology/gynecologic oncology group study. *Gynecol Oncol* 139(3):541–545. <https://doi.org/10.1016/j.ygyno.2015.10.003>
- Hosseini SM, Koovakkattu D, Kesler SR (2012) Altered small-world properties of gray matter networks in breast cancer. *BMC Neurol* 12:28. <https://doi.org/10.1186/1471-2377-12-28>
- Huo X, Reyes TM, Heijnen CJ, Kavelaars A (2018) Cisplatin treatment induces attention deficits and impairs synaptic integrity in the prefrontal cortex in mice. *Sci Rep* 8(1):17400. <https://doi.org/10.1038/s41598-018-35919-x>
- Hurria A, Goldfarb S, Rosen C et al (2006a) Effect of adjuvant breast cancer chemotherapy on cognitive function from the older patient’s perspective. *Breast Cancer Res Treat* 98(3):343–348. <https://doi.org/10.1007/s10549-006-9171-6>
- Hurria A, Rosen C, Hudis C et al (2006b) Cognitive function of older patients receiving adjuvant chemotherapy for breast cancer: a pilot prospective longitudinal study. *J Am Geriatr Soc* 54(6):925–931. <https://doi.org/10.1111/j.1532-5415.2006.00732.x>
- Iconomou G, Koutras A, Karaivazoglou K et al (2008) Effect of epoetin alpha therapy on cognitive function in anaemic patients with solid tumours undergoing chemotherapy. *Eur J Cancer Care (engl)* 17(6):535–541. <https://doi.org/10.1111/j.1365-2354.2007.00857.x>
- Jacobs S, McCully C, Murphy R, Bacher J, Balis F, Fox E (2009) Extracellular fluid concentrations of cisplatin, carboplatin, and oxaliplatin in brain, muscle, and blood measured using microdialysis in nonhuman primates. *Cancer Chemotherapy Pharmacol* 65:817–824. <https://doi.org/10.1007/s00280-009-1085-7>
- Jacobs S, McCully CL, Murphy RF, Bacher J, Balis FM, Fox E (2010) Extracellular fluid concentrations of cisplatin, carboplatin, and oxaliplatin in brain, muscle, and blood measured using microdialysis in nonhuman primates. *Cancer Chemotherapy Pharmacol* 65(5):817–824. <https://doi.org/10.1007/s00280-009-1085-7>
- Janelins MC, Heckler CE, Thompson BD, Gross RA, Opanashuk LA, Cory-Slechta DA (2016) A clinically relevant dose of

- cyclophosphamide chemotherapy impairs memory performance on the delayed spatial alternation task that is sustained over time as mice age. *Neurotoxicology* 56:287–293. <https://doi.org/10.1016/j.neuro.2016.06.013>
- Janelins MC, Roscoe JA, Berg MJ et al (2010) IGF-1 partially restores chemotherapy-induced reductions in neural cell proliferation in adult C57BL/6 mice. *Cancer Invest* 28(5):544–553. <https://doi.org/10.3109/07357900903405942>
- Jansen C, Miaskowski C, Dodd M, Dowling G, Kramer J (2005) Potential mechanisms for chemotherapy-induced impairments in cognitive function. *Oncol Nurs Forum* 32(6):1151–1163. <https://doi.org/10.1188/05.onf.1151-1163>
- Jansen CE, Cooper BA, Dodd MJ, Miaskowski CA (2011) A prospective longitudinal study of chemotherapy-induced cognitive changes in breast cancer patients. *Support Care Cancer* 19(10):1647–1656. <https://doi.org/10.1007/s00520-010-0997-4>
- Jansen CE, Dodd MJ, Miaskowski CA, Dowling GA, Kramer J (2008) Preliminary results of a longitudinal study of changes in cognitive function in breast cancer patients undergoing chemotherapy with doxorubicin and cyclophosphamide. *Psychooncology* 17(12):1189–1195. <https://doi.org/10.1002/pon.1342>
- Jansen CE, Miaskowski CA, Dodd MJ, Dowling GA (2007) A meta-analysis of the sensitivity of various neuropsychological tests used to detect chemotherapy-induced cognitive impairment in patients with breast cancer. *Oncol Nurs Forum* 34(5):997–1005. <https://doi.org/10.1188/07.onf.997-1005>
- Jarmolowicz DP, Gehringer R, Lemley SM, Sofis MJ, Kaplan S, Johnson MA (2019) 5-Fluorouracil impairs attention and dopamine release in rats. *Behav Brain Res* 362:319–322. <https://doi.org/10.1016/j.bbr.2019.01.007>
- Jenkins V, Shilling V, Deutsch G et al (2006) A 3-year prospective study of the effects of adjuvant treatments on cognition in women with early stage breast cancer. *Br J Cancer* 94(6):828–834. <https://doi.org/10.1038/sj.bjc.6603029>
- Jim HS, Donovan KA, Small BJ, Andrykowski MA, Munster PN, Jacobsen PB (2009) Cognitive functioning in breast cancer survivors: a controlled comparison. *Cancer* 115(8):1776–1783. <https://doi.org/10.1002/cncr.24192>
- Joly F, Espie M, Marty M, Heron JF, Henry-Amar M (2000) Long-term quality of life in premenopausal women with node-negative localized breast cancer treated with or without adjuvant chemotherapy. *Br J Cancer* 83(5):577–582. <https://doi.org/10.1054/bjoc.2000.1337>
- Joshi G, Hardas S, Sultana R, St Clair DK, Vore M, Butterfield DA (2007) Glutathione elevation by gamma-glutamyl cysteine ethyl ester as a potential therapeutic strategy for preventing oxidative stress in brain mediated by in vivo administration of adriamycin: Implication for chemobrain. *J Neurosci Res* 85(3):497–503. <https://doi.org/10.1002/jnr.21158>
- Jung MS, Cimprich B (2014) Cognitive deficits in Korean women treated with chemotherapy for breast cancer. *Cancer Nurs* 37(3):E31–42. <https://doi.org/10.1097/NCC.0b013e3182980383>
- Jung MS, Zhang M, Askren MK et al (2017) Cognitive dysfunction and symptom burden in women treated for breast cancer: a prospective behavioral and fMRI analysis. *Brain Imaging Behav* 11(1):86–97. <https://doi.org/10.1007/s11682-016-9507-8>
- Kardan O, Reuter-Lorenz PA, Peltier S et al (2019) Brain connectivity tracks effects of chemotherapy separately from behavioral measures. *Neuroimage Clin* 21:101654. <https://doi.org/10.1016/j.nicl.2019.101654>
- Kassmann CM, Lappe-Siefke C, Baes M et al (2007) Axonal loss and neuroinflammation caused by peroxisome-deficient oligodendrocytes. *Nat Genet* 39(8):969–976. <https://doi.org/10.1038/ng2070>
- Keeney JTR, Ren X, Warriar G et al (2018) Doxorubicin-induced elevated oxidative stress and neurochemical alterations in brain and cognitive decline: protection by MESNA and insights into mechanisms of chemotherapy-induced cognitive impairment (“chemobrain”). *Oncotarget* 9(54):30324–30339. <https://doi.org/10.18632/oncotarget.25718>
- Kesler S, Janelins M, Koovakkattu D et al (2013a) Reduced hippocampal volume and verbal memory performance associated with interleukin-6 and tumor necrosis factor-alpha levels in chemotherapy-treated breast cancer survivors. *Brain Behav Immun* 30(Suppl):S109–S116. <https://doi.org/10.1016/j.bbi.2012.05.017>
- Kesler SR, Wefel JS, Hosseini SM, Cheung M, Watson CL, Hoeft F (2013b) Default mode network connectivity distinguishes chemotherapy-treated breast cancer survivors from controls. *Proc Natl Acad Sci USA* 110(28):11600–11605. <https://doi.org/10.1073/pnas.1214551110>
- Kesler SR, Blayney DW (2016) Neurotoxic effects of anthracycline vs nonanthracycline-based chemotherapy on cognition in breast cancer survivors. *JAMA Oncol* 2(2):185–192. <https://doi.org/10.1001/jamaoncol.2015.4333>
- Kettenmann H, Verkhratsky A (2013) Glial cells. *Neuroscience in the 21st Century: From Basic to Clinical*:475–506
- Khan HA, Khan I, Lee Y-k (2018) Role of immune factors in bio-availability and toxicity of carbon nanomaterials Fullerenes, Graphenes and Nanotubes. Elsevier, p 601–630
- Khan OF, Cusano E, Raissouni S et al (2019) Immediate-term cognitive impairment following intravenous (IV) chemotherapy: a prospective pre-post design study. *BMC Cancer* 19(1):150. <https://doi.org/10.1186/s12885-019-5349-2>
- Kim HG, Shin NY, Bak Y et al (2017) Altered intrinsic brain activity after chemotherapy in patients with gastric cancer: a preliminary study. *Eur Radiol* 27(7):2679–2688. <https://doi.org/10.1007/s00330-016-4578-x>
- Kim YA, Chung HC, Choi HJ, Rha SY, Seong JS, Jeung HC (2006) Intermediate dose 5-fluorouracil-induced encephalopathy. *Jpn J Clin Oncol* 36(1):55–59. <https://doi.org/10.1093/jjco/hyi214>
- Kitamura Y, Hattori S, Yoneda S et al (2015) Doxorubicin and cyclophosphamide treatment produces anxiety-like behavior and spatial cognition impairment in rats: possible involvement of hippocampal neurogenesis via brain-derived neurotrophic factor and cyclin D1 regulation. *Behav Brain Res* 292:184–193. <https://doi.org/10.1016/j.bbr.2015.06.007>
- Kohli S, Fisher SG, Tra Y et al (2009) The effect of modafinil on cognitive function in breast cancer survivors. *Cancer* 115(12):2605–2616. <https://doi.org/10.1002/cncr.24287>
- Koll TT, Sheese AN, Semin J et al (2020) Screening for cognitive impairment in older adults with hematological malignancies using the montreal cognitive assessment and neuropsychological testing. *J Geriatr Oncol* 11(2):297–303. <https://doi.org/10.1016/j.jgo.2019.11.007>
- Koppelmans V, Breteler MM, Boogerd W, Seynaeve C, Gundy C, Schagen SB (2012a) Neuropsychological performance in survivors of breast cancer more than 20 years after adjuvant chemotherapy. *J Clin Oncol* 30(10):1080–1086. <https://doi.org/10.1200/jco.2011.37.0189>
- Koppelmans V, de Ruiter MB, van der Lijn F et al (2012b) Global and focal brain volume in long-term breast cancer survivors exposed to adjuvant chemotherapy. *Breast Cancer Res Treat* 132(3):1099–1106. <https://doi.org/10.1007/s10549-011-1888-1>
- Kovalchuk A, Illytskyy Y, Rodriguez-Juarez R et al (2017) Growth of malignant extracranial tumors alters microRNAome in the prefrontal cortex of TumorGraft mice. *Oncotarget* 8(51):88276–88293. <https://doi.org/10.18632/oncotarget.19835>
- Kreukels BP, Schagen SB, Ridderinkhof KR et al (2006) Effects of high-dose and conventional-dose adjuvant chemotherapy on long-term cognitive sequelae in patients with breast cancer: an electrophysiologic study. *Clin Breast Cancer* 7(1):67–78. <https://doi.org/10.3816/CBC.2006.n.015>

- Kreukels BP, Schagen SB, Ridderinkhof KR, Boogerd W, Hamburger HL, van Dam FS (2005) Electrophysiological correlates of information processing in breast-cancer patients treated with adjuvant chemotherapy. *Breast Cancer Res Treat* 94(1):53–61. <https://doi.org/10.1007/s10549-005-7093-3>
- Kurita K, Meyerowitz BE, Hall P, Gatz M (2011) Long-term cognitive impairment in older adult twins discordant for gynecologic cancer treatment. *J Gerontol A Biol Sci Med Sci* 66(12):1343–1349. <https://doi.org/10.1093/gerona/glr140>
- Kwatra M, Jangra A, Mishra M et al (2016) Naringin and sertraline ameliorate doxorubicin-induced behavioral deficits through modulation of serotonin level and mitochondrial complexes protection pathway in rat hippocampus. *Neurochem Res* 41(9):2352–2366. <https://doi.org/10.1007/s11064-016-1949-2>
- Kwon KA, Kwon HC, Kim MC et al (2010) A case of 5-fluorouracil induced encephalopathy. *Cancer Res Treat* 42(2):118–120. <https://doi.org/10.4143/crt.2010.42.2.118>
- Lange M, Heutte N, Rigal O et al (2016) Decline in cognitive function in older adults with early-stage breast cancer after adjuvant treatment. *Oncologist* 21(11):1337–1348. <https://doi.org/10.1634/theoncologist.2016-0014>
- Lepage C, Smith AM, Moreau J et al (2014) A prospective study of grey matter and cognitive function alterations in chemotherapy-treated breast cancer patients. *Springerplus* 3:444. <https://doi.org/10.1186/2193-1801-3-444>
- Levine PM, Silberfarb PM, Lipowski Z (1978) Mental disorders in cancer patients. A study of 100 psychiatric referrals. *Cancer* 42(3):1385–1391
- Li TY, Chen VC, Yeh DC et al (2018a) Investigation of chemotherapy-induced brain structural alterations in breast cancer patients with generalized q-sampling MRI and graph theoretical analysis. *BMC Cancer* 18(1):1211. <https://doi.org/10.1186/s12885-018-5113-z>
- Li X, Chen H, Lv Y et al (2018b) Diminished gray matter density mediates chemotherapy dosage-related cognitive impairment in breast cancer patients. *Sci Rep* 8(1):13801. <https://doi.org/10.1038/s41598-018-32257-w>
- Libert Y, Dubruille S, Borghgraef C et al (2016) Vulnerabilities in older patients when cancer treatment is initiated: does a cognitive impairment impact the two-year survival? *PLoS ONE* 11(8):e0159734. <https://doi.org/10.1371/journal.pone.0159734>
- Liedke PE, Reolon GK, Kilpp B, Brunetto AL, Roesler R, Schwartzmann G (2009) Systemic administration of doxorubicin impairs aversively motivated memory in rats. *Pharmacol Biochem Behav* 94(2):239–243. <https://doi.org/10.1016/j.pbb.2009.09.001>
- Linden DE (2005) The p300: where in the brain is it produced and what does it tell us? *Neuroscientist* 11(6):563–576. <https://doi.org/10.1177/1073858405280524>
- Lomeli N, Di K, Czerniawski J, Guzowski JF, Bota DA (2017) Cisplatin-induced mitochondrial dysfunction is associated with impaired cognitive function in rats. *Free Radic Biol Med* 102:274–286. <https://doi.org/10.1016/j.freeradbiomed.2016.11.046>
- Long JM, Lee GD, Kelley-Bell B et al (2011) Preserved learning and memory following 5-fluorouracil and cyclophosphamide treatment in rats. *Pharmacol Biochem Behav* 100(1):205–211. <https://doi.org/10.1016/j.pbb.2011.08.012>
- Lopes MÂ, Meisel A, Carvalho F, Bastos ML (2011) Neuronal nitric oxide synthase is a key factor in doxorubicin-induced toxicity to rat-isolated cortical neurons. *Neurotoxicity Res* 19(1):14–22. <https://doi.org/10.1007/s12640-009-9135-9>
- Lopes MÂ, Meisel A, Dirnagl U, Carvalho F, Bastos ML (2008) Doxorubicin induces biphasic neurotoxicity to rat cortical neurons. *Neurotoxicology* 29(2):286–293. <https://doi.org/10.1016/j.neuro.2007.12.003>
- Lopez Zunini RA, Scherling C, Wallis N et al (2013) Differences in verbal memory retrieval in breast cancer chemotherapy patients compared to healthy controls: a prospective fMRI study. *Brain Imaging Behav* 7(4):460–477. <https://doi.org/10.1007/s11682-012-9213-0>
- Löscher W, Potschka H (2005) Blood-brain barrier active efflux transporters: ATP-binding cassette gene family. *NeuroRx* 2(1):86–98. <https://doi.org/10.1602/neurorx.2.1.86>
- Luqmani Y (2005) Mechanisms of drug resistance in cancer chemotherapy. *Med Principles Pract* 14(Suppl. 1):35–48. <https://doi.org/10.1159/000086183>
- Lyon DE, Cohen R, Chen H et al (2016) Relationship of systemic cytokine concentrations to cognitive function over two years in women with early stage breast cancer. *J Neuroimmunol* 301:74–82. <https://doi.org/10.1016/j.jneuroim.2016.11.002>
- Lyons L, Elbeltagy M, Bennett G, Wigmore P (2011) The effects of cyclophosphamide on hippocampal cell proliferation and spatial working memory in rat. *PLoS ONE* 6(6):e21445. <https://doi.org/10.1371/journal.pone.0021445>
- Lyons L, Elbeltagy M, Bennett G, Wigmore P (2012) Fluoxetine counteracts the cognitive and cellular effects of 5-fluorouracil in the rat hippocampus by a mechanism of prevention rather than recovery. *PLoS ONE* 7(1):e30010. <https://doi.org/10.1371/journal.pone.0030010>
- Maiese K (2019) Neurotransmissão. In. Accessed 22–01–2021
- Mar Fan HG, Clemons M, Xu W et al (2008) A randomised, placebo-controlled, double-blind trial of the effects of d-methylphenidate on fatigue and cognitive dysfunction in women undergoing adjuvant chemotherapy for breast cancer. *Support Care Cancer* 16(6):577–583. <https://doi.org/10.1007/s00520-007-0341-9>
- Mars RB, Neubert F-X, Noonan MP, Sallet J, Toni I, Rushworth MF (2012) On the relationship between the “default mode network” and the “social brain”. *Front Hum Neurosci* 6:189. <https://doi.org/10.3389/fnhum.2012.00189>
- Matsos A, Johnston IN (2019) Chemotherapy-induced cognitive impairments: a systematic review of the animal literature. *Neurosci Biobehav Rev* 102:382–399. <https://doi.org/10.1016/j.neubiorev.2019.05.001>
- Mayer DK, Nasso SF, Earp JA (2017) Defining cancer survivors, their needs, and perspectives on survivorship health care in the USA. *Lancet Oncol* 18(1):e11–e18. [https://doi.org/10.1016/S1470-2045\(16\)30573-3](https://doi.org/10.1016/S1470-2045(16)30573-3)
- McDonald BC, Conroy SK, Ahles TA, West JD, Saykin AJ (2012) Alterations in brain activation during working memory processing associated with breast cancer and treatment: a prospective functional magnetic resonance imaging study. *J Clin Oncol* 30(20):2500–2508. <https://doi.org/10.1200/jco.2011.38.5674>
- McDonald BC, Conroy SK, Smith DJ, West JD, Saykin AJ (2013) Frontal gray matter reduction after breast cancer chemotherapy and association with executive symptoms: a replication and extension study. *Brain Behav Immun* 30(Suppl):S117–S125. <https://doi.org/10.1016/j.bbi.2012.05.007>
- Megari K (2020) Neuropsychological functioning among patients with different types of cancer: Postchemotherapy cognitive impairment and implications for rehabilitation. *Neuropsychiatr*. <https://doi.org/10.1007/s40211-020-00345-x>
- Mehlsen M, Pedersen AD, Jensen AB, Zachariae R (2009) No indications of cognitive side-effects in a prospective study of breast cancer patients receiving adjuvant chemotherapy. *Psychooncology* 18(3):248–257. <https://doi.org/10.1002/pon.1398>
- Menning S, de Ruiter MB, Kieffer JM et al (2016) Cognitive impairment in a subset of breast cancer patients after systemic therapy—results from a longitudinal study. *J Pain Symptom Manage* 52(4):560–569.e1. <https://doi.org/10.1016/j.jpainsymman.2016.04.012>

- Menning S, de Ruiter MB, Veltman DJ et al (2017) Changes in brain activation in breast cancer patients depend on cognitive domain and treatment type. *PLoS ONE* 12(3):e0171724. <https://doi.org/10.1371/journal.pone.0171724>
- Menning S, de Ruiter MB, Veltman DJ et al (2018) Changes in brain white matter integrity after systemic treatment for breast cancer: a prospective longitudinal study. *Brain Imaging Behav* 12(2):324–334. <https://doi.org/10.1007/s11682-017-9695-x>
- Meyers CA, Albitar M, Estey E (2005) Cognitive impairment, fatigue, and cytokine levels in patients with acute myelogenous leukemia or myelodysplastic syndrome. *Cancer* 104(4):788–793. <https://doi.org/10.1002/cncr.21234>
- Miao H, Chen X, Yan Y et al (2016) Functional connectivity change of brain default mode network in breast cancer patients after chemotherapy. *Neuroradiology* 58(9):921–928. <https://doi.org/10.1007/s00234-016-1708-8>
- Michalak S, Rybacka-Mossakowska J, Ambrosius W et al (2016) The markers of glutamate metabolism in peripheral blood mononuclear cells and neurological complications in lung cancer patients. *Dis Markers* 2016:2895972. <https://doi.org/10.1155/2016/2895972>
- Miller KD, Nogueira L, Mariotto AB et al (2019) (2019) Cancer treatment and survivorship statistics, 2019. *CA Cancer J Clin* 69(5):363–385. <https://doi.org/10.3322/caac.21565>
- Minagar A, Alexander JS (2003) Blood-brain barrier disruption in multiple sclerosis. *Multiple Sclerosis J* 9(6):540–549. <https://doi.org/10.1191/1352458503ms9650a>
- Minisini AM, De Faccio S, Ermacora P et al (2008) Cognitive functions and elderly cancer patients receiving anticancer treatment: a prospective study. *Crit Rev Oncol Hematol* 67(1):71–79. <https://doi.org/10.1016/j.critrevonc.2008.02.004>
- Mo C, Lin H, Fu F et al (2017) Chemotherapy-induced changes of cerebral activity in resting-state functional magnetic resonance imaging and cerebral white matter in diffusion tensor imaging. *Oncotarget* 8(46):81273–81284. <https://doi.org/10.18632/oncotarget.18111>
- Moore HC, Parsons MW, Yue GH, Rybicki LA, Siemionow W (2014) Electroencephalogram power changes as a correlate of chemotherapy-associated fatigue and cognitive dysfunction. *Supp Care Cancer* 22(8):2127–2131. <https://doi.org/10.1007/s00520-014-2197-0>
- Morin RT, Midlarsky E (2018) Treatment with chemotherapy and cognitive functioning in older adult cancer survivors. *Arch Phys Med Rehabil* 99(2):257–263. <https://doi.org/10.1016/j.apmr.2017.06.016>
- Morse R, Rodgers J, Verrill M, Kendell K (2003) Neuropsychological functioning following systemic treatment in women treated for breast cancer: a review. *Eur J Cancer* 39(16):2288–2297. [https://doi.org/10.1016/s0959-8049\(03\)00600-2](https://doi.org/10.1016/s0959-8049(03)00600-2)
- Moruno-Manchon JF, Uzor NE, Kesler SR et al (2016) TFEB ameliorates the impairment of the autophagy-lysosome pathway in neurons induced by doxorubicin. *Aging (alban NY)* 8(12):3507–3519. <https://doi.org/10.18632/aging.101144>
- Mu L, Wang J, Cao B et al (2015) Impairment of cognitive function by chemotherapy: association with the disruption of phase-locking and synchronization in anterior cingulate cortex. *Mol Brain* 8:32. <https://doi.org/10.1186/s13041-015-0125-y>
- Mustafa S, Walker A, Bennett G, Wigmore PM (2008) 5-Fluorouracil chemotherapy affects spatial working memory and newborn neurons in the adult rat hippocampus. *Eur J Neurosci* 28(2):323–330. <https://doi.org/10.1111/j.1460-9568.2008.06325.x>
- Muzzatti B, Giovannini L, Flaiban C, Cattaruzza N, Annunziata MA (2017) Cognitive functioning in long-term cancer survivorship: a survey utilizing both standardized neuropsychological and self-report measures. *Appl Neuropsychol Adult*:1–8. <https://doi.org/10.1080/23279095.2017.1387551>
- Myers JS (2010) Neuropsychologic testing for chemotherapy-related cognitive impairment. *Chemo Fog. Adv Exp Med Biol* 2010:678:55–69. https://doi.org/10.1007/978-1-4419-6306-2_9
- Nandakumar K, Ramalingaya G, Nayak P, et al. (2018) Naringin ameliorates doxorubicin-induced neurotoxicity In vitro and cognitive dysfunction In vivo. *Pharmacognosy Magazine* 14(55):https://doi.org/10.4103/pm.pm_364_17
- Natori A, Ogata T, Sumitani M, Kogure T, Yamauchi T, Yamauchi H (2015) Potential role of pNF-H, a biomarker of axonal damage in the central nervous system, as a predictive marker of chemotherapy-induced cognitive impairment. *Clin Cancer Res* 21(6):1348–1352. <https://doi.org/10.1158/1078-0432.ccr-14-2775>
- Nelson DL, Cox MM (2013) Bioenergetics and metabolism lehrninger principles of biochemistry, 6th edn. Macmillan Higher Education, pp 764–765
- Ng T, Teo SM, Yeo HL et al (2016) Brain-derived neurotrophic factor genetic polymorphism (rs6265) is protective against chemotherapy-associated cognitive impairment in patients with early-stage breast cancer. *Neuro Oncol* 18(2):244–251. <https://doi.org/10.1093/neuonc/nov162>
- Nguyen CM, Yamada TH, Beglinger LJ, Cavanaugh JE, Denburg NL, Schultz SK (2013) Cognitive features 10 or more years after successful breast cancer survival: comparisons across types of cancer interventions. *Psychooncology* 22(4):862–868. <https://doi.org/10.1002/pon.3086>
- Nguyen LD, Ehrlich BE (2020) Cellular mechanisms and treatments for chemobrain: insight from aging and neurodegenerative diseases. *EMBO Mol Med* 12(6):e12075. <https://doi.org/10.15252/emmm.202012075>
- Noh T, Walbert T (2018) Brain metastasis: clinical manifestations, symptom management, and palliative care. *Handbook Clin Neurol* 149:75–88. <https://doi.org/10.1016/b978-0-12-811161-1.00006-2>
- Nudelman KN, Wang Y, McDonald BC et al (2014) Altered cerebral blood flow one month after systemic chemotherapy for breast cancer: a prospective study using pulsed arterial spin labeling MRI perfusion. *PLoS ONE* 9(5):e96713. <https://doi.org/10.1371/journal.pone.0096713>
- Ohnishi T, Tamai I, Sakanaka K et al (1995) In vivo and in vitro evidence for ATP-dependency of P-glycoprotein-mediated efflux of doxorubicin at the blood-brain barrier. *Biochem Pharmacol* 49(10):1541–1544
- Olver I, Carey M, Boyes A et al (2018) The timeliness of patients reporting the side effects of chemotherapy. *Supportive Care Cancer* 26(10):3579–3586. <https://doi.org/10.1007/s00520-018-4225-y>
- Oxman TE, Schnurr PP, Silberfarb PM (1986) Assessment of cognitive function in cancer patients. *Hosp J* 2(3):99–128
- Oxman TE, Silberfarb PM (1980) Serial cognitive testing in cancer patients receiving chemotherapy. *Am J Psychiatry* 137(10):1263–1265. <https://doi.org/10.1176/ajp.137.10.1263>
- Palesh O, Scheiber C, Kesler S, Mustian K, Koopman C, Schapira L (2018) Management of side effects during and post-treatment in breast cancer survivors. *Breast J* 24(2):167–175. <https://doi.org/10.1111/tbj.12862>
- Park SH, Lee WK, Chung M et al (2006) Quality of life in patients with advanced gastric cancer treated with second-line chemotherapy. *Cancer Chemother Pharmacol* 57(3):289–294. <https://doi.org/10.1007/s00280-005-0055-y>
- Parsons MW, Dietrich J (2019) Assessment and management of cognitive changes in patients with cancer. *Cancer*. <https://doi.org/10.1002/cncr.31905>

- Pedersen AD, Rossen P, Mehlsen MY, Pedersen CG, Zachariae R, von der Maase H (2009) Long-term cognitive function following chemotherapy in patients with testicular cancer. *J Int Neuropsychol Soc* 15(2):296–301. <https://doi.org/10.1017/s1355617709090316>
- Pendergrass JC, Targum SD, Harrison JE (2018) Cognitive impairment associated with cancer: a brief review. *Innov Clin Neurosci* 15(1–2):36–44
- Pereira J, Hanson J, Bruera E (1997) The frequency and clinical course of cognitive impairment in patients with terminal cancer. *Cancer* 79(4):835–842
- Pergolizzi D, Root JC, Pan H et al (2019) Episodic memory for visual scenes suggests compensatory brain activity in breast cancer patients: a prospective longitudinal fMRI study. *Brain Imaging Behav*. <https://doi.org/10.1007/s11682-019-00038-2>
- Piccirillo JF, Hardin FM, Nicklaus J et al (2015) Cognitive impairment after chemotherapy related to atypical network architecture for executive control. *Oncology* 88(6):360–368. <https://doi.org/10.1159/000370117>
- Pomykala KL, Ganz PA, Bower JE et al (2013) The association between pro-inflammatory cytokines, regional cerebral metabolism, and cognitive complaints following adjuvant chemotherapy for breast cancer. *Brain Imaging Behav* 7(4):511–523. <https://doi.org/10.1007/s11682-013-9243-2>
- Ponto LL, Menda Y, Magnotta VA, Yamada TH, Denburg NL, Schultz SK (2015) Frontal hypometabolism in elderly breast cancer survivors determined by [(18)F]fluorodeoxyglucose (FDG) positron emission tomography (PET): a pilot study. *Int J Geriatr Psychiatry* 30(6):587–594. <https://doi.org/10.1002/gps.4189>
- Prokasheva S, Faran Y, Cwikel J, Geffen DB (2011) Analysis of memory deficits following chemotherapy in breast cancer survivors: evidence from the doors and people test. *J Psychosoc Oncol* 29(5):499–514. <https://doi.org/10.1080/07347332.2011.600751>
- Purves D (2004) Neural signaling neuroscience, 3rd edn. Sinauer Associates, pp 129–142
- Quesnel C, Savard J, Ivers H (2009) Cognitive impairments associated with breast cancer treatments: results from a longitudinal study. *Breast Cancer Res Treat* 116(1):113–123. <https://doi.org/10.1007/s10549-008-0114-2>
- Ramvalho M, Fontes F, Ruano L, Pereira S, Lunet N (2017) Cognitive impairment in the first year after breast cancer diagnosis: a prospective cohort study. *Breast* 32:173–178. <https://doi.org/10.1016/j.breast.2017.01.018>
- Reid-Arndt SA, Yee A, Perry MC, Hsieh C (2009) Cognitive and psychological factors associated with early posttreatment functional outcomes in breast cancer survivors. *J Psychosoc Oncol* 27(4):415–434. <https://doi.org/10.1080/07347330903183117>
- Ren X, Boriero D, Chaiswing L, Bondada S, St Clair DK (1865) Butterfield DA (2019a) Plausible biochemical mechanisms of chemotherapy-induced cognitive impairment (“chemobrain”), a condition that significantly impairs the quality of life of many cancer survivors. *Biochim Biophys Acta Mol Basis Dis* 6:1088–1097. <https://doi.org/10.1016/j.bbadis.2019.02.007>
- Ren X, Keeney JTR, Miriyala S et al (2019b) The triangle of death of neurons: Oxidative damage, mitochondrial dysfunction, and loss of choline-containing biomolecules in brains of mice treated with doxorubicin. Advanced insights into mechanisms of chemotherapy induced cognitive impairment (“chemobrain”) involving TNF- α . *Free Radic Biol Med* 134:1–8. <https://doi.org/10.1016/j.freeradbiomed.2018.12.029>
- Ren X, St Clair DK, Butterfield DA (2017) Dysregulation of cytokine mediated chemotherapy induced cognitive impairment. *Pharmacol Res* 117:267–273. <https://doi.org/10.1016/j.phrs.2017.01.001>
- Rice A, Michaelis ML, Georg G, Liu Y, Turunen B, Audus KL (2003) Overcoming the blood-brain barrier to taxane delivery for neurodegenerative diseases and brain tumors. *J Mol Neurosci* 20(3):339–343. <https://doi.org/10.1385/JMN:20:3:339>
- Richardson J, Smith J, McCall G, Richardson A, Pilkington K, Kirsch I (2007) Hypnosis for nausea and vomiting in cancer chemotherapy: a systematic review of the research evidence. *Euro J Cancer Care* 16(5):402–412. <https://doi.org/10.1111/j.1365-2354.2006.00736.x>
- Ruiz-Ojeda FJ, Olza J, Gil Á, Aguilera CM (2018) Oxidative Stress and Inflammation in Obesity and Metabolic Syndrome Obesity. p 1–15
- Sadighi S, Mohagheghi MA, Montazeri A, Sadighi Z (2006) Quality of life in patients with advanced gastric cancer: a randomized trial comparing docetaxel, cisplatin, 5-FU (TCF) with epirubicin, cisplatin, 5-FU (ECF). *BMC Cancer* 6:274. <https://doi.org/10.1186/1471-2407-6-274>
- Sales MVC, Suemoto CK, Apolinario D et al (2019) Effects of adjuvant chemotherapy on cognitive function of patients with early-stage colorectal cancer. *Clin Colorectal Cancer* 18(1):19–27. <https://doi.org/10.1016/j.clcc.2018.09.002>
- Sardi I, La Marca G, Giovannini MG et al (2011) Detection of doxorubicin hydrochloride accumulation in the rat brain after morphine treatment by mass spectrometry. *Cancer Chemother Pharmacol* 67(6):1333–1340. <https://doi.org/10.1007/s00280-010-1429-3>
- Schagen SB, Boogerd W, Muller MJ et al (2008) Cognitive complaints and cognitive impairment following BEP chemotherapy in patients with testicular cancer. *Acta Oncol* 47(1):63–70. <https://doi.org/10.1080/02841860701518058>
- Schagen SB, Hamburger HL, Muller MJ, Boogerd W, van Dam FS (2001) Neurophysiological evaluation of late effects of adjuvant high-dose chemotherapy on cognitive function. *J Neurooncol* 51(2):159–165. <https://doi.org/10.1023/a:1010635229762>
- Schagen SB, Muller MJ, Boogerd W, Mellenbergh GJ, Van Dam FS (2006) Change in cognitive function after chemotherapy: a prospective longitudinal study in breast cancer patients. *J Natl Cancer Inst* 98(23):1742–1745. <https://doi.org/10.1093/jnci/djj470>
- Schagen SB, Muller MJ, Boogerd W et al (2002) Late effects of adjuvant chemotherapy on cognitive function: a follow-up study in breast cancer patients. *Ann Oncol* 13(9):1387–1397. <https://doi.org/10.1023/a:101063522976>
- Schagen SB, van Dam FS, Muller MJ, Boogerd W, Lindeboom J, Bruning PF (1999) Cognitive deficits after postoperative adjuvant chemotherapy for breast carcinoma. *Cancer* 85(3):640–650
- Scherling CS, Smith A (2013) Opening up the window into “chemobrain”: a neuroimaging review. *Sensors (basel)* 13(3):3169–3203. <https://doi.org/10.3390/s130303169>
- Schrauwen W, Van de Cavey J, Vingerhoets G, Vanheule S, Van den Broecke R, Denys H (2020) Heterogeneous response of chemotherapy-related cognitive decline in patients with breast cancer: a prospective study. *J Int Neuropsychol Soc* 26(8):806–814. <https://doi.org/10.1017/s1355617720000296>
- Seigers R, Fardell JE (2011) Neurobiological basis of chemotherapy-induced cognitive impairment: a review of rodent research. *Neurosci Biobehav Rev* 35(3):729–741. <https://doi.org/10.1016/j.neubiorev.2010.09.006>
- Seigers R, Loos M, Van Tellingen O, Boogerd W, Smit AB, Schagen SB (2016) Neurobiological changes by cytotoxic agents in mice. *Behav Brain Res* 299:19–26. <https://doi.org/10.1016/j.bbr.2015.10.057>
- Seigers R, Schagen SB, Beerling W et al (2008) Long-lasting suppression of hippocampal cell proliferation and impaired cognitive performance by methotrexate in the rat. *Behav Brain Res* 186(2):168–175. <https://doi.org/10.1016/j.bbr.2007.08.004>
- Seigers R, Schagen SB, Coppens CM et al (2009) Methotrexate decreases hippocampal cell proliferation and induces memory deficits in rats. *Behav Brain Res* 201(2):279–284. <https://doi.org/10.1016/j.bbr.2009.02.025>

- Seigers R, Schagen SB, Van Tellingen O, Dietrich J (2013) Chemotherapy-related cognitive dysfunction: current animal studies and future directions. *Brain Imaging Behav* 7(4):453–459. <https://doi.org/10.1007/s11682-013-9250-3>
- Seiter K (2005) Toxicity of the topoisomerase II inhibitors. *Expert Opin Drug Safety* 4(2):219–234. <https://doi.org/10.1517/14740338.4.2.219>
- Seo EJ, Klauck SM, Efferth T, Panossian A (2018) Adaptogens in chemobrain (Part I): Plant extracts attenuate cancer chemotherapy-induced cognitive impairment—Transcriptome-wide microarray profiles of neuroglia cells. *Phytomedicine* 55:80–91. <https://doi.org/10.1016/j.phymed.2018.10.022>
- Shen CY, Chen VC, Yeh DC et al (2019) Association of functional dorsal attention network alterations with breast cancer and chemotherapy. *Sci Rep* 9(1):104. <https://doi.org/10.1038/s41598-018-36380-6>
- Shih YC, Smieliauskas F, Geynisman DM, Kelly RJ, Smith TJ (2015) Trends in the cost and use of targeted cancer therapies for the privately insured nonelderly: 2001 to 2011. *J Clin Oncol* 33(19):2190–2196. <https://doi.org/10.1200/jco.2014.58.2320>
- Shilling V, Jenkins V (2007) Self-reported cognitive problems in women receiving adjuvant therapy for breast cancer. *Eur J Oncol Nurs* 11(1):6–15. <https://doi.org/10.1016/j.ejon.2006.02.005>
- Shilling V, Jenkins V, Morris R, Deutsch G, Bloomfield D (2005) The effects of adjuvant chemotherapy on cognition in women with breast cancer—preliminary results of an observational longitudinal study. *Breast* 14(2):142–150. <https://doi.org/10.1016/j.breast.2004.10.004>
- Siddiqui T, Deshmukh VD, Karimjee N (1992) Subclinical cognitive deficits in cancer patients: a preliminary P300 study. *Clin Electroencephalogr* 23(3):132–136
- Silverman DH, Dy CJ, Castellon SA et al (2007) Altered frontocortical, cerebellar, and basal ganglia activity in adjuvant-treated breast cancer survivors 5–10 years after chemotherapy. *Breast Cancer Res Treat* 103(3):303–311. <https://doi.org/10.1007/s10549-006-9380-z>
- Simo M, Gurtubay-Antolin A, Vaquero L, Bruna J, Rodriguez-Fornells A (2018) Performance monitoring in lung cancer patients pre- and post-chemotherapy using fine-grained electrophysiological measures. *Neuroimage Clin* 18:86–96. <https://doi.org/10.1016/j.nicl.2017.12.032>
- Simo M, Rifa-Ros X, Rodriguez-Fornells A, Bruna J (2013) Chemobrain: a systematic review of structural and functional neuroimaging studies. *Neurosci Biobehav Rev* 37(8):1311–1321. <https://doi.org/10.1016/j.neubiorev.2013.04.015>
- Skaali T, Fossa SD, Andersson S et al (2011) A prospective study of neuropsychological functioning in testicular cancer patients. *Ann Oncol* 22(5):1062–1070. <https://doi.org/10.1093/annonc/mdq553>
- Small BJ, Rawson KS, Walsh E et al (2011) Catechol-O-methyltransferase genotype modulates cancer treatment-related cognitive deficits in breast cancer survivors. *Cancer* 117(7):1369–1376. <https://doi.org/10.1002/cncr.25685>
- Smith JA, Das A, Ray SK, Banik NL (2012) Role of pro-inflammatory cytokines released from microglia in neurodegenerative diseases. *Brain Res Bull* 87(1):10–20. <https://doi.org/10.1016/j.brainresbull.2011.10.004>
- Snell RS (2010) Clinical neuroanatomy. Lippincott Williams & Wilkins
- Sorokin J, Saboury B, Ahn JA, Moghbel M, Basu S, Alavi A (2014) Adverse functional effects of chemotherapy on whole-brain metabolism: a PET/CT quantitative analysis of FDG metabolic pattern of the “chemo-brain.” *Clin Nucl Med* 39(1):e35–e39. <https://doi.org/10.1097/RLU.0b013e318292aa81>
- Stewart A, Collins B, Mackenzie J, Tomiak E, Verma S, Bielajew C (2008) The cognitive effects of adjuvant chemotherapy in early stage breast cancer: a prospective study. *Psychooncology* 17(2):122–130. <https://doi.org/10.1002/pon.1210>
- Stewart DJ, Grewaal D, Green RM et al (1993) Concentrations of doxorubicin and its metabolites in human autopsy heart and other tissues. *Anticancer Res* 13(6a):1945–1952
- Stouten-Kemperman MM, de Ruiter MB, Caan MW et al (2015) Lower cognitive performance and white matter changes in testicular cancer survivors 10 years after chemotherapy. *Hum Brain Mapp* 36(11):4638–4647. <https://doi.org/10.1002/hbm.22942>
- Tacar O, Sriamornsak P, Dass CR (2013) Doxorubicin: an update on anticancer molecular action, toxicity and novel drug delivery systems. *J Pharm Pharmacol* 65(2):157–170. <https://doi.org/10.1111/j.2042-7158.2012.01567.x>
- Tager FA, McKinley PS, Schnabel FR et al (2010) The cognitive effects of chemotherapy in post-menopausal breast cancer patients: a controlled longitudinal study. *Breast Cancer Res Treat* 123(1):25–34. <https://doi.org/10.1007/s10549-009-0606-8>
- Tangpong J, Cole MP, Sultana R et al (2006) Adriamycin-induced, TNF- α -mediated central nervous system toxicity. *Neurobiol Dis* 23(1):127–139. <https://doi.org/10.1016/j.nbd.2006.02.013>
- Tangpong J, Miriyala S, Noel T, Sinthupibulyakit C, Jungsuwadee P, St Clair DK (2011) Doxorubicin-induced central nervous system toxicity and protection by xanthone derivative of *Garcinia mangostana*. *Neuroscience* 175:292–299. <https://doi.org/10.1016/j.neuroscience.2010.11.007>
- Tao L, Lin H, Yan Y, et al. (2017) Impairment of the executive function in breast cancer patients receiving chemotherapy treatment: a functional MRI study. *Eur J Cancer Care (Engl)* 26(6). doi:<https://doi.org/10.1111/ecc.12553>
- Tao L, Wang L, Chen X et al (2020) Modulation of Interhemispheric functional coordination in breast cancer patients receiving chemotherapy. *Front Psychol* 11:1689. <https://doi.org/10.3389/fpsyg.2020.01689>
- Tchen N, Juffs HG, Downie FP et al (2003) Cognitive function, fatigue, and menopausal symptoms in women receiving adjuvant chemotherapy for breast cancer. *J Clin Oncol* 21(22):4175–4183. <https://doi.org/10.1200/jco.2003.01.119>
- Thomas TC, Beitchman JA, Pomerleau F et al (2017) Acute treatment with doxorubicin affects glutamate neurotransmission in the mouse frontal cortex and hippocampus. *Brain Res* 1672:10–17. <https://doi.org/10.1016/j.brainres.2017.07.003>
- Tong Y, Wang K, Sheng S, Cui J (2020) Polydatin ameliorates chemotherapy-induced cognitive impairment (chemobrain) by inhibiting oxidative stress, inflammatory response, and apoptosis in rats. *Biosci Biotechnol Biochem* 84(6):1201–1210. <https://doi.org/10.1080/09168451.2020.1722057>
- Trachootham D, Lu W, Ogasawara MA, Nilsa RD, Huang P (2008) Redox regulation of cell survival. *Antioxid Redox Signal* 10(8):1343–1374. <https://doi.org/10.1089/ars.2007.1957>
- Urruticoechea A, Alemany R, Balart J, Villanueva A, Viñals F, Capella G (2010) Recent advances in cancer therapy: an overview. *Curr Pharmaceutical Des* 16(1):3–10. <https://doi.org/10.2174/138161210789941847>
- Utne I, Loyland B, Grov EK et al (2018) Distinct attentional function profiles in older adults receiving cancer chemotherapy. *Eur J Oncol Nurs* 36:32–39. <https://doi.org/10.1016/j.ejon.2018.08.006>
- van Dam FS, Schagen SB, Muller MJ et al (1998) Impairment of cognitive function in women receiving adjuvant treatment for high-risk breast cancer: high-dose versus standard-dose chemotherapy. *J Natl Cancer Inst* 90(3):210–218
- Van Dyk K, Bower JE, Crespi CM, Petersen L, Ganz PA (2018) Cognitive function following breast cancer treatment and associations with concurrent symptoms. *NPJ Breast Cancer* 4:25. <https://doi.org/10.1038/s41523-018-0076-4>
- Vearncombe KJ, Rolfe M, Wright M, Pachana NA, Andrew B, Beadle G (2009) Predictors of cognitive decline after chemotherapy in

- breast cancer patients. *J Int Neuropsychol Soc* 15(6):951–962. <https://doi.org/10.1017/s1355617709900567>
- Vitor T, Kozasa EH, Bressan RA et al (2019) Impaired brain dopamine transporter in chemobrain patients submitted to brain SPECT imaging using the technetium-99m labeled tracer TRODAT-1. *Ann Nucl Med* 33(4):269–279. <https://doi.org/10.1007/s12149-019-01331-2>
- Von Ah D, Harvison KW, Monahan PO et al (2009) Cognitive function in breast cancer survivors compared to healthy age- and education-matched women. *Clin Neuropsychol* 23(4):661–674. <https://doi.org/10.1080/13854040802541439>
- Wahdan SA, El-Derany MO, Abdel-Maged AE, Azab SS (2020) Abrogating doxorubicin-induced chemobrain by immunomodulators IFN-beta 1a or infliximab: Insights to neuroimmune mechanistic hallmarks. *Neurochem Int* 138:104777. <https://doi.org/10.1016/j.neuint.2020.104777>
- Wang C, Zhao Y, Wang L et al (2021) C-phycocyanin mitigates cognitive impairment in doxorubicin-induced chemobrain: impact on neuroinflammation, oxidative stress, and brain mitochondrial and synaptic alterations. *Neurochem Res* 46(2):149–158. <https://doi.org/10.1007/s11064-020-03164-2>
- Wang D, Wang B, Liu Y, Dong X, Su Y, Li S (2019) Protective effects of ACY-1215 against chemotherapy-related cognitive impairment and brain damage in mice. *Neurochem Res* 44(11):2460–2469. <https://doi.org/10.1007/s11064-019-02882-6>
- Wang L, Apple AC, Schroeder MP et al (2016) Reduced prefrontal activation during working and long-term memory tasks and impaired patient-reported cognition among cancer survivors postchemotherapy compared with healthy controls. *Cancer* 122(2):258–268. <https://doi.org/10.1002/cncr.29737>
- Wang L, Zou L, Chen Q et al (2020) Gray matter structural network disruptions in survivors of acute lymphoblastic leukemia with chemotherapy treatment. *Acad Radiol* 27(3):e27–e34. <https://doi.org/10.1016/j.acra.2019.04.010>
- Wang XM, Walitt B, Saligan L, Tiwari AF, Cheung CW, Zhang ZJ (2015) Chemobrain: a critical review and causal hypothesis of link between cytokines and epigenetic reprogramming associated with chemotherapy. *Cytokine* 72(1):86–96. <https://doi.org/10.1016/j.cyto.2014.12.006>
- Wefel JS, Lenzi R, Theriault R, Buzdar AU, Cruickshank S, Meyers CA (2004a) “Chemobrain” in breast carcinoma? a prologue. *Cancer* 101(3):466–475. <https://doi.org/10.1002/cncr.20393>
- Wefel JS, Lenzi R, Theriault RL, Davis RN, Meyers CA (2004b) The cognitive sequelae of standard-dose adjuvant chemotherapy in women with breast carcinoma: results of a prospective, randomized, longitudinal trial. *Cancer* 100(11):2292–2299. <https://doi.org/10.1002/cncr.20272>
- Wefel JS, Saleeba AK, Buzdar AU, Meyers CA (2010) Acute and late onset cognitive dysfunction associated with chemotherapy in women with breast cancer. *Cancer* 116(14):3348–3356. <https://doi.org/10.1002/cncr.25098>
- Weiss B (2008) Chemobrain: a translational challenge for neurotoxicology. *Neurotoxicology* 29(5):891–898. <https://doi.org/10.1016/j.neuro.2008.03.009>
- Weiss HD, Walker MD, Wiernik PH (1974a) Neurotoxicity of commonly used antineoplastic agents. *N Engl J Med* 291(3):127–133
- Weiss HD, Walker MD, Wiernik PH (1974b) Neurotoxicity of commonly used antineoplastic agents (second of two parts). *New Engl J Med* 291(3):127–133. <https://doi.org/10.1056/nejm197407182910305>
- Whitford HS, Kalinowski P, Schembri A et al (2019) The impact of chemotherapy on cognitive function: a multicentre prospective cohort study in testicular cancer. *Support Care Cancer*. <https://doi.org/10.1007/s00520-019-05095-3>
- Whitford HS, Kalinowski P, Schembri A et al (2020) The impact of chemotherapy on cognitive function: a multicentre prospective cohort study in testicular cancer. *Support Care Cancer* 28(7):3081–3091. <https://doi.org/10.1007/s00520-019-05095-3>
- Wieneke MH, Dienst ER (1995) Neuropsychological assessment of cognitive functioning following chemotherapy for breast cancer. *Psycho-Oncol* 4(1):61–66
- Wigmore PM, Mustafa S, El-Beltagy M, Lyons L, Umka J, Bennett G (2010) Effects of 5-FU. *Adv Exp Med Biol* 678:157–164. https://doi.org/10.1007/978-1-4419-6306-2_20
- Williams AM, Shah R, Shayne M et al (2018) Associations between inflammatory markers and cognitive function in breast cancer patients receiving chemotherapy. *J Neuroimmunol* 314:17–23. <https://doi.org/10.1016/j.jneuroim.2017.10.005>
- Williams AM, van Wijngaarden E, Seplaki CL et al (2020) Cognitive function in patients with chronic lymphocytic leukemia: a cross-sectional study examining effects of disease and treatment. *Leuk Lymphoma* 61(7):1627–1635. <https://doi.org/10.1080/10428194.2020.1728748>
- Wouters H, Baars JW, Schagen SB (2016) Neurocognitive function of lymphoma patients after treatment with chemotherapy. *Acta Oncol* 55(9–10):1121–1125. <https://doi.org/10.1080/0284186x.2016.1189092>
- Xu Z, Luo F, Wang Y et al (2020) Cognitive impairments in breast cancer survivors treated with chemotherapy: a study based on event-related potentials. *Cancer Chemother Pharmacol* 85(1):61–67. <https://doi.org/10.1007/s00280-019-03994-0>
- Yamada TH, Denburg NL, Beglinger LJ, Schultz SK (2010) Neuropsychological outcomes of older breast cancer survivors: cognitive features ten or more years after chemotherapy. *J Neuropsychiatry Clin Neurosci* 22(1):48–54. <https://doi.org/10.1176/appi.neuropsych.22.1.48>
- Yang GS, Mi X, Jackson-Cook CK et al (2020) Differential DNA methylation following chemotherapy for breast cancer is associated with lack of memory improvement at one year. *Epigenetics* 15(5):499–510. <https://doi.org/10.1080/15592294.2019.1699695>
- Yang H, Villani RM, Wang H et al (2018) The role of cellular reactive oxygen species in cancer chemotherapy. *J Exp Clin Cancer Res* 37(1):266–266. <https://doi.org/10.1186/s13046-018-0909-x>
- Yang M, Kim JS, Kim J et al (2012) Acute treatment with methotrexate induces hippocampal dysfunction in a mouse model of breast cancer. *Brain Res Bull* 89(1–2):50–56. <https://doi.org/10.1016/j.brainresbull.2012.07.003>
- Yarlagadda A, Alfson E, Clayton AH (2009) The blood brain barrier and the role of cytokines in neuropsychiatry. *Psychiatry (edgmont)* 6(11):18–22
- Yokoyama C, Sueyoshi Y, Ema M, Mori Y, Takaishi K, Hisatomi H (2017) Induction of oxidative stress by anticancer drugs in the presence and absence of cells. *Oncol Lett* 14(5):6066–6070. <https://doi.org/10.3892/ol.2017.6931>
- Zeng Y, Cheng ASK, Song T et al (2019) Changes in functional brain networks and neurocognitive function in Chinese gynecological cancer patients after chemotherapy: a prospective longitudinal study. *BMC Cancer* 19(1):386. <https://doi.org/10.1186/s12885-019-5576-6>
- Zhang J-M, An J (2007) Cytokines, inflammation, and pain. *Int Anesthesiol Clin* 45(2):27–37. <https://doi.org/10.1097/AIA.0b013e318034194e>
- Zhang J, Zhang L, Yan Y et al (2015) Are capecitabine and the active metabolite 5-FU CNS penetrable to treat breast cancer brain metastasis? *Drug Metab Disposition* 43(3):411–417. <https://doi.org/10.1124/dmd.114.061820>
- Zhang Y, Chen YC, Hu L et al (2019) Chemotherapy-induced functional changes of the default mode network in patients with lung cancer. *Brain Imaging Behav*. <https://doi.org/10.1007/s11682-018-0030-y>

- Zheng Y, Luo J, Bao P et al (2014) Long-term cognitive function change among breast cancer survivors. *Breast Cancer Res Treat* 146(3):599–609. <https://doi.org/10.1007/s10549-014-3044-1>
- Zhou W, Kavelaars A, Heijnen CJ (2016) Metformin prevents cisplatin-induced cognitive impairment and brain damage in mice. *PLoS ONE* 11(3):e0151890. <https://doi.org/10.1371/journal.pone.0151890>
- Zhou X, Xu P, Dang R et al (2018) The involvement of autophagic flux in the development and recovery of doxorubicin-induced neurotoxicity. *Free Radical Biol Med* 129:440–445. <https://doi.org/10.1016/j.freeradbiomed.2018.10.418>
- Zimberg M, Berenson S (1990) Delirium in patients with cancer: nursing assessment and intervention. *Oncol Nurs Forum* 17(4):529–538
- Zuccato C, Cattaneo E (2009) Brain-derived neurotrophic factor in neurodegenerative diseases. *Nat Rev Neurol* 5(6):311. <https://doi.org/10.1038/nrneurol.2009.54>
- Zuniga KE, Bishop NJ (2018) Recent cancer treatment and memory decline in older adults: An analysis of the 2002–2012 Health and Retirement Study. *J Geriatr Oncol* 9(3):186–193. <https://doi.org/10.1016/j.jgo.2017.10.004>

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